ADVANCED OLYMPIAD SERIES

Olympiad Geometry: Collected Problems

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Research edition • 6 November 2026

Definitions. Certificates. Complete arguments.

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Scope and attribution

This volume contains sixteen restated problems from the checked core, followed by 493 source-provided statements. It is a partial collection. Source classifications, numbering and transcriptions are not all independently verified. Mixed-domain records may occur in other subject volumes. The corresponding proofs are in the separate worked volume and the individually labeled source-solution files.

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MathNet: Shaden Alshammari, Kevin Wen, Abrar Zainal, Mark Hamilton, Navid Safaei, Sultan Albarakati, William T. Freeman and Antonio Torralba, *MathNet: A Global Multimodal Benchmark for Mathematical Reasoning and Retrieval*, ICLR 2026. Sources: https://mathnet.mit.edu/ ; https://huggingface.co/datasets/ShadenA/MathNet ; https://creativecommons.org/licenses/by/4.0/ . Original source contributors are credited in the ledger and individual problem files.

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The sixteen checked-core problems

G01. Two similarities sharing a secant

Source: IMO shortlist 2015, G1. **Record:** `imo_2015_29f541`.

In an acute triangle ABC with orthocenter H, complete the parallelogram $ABGH$. On line GH choose I so that AC passes through the midpoint of HI. Let $J \neq C$ be the other intersection of AC with (GCI). Prove $IJ = AH$.

G02. Reflect the two lines

Source: IMO shortlist 2015, G2. **Record:** `imo_2015_7c01e3`.

Triangle ABC lies on a circle Ω with center O. A circle centered at A cuts BC at D, E in the order B, D, E, C, and cuts the arcs AB and AC of Ω not containing the third vertex at F, G, respectively. Let $K \neq B$ be the intersection of AB with (BDF), and $L \neq C$ the intersection of AC with (CEG). If distinct lines FK, GL meet at X, prove A, X, O collinear.

G03. An angle condition is a distance locus

Source: IMO shortlist 2006, G1. **Record:** `imo_2006_e64448`.

Let I be the incenter of ABC. An interior point P satisfies $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. Prove $AP \geq AI$, with equality only at $P = I$.

G04. Reflection completes a cyclicity proof

Source: IMO shortlist 2012, G2. **Record:** `imo_2012_758467`.

In cyclic quadrilateral $ABCD$, the diagonals meet at E. The extensions of AD, BC beyond A, B meet at F. Complete parallelogram $ECGD$, and reflect E in AD to obtain H. Prove D, H, F, G concyclic.

G05. Reflected lines meet on the mirror

Source: IMO shortlist 2013, G2. **Record:** `imo_2013_22bd4a`.

Let M, N be the midpoints of AB, AC, and T the midpoint of the arc BC not containing A on (ABC). The circle (AMT) meets the perpendicular bisector of AC at an interior point X of ABC; similarly (ANT) meets the perpendicular bisector of AB at an interior point Y. If $K = MN \cap XY$, prove $KA = KT$.

G06. Double the cevians

Source: IMO shortlist 2014, G1. **Record:** `imo_2014_a0b822`.

In an acute triangle, choose P, Q on BC with $\angle PAB = C$ and $\angle QAC = B$. Extend AP, AQ to M, N so that $PM = PA$ and $QN = QA$. Show that BM and CN meet on (ABC).

G07. Pack two similar triangles

Source: IMO shortlist 2014, G2. **Record:** `imo_2014_32e67b`.

Points K, L, M on BC, CA, AB have concurrent cevians AK, BL, CM. Prove that two of ALM, BMK, CKL have inradii whose sum is at least the inradius r of ABC.

G08. Convert intersection ratios into half angles

Source: IMO shortlist 2016, G2. **Record:** `imo_2016_022acb`.

Let I be the incenter, M the midpoint of BC, and D the foot from I to BC. The perpendicular to AI through I meets AB, AC at F, E. Let $X \neq A$ be the other common point of (AEF) and (ABC). Prove that XD and AM meet on (ABC).

G09. The hidden orthocenter of a pentagon

Source: IMO shortlist 2017, G1. **Record:** `imo_2017_e0103e`.

A convex pentagon $ABCDE$ satisfies $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that AC, BD and the perpendicular from E to BC concur.

G10. A half-turn creates the tangent

Source: IMO 2017, Problem 4; shortlist G2. **Record:** `imo_2017_bb6ddb`.

Points R, S lie on Ω, and $R' = 2S - R$. Let I lie on the smaller arc RS. The circle $\Gamma = (ISR')$ meets the tangent to Ω at R twice; call the nearer intersection to R point A. If AI meets Ω again at J, prove JR' tangent to Γ.

G11. Two tangents encode a reflection

Source: IMO shortlist 2019, G1. **Record:** `imo_2019_be46cd`.

A circle through A meets AB, AC again at D, E and crosses BC at F, G in that order. Tangents at F to (BDF) and at G to (CEG) meet at $T \neq A$. Prove $AT \parallel BC$.

G12. Close two circles around the target point

Source: IMO shortlist 2021, G1. **Record:** `imo_2021_18d688`.

In parallelogram $ABCD$, assume $AC = BC$. Take P beyond B on AB. Let $Q \neq D$ be the intersection of PD with (ACD), and $R \neq P$ the intersection of PC with (APQ). Prove CD, AQ, BR concurrent.

G13. A circle equation linearizes the ratio

Source: IMO shortlist 2015, G4. **Record:** `imo_2015_b11e64`.

In an acute triangle ABC, let M be the midpoint of AC. A circle through B, M meets BA, BC again at P, Q. Complete parallelogram $BPTQ$, and assume $T \in (ABC)$. Determine BT/BM.

G14. Transfer a tangent through equal powers

Source: IMO shortlist 2015, G6. **Record:** `imo_2015_d499b1`.

An acute triangle ABC has $AB > AC$, orthocenter H, base midpoint M, and altitude foot F on BC. On its circumcircle Ω, choose Q, K so that $\angle AQH = \angle QKH = 90^\circ$. Prove that (KQH) and (KFM) are tangent at K.

G15. Equal segments turn spiral similarity into rotation

Source: IMO shortlist 2024, G1. **Record:** `imo_2024_a60b6c`.

A cyclic quadrilateral satisfies $AC < BD < AD$ and $\angle DBA < 90^\circ$. On the side of AD opposite C, choose E with $DE \parallel AB$, $DE = AC$, and F with $AF \parallel CD$, $AF = BD$. Prove that the perpendicular bisectors of BC, EF meet on $(ABCD)$.

G16. Reflect a vertex in the incenter

Source: IMO shortlist 2024, G2. **Record:** `imo_2024_7db46b`.

A triangle has $AB < AC < BC$, incenter I, and incircle. The other tangent parallel to AC meets the interior of BC at X; the other tangent parallel to AB meets it at Y. Let $P \neq A$ be the intersection of AI with (ABC), and let K, L be the midpoints of AB, AC. Prove $\angle KIL + \angle YPX = 180^\circ$.

APMO source statements

1989 — 3

Record: `apm_1989_6cbaa2`. **Category:** contest. Source.

Let A_1, A_2, A_3 be three points in the plane, and for convenience, let $A_4 = A_1$, $A_5 = A_2$. For $n = 1, 2$, and 3, suppose that B_n is the midpoint of A_nA_{n+1}, and suppose that C_n is the midpoint of A_nB_n. Suppose that A_nC_{n+1} and B_nA_{n+2} meet at D_n, and that A_nB_{n+1} and C_nA_{n+2} meet at E_n. Calculate the ratio of the area of triangle $D_1D_2D_3$ to the area of triangle $E_1E_2E_3$. Answer: $\frac{25}{49}$.

1991 — 1

Record: `apm_1991_942510`. **Category:** contest. Source.

Let G be the centroid of triangle ABC and M be the midpoint of BC. Let X be on AB and Y on AC such that the points X, Y, and G are collinear and XY and BC are parallel. Suppose that XC and GB intersect at Q and YB and GC intersect at P. Show that triangle MPQ is similar to triangle ABC.

1991 — 5

Record: `apm_1991_ab7085`. **Category:** contest. Source.

Given are two tangent circles and a point P on their common tangent perpendicular to the lines joining their centres. Construct with ruler and compass all the circles that are tangent to these two circles and pass through the point P.

1991 — 2

Record: `apm_1991_b65647`. **Category:** contest. Source.

Suppose there are 997 points given in a plane. If every two points are joined by a line segment with its midpoint coloured in red, show that there are at least 1991 red points in the plane. Can you find a special case with exactly 1991 red points?

1992 — 4

Record: `apm_1992_0e43a2`. **Category:** contest. Source.

Determine all pairs (h, s) of positive integers with the following property: If one draws h horizontal lines and another s lines which satisfy (i) they are not horizontal, (ii) no two of them are parallel, (iii) no three of the $h + s$ lines are concurrent, then the number of regions formed by these $h + s$ lines is 1992.

Answer: $(995, 1)$, $(176, 10)$, and $(80, 21)$.

1993 — 5

Record: `apm_1993_2c32ee`. **Category:** contest. Source.

Let $P_1, P_2, \ldots, P_{1993} = P_0$ be distinct points in the xy-plane with the following properties: (i) both coordinates of P_i are integers, for $i = 1, 2, \ldots, 1993$; (ii) there is no point other than P_i and P_{i+1} on the line segment joining P_i with P_{i+1} whose coordinates are both integers, for $i = 0, 1, \ldots, 1992$. Prove that for some $i, 0 \leq i \leq 1992$, there exists a point Q with coordinates (q_x, q_y) on the line segment joining P_i with P_{i+1} such that both $2q_x$ and $2q_y$ are odd integers.

1993 — 1

Record: `apm_1993_f901a7`. **Category:** contest. Source.

Let $ABCD$ be a quadrilateral such that all sides have equal length and angle $\angle ABC$ is 60 degrees. Let ℓ be a line passing through D and not intersecting the quadrilateral (except at D). Let E and F be the points of intersection of ℓ with AB and BC respectively. Let M be the point of intersection of CE and AF. Prove that $CA^2 = CM \times CE$.

1994 — 4

Record: `apm_1994_74e082`. **Category:** contest. Source.

Is there an infinite set of points in the plane such that no three points are collinear, and the distance between any two points is rational?

1994 — 2

Record: `apm_1994_82c5be`. **Category:** contest. Source.

Given a nondegenerate triangle ABC, with circumcentre O, orthocentre H, and circumradius R, prove that $|OH| < 3R$.

1999 — 5

Record: `apm_1999_563b73`. **Category:** contest. Source.

Let S be a set of $2n + 1$ points in the plane such that no three are collinear and no four concyclic. A circle will be called good if it has 3 points of S on its circumference, $n - 1$ points in its interior and $n - 1$ in its exterior. Prove that the number of good circles has the same parity as n.

2003 — 2

Record: `apm_2003_423680`. **Category:** contest. Source.

Suppose $ABCD$ is a square piece of cardboard with side length a. On a plane are two parallel lines ℓ_1 and ℓ_2, which are also a units apart. The square $ABCD$ is placed on the plane so that sides AB and AD intersect ℓ_1 at E and F respectively. Also, sides CB and CD intersect ℓ_2 at G and H respectively. Let the perimeters of $\triangle AEF$ and $\triangle CGH$ be m_1 and m_2 respectively. Prove that no matter how the square was placed, $m_1 + m_2$ remains constant.

2003 — 4

Record: `apm_2003_7dec2d`. **Category:** contest. Source.

Let a, b, c be the sides of a triangle, with $a+b+c=1$, and let $n \geq 2$ be an integer. Show that

$$\sqrt[n]{a^n+b^n} + \sqrt[n]{b^n+c^n} + \sqrt[n]{c^n+a^n} < 1 + \frac{\sqrt[n]{2}}{2}$$

2004 — 3

Record: `apm_2004_f5148e`. **Category:** contest. Source.

Let a set S of 2004 points in the plane be given, no three of which are collinear. Let $\mathcal{L}$ denote the set of all lines (extended indefinitely in both directions) determined by pairs of points from the set. Show that it is possible to colour the points of S with at most two colours, such that for any points p, q of S, the number of lines in $\mathcal{L}$ which separate p from q is odd if and only if p and q have the same colour.

Note: A line ℓ separates two points p and q if p and q lie on opposite sides of ℓ with neither point on ℓ.

2005 — Problem 3

Record: `apm_2005_252bb2`. **Category:** contest. Source.

Prove that there exists a triangle which can be cut into 2005 congruent triangles.

2005 — Problem 5

Record: `apm_2005_7d0af9`. **Category:** contest. Source.

In a triangle ABC, points M and N are on sides AB and AC, respectively, such that $MB = BC = CN$. Let R and r denote the circumradius and the inradius of the triangle ABC, respectively. Express the ratio MN/BC in terms of R and r.

2007 — Problem 3

Record: `apm_2007_7a497c`. **Category:** contest. Source.

Consider n disks $C_1, C_2, \ldots, C_n$ in a plane such that for each $1 \leq i < n$, the center of C_i is on the circumference of C_{i+1}, and the center of C_n is on the circumference of C_1. Define the score of such an arrangement of n disks to be the number of pairs (i, j) for which C_i properly contains C_j. Determine the maximum possible score.

2008 — Problem 3

Record: `apm_2008_6c84cd`. **Category:** contest. Source.

Let Γ be the circumcircle of a triangle ABC. A circle passing through points A and C meets the sides BC and BA at D and E, respectively. The lines AD and CE meet Γ again at G and H, respectively. The tangent lines of Γ at A and C meet the line DE at L and M, respectively. Prove that the lines LH and MG meet at Γ.

2008 — Problem 1

Record: `apm_2008_f38549`. **Category:** contest. Source.

Let ABC be a triangle with $\angle A < 60^\circ$. Let X and Y be the points on the sides AB and AC, respectively, such that $CA + AX = CB + BX$ and $BA + AY = BC + CY$. Let P be the point in the plane such that the lines PX and PY are perpendicular to AB and AC, respectively. Prove that $\angle BPC < 120^\circ$.

2010 — 1

Record: `apm_2010_4c0e80`. **Category:** contest. Source.

Let ABC be a triangle with $\angle BAC \neq 90^\circ$. Let O be the circumcenter of the triangle ABC and let Γ be the circumcircle of the triangle BOC. Suppose that Γ intersects the line segment AB at P different from B, and the line segment AC at Q different from C. Let ON be a diameter of the circle Γ. Prove that the quadrilateral $APNQ$ is a parallelogram.

2010 — 4

Record: `apm_2010_de8361`. **Category:** contest. Source.

Let ABC be an acute triangle satisfying the condition $AB > BC$ and $AC > BC$. Denote by O and H the circumcenter and the orthocenter, respectively, of the triangle ABC. Suppose that the circumcircle of the triangle AHC intersects the line AB at M different from A, and that the circumcircle of the triangle AHB intersects the line AC at N different from A. Prove that the circumcenter of the triangle MNH lies on the line OH.

2013 — Problem 1

Record: `apm_2013_4c0b59`. **Category:** contest. Source.

Let ABC be an acute triangle with altitudes AD, BE and CF, and let O be the center of its circumcircle. Show that the segments OA, OF, OB, OD, OC, OE dissect the triangle ABC into three pairs of triangles that have equal areas.

2013 — Problem 5

Record: `apm_2013_8a2e44`. **Category:** contest. Source.

Let $ABCD$ be a quadrilateral inscribed in a circle ω, and let P be a point on the extension of AC such that PB and PD are tangent to ω. The tangent at C intersects PD at Q and the line AD at R. Let E be the second point of intersection between AQ and ω. Prove that B, E, R are collinear.

2016 — 1

Record: `apm_2016_703ac5`. **Category:** contest. Source.

We say that a triangle ABC is great if the following holds: for any point D on the side BC, if P and Q are the feet of the perpendiculars from D to the lines AB and AC, respectively, then the reflection of D in the line PQ lies on the circumcircle of the triangle ABC. Prove that triangle ABC is great if and only if $\angle A = 90^\circ$ and $AB = AC$.

2016 — 3

Record: apm_2016_7112d6. **Category:** contest. Source.

Let AB and AC be two distinct rays not lying on the same line, and let ω be a circle with center O that is tangent to ray AC at E and ray AB at F. Let R be a point on segment EF. The line through O parallel to EF intersects line AB at P. Let N be the intersection of lines PR and AC, and let M be the intersection of line AB and the line through R parallel to AC. Prove that line MN is tangent to ω.

2018 — 3

Record: mkd_2018_77a86a. **Category:** contest. Source.

A collection of n squares on the plane is called tri-connected if the following criteria are satisfied: (i) All the squares are congruent (ii) If two squares have a point P in common, then P is a vertex of each of the squares. (iii) Each square touches exactly three other squares. How many positive integers n are there with $2018 \leq n \leq 3018$ such that there exists a collection of n squares that is tri-connected?

2018 — 1

Record: mkd_2018_8e711e. **Category:** contest. Source.

Let H be the orthocenter of triangle ABC. Let M and N be the midpoints of the sides AB and AC respectively. Assume that H lies inside the quadrilateral $BMNC$ and that the circumcircles of triangles BMH and CNH are tangent to each other. The line through H parallel to BC intersects the circumcircles of the triangles BMH and CNH in the points K and L, respectively. Let F be the intersection point of MK and NL and let J be the incenter of triangle MHN. Prove that $FJ = FA$.

2018 — 4

Record: mkd_2018_8ede86. **Category:** contest. Source.

Let ABC be an equilateral triangle. From the vertex A we draw a ray towards the interior of the triangle such that the ray reaches one of the sides of the triangle. When the ray reaches a side, it then bounces off following the law of reflection, that is, if it arrives with a directed angle α, it leaves with a directed angle $180^\circ - \alpha$. After n bounces, the ray returns to A without ever landing on any of the other vertices. Find all possible values of n.

2020 — 1

Record: apm_2020_1a86f9. **Category:** contest. Source.

Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC. The tangent to Γ at A intersects the parallel line to BA through D at point E. The segment CE intersects Γ again at F. Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

2023 — 5

Record: apm_2023_effff8. **Category:** contest. Source.

There are n line segments on the plane, no three intersecting at a point, and each pair intersecting once in their respective interiors. Tony and his $2n-1$ friends each stand at a distinct endpoint of a line segment. Tony wishes to send Christmas presents to each of his friends as follows: First, he chooses an endpoint of each segment as a "sink". Then he places the present at the endpoint of the segment he is at. The present moves as follows: - If it is on a line segment, it moves towards the sink. - When it reaches an intersection of two segments, it changes the line segment it travels on and starts moving towards the new sink. If the present reaches an endpoint, the friend on that endpoint can receive their present. Prove Tony can send presents to exactly n of his $2n-1$ friends.

2023 — 1

Record: `apm_2023_fb45d1`. **Category:** contest. Source.

Let $n \geq 5$ be an integer. Consider n squares with side lengths $1, 2, \ldots, n$, respectively. The squares are arranged in the plane with their sides parallel to the x and y axes. Suppose that no two squares touch, except possibly at their vertices. Show that it is possible to arrange these squares in a way such that every square touches exactly two other squares.

2024 — 1

Record: `apm_2024_455746`. **Category:** contest. Source.

Let ABC be an acute triangle. Let D be a point on side AB and E be a point on side AC such that lines BC and DE are parallel. Let X be an interior point of $BCED$. Suppose rays DX and EX meet side BC at points P and Q, respectively, such that both P and Q lie between B and C. Suppose that the circumcircles of triangles BQX and CPX intersect at a point $Y \neq X$. Prove that points A, X, and Y are collinear.

2024 — 5

Record: `apm_2024_5b1ccf`. **Category:** contest. Source.

Line ℓ intersects sides BC and AD of cyclic quadrilateral $ABCD$ in its interior points R and S respectively, and intersects ray DC beyond point C at Q, and ray BA beyond point A at P. Circumcircles of the triangles QCR and QDS intersect at $N \neq Q$, while circumcircles of the triangles PAS and PBR intersect at $M \neq P$. Let lines MP and NQ meet at point X, lines AB and CD meet at point K and lines BC and AD meet at point L. Prove that point X lies on line KL.

2025 — 2

Record: `apm_2025_0453f0`. **Category:** contest. Source.

Let α and β be positive real numbers. Emerald makes a trip in the coordinate plane, starting off from the origin $(0,0)$. Each minute she moves one unit up or one unit to the right, restricting herself to the region $|x-y| < 2025$, in the coordinate plane. By the time she visits a point (x, y) she writes down the integer $\lfloor x\alpha + y\beta \rfloor$ on it. It turns out that Emerald wrote each non-negative integer exactly once. Find all the possible pairs (α, β) for which such a trip would be possible.

BMO source statements

2004 — 3.

Record: `rou_2004_6e709c`. **Category:** contest. Source.

Fie O un punct interior triunghiului ascuţitunghic ABC. Cercurile centrate în mijloacele laturilor triunghiului şi care trec prin O, se intersectează a doua oară în K, L şi M. Demonstraţi că O este centrul cercului înscris în triunghiul KLM dacă şi numai dacă O este centrul cercului circumscris triunghiului ABC.

2004 — 4.

Record: `rou_2004_775bce`. **Category:** contest. Source.

O mulţime finită de drepte, oricare trei neconcurente, partiţionează planul într-un număr de regiuni. Două regiuni se numesc “vecine” dacă frontierele lor au în comun mai mult decât un segment nedegenerat.

În fiecare regiune trebuie scris un număr întreg astfel încât: (i) produsul numerelor scrise în oricare două regiuni vecine este strict mai mic decât suma lor; (ii) pentru fiecare dintre dreptele date, suma numerelor scrise în regiunile aflate de aceeaşi parte a ei este egală cu 0.

Demonstraţi că acesta este posibil dacă şi numai dacă nu toate dreptele sunt paralele.

2004 — Problema 1

Record: `rou_2004_f162c5`. **Category:** contest. Source.

Fie I centrul cercului înscris în triunghiul ABC şi A_1, B_1 şi C_1 puncte arbitrare pe segmentele (AI), (BI), respectiv (CI). Mediatoarele segmentelor AA_1, BB_1 şi CC_1 se intersectează în A_2, B_2 şi respectiv C_2. Arătați că centrul cercului circumscris triunghiului $A_2B_2C_2$ coincide cu centrul cercului circumscris triunghiului ABC dacă şi numai dacă I este ortocentrul triunghiului $A_1B_1C_1$.

2007 — 4

Record: `grc_2007_31776c`. **Category:** contest. Source.

For a given positive integer $n > 2$, let C_1, C_2, C_3 be the boundaries of three convex n-gons in the plane such that the sets $C_1 \cap C_2$, $C_2 \cap C_3$, $C_3 \cap C_1$ are finite. Find the maximum number of points of the set $C_1 \cap C_2 \cap C_3$.

2007 — 1

Record: `grc_2007_cc57c7`. **Category:** contest. Source.

Let $ABCD$ be a convex quadrilateral with $AB = BC = CD$, $AC \neq BD$ and let E be the intersection point of its diagonals. Prove that $AE = DE$ if and only if $\angle BAD + \angle ADC = 120^\circ$.

2008 — 3

Record: mkd_2008_010f94. **Category:** contest. Source.

In the isosceles triangle ABC let D be the base point for the altitude from C to the base AB of ABC. Calculate the length of the segment that connects the middle point of AD and the middle point of AC if it is known that the perimeter of the triangle ABC is $36\,extcm$ and the perimeter of the triangle ADC is $29\,extcm$.

2008 — 1

Record: mkd_2008_017711. **Category:** contest. Source.

In the right triangle $\lg \frac{a-b}{2} = \frac{1}{2}(\lg a + \lg b - \lg 2)$, where $a > b$, a, b are the sides of the triangle. Find the angles of the triangle.

2008 — 1

Record: mkd_2008_031f91. **Category:** contest. Source.

Let a and b be two side lengths of the triangle ΔABC. Their two corresponding medians are perpendicular. Evaluate the third side length using only a and b.

2008 — 4

Record: mkd_2008_093366. **Category:** contest. Source.

In a circle with diameter d the chords AB and CD are perpendicular and intersect at a point E distinct from the centre of the circle. Prove that $\overline{AE}^2 + \overline{BE}^2 + \overline{CE}^2 + \overline{DE}^2 = d^2$.

2008 — 2. (A)

Record: mkd_2008_0c48b6. **Category:** contest. Source.

The sides of the triangle are consecutive terms in the arithmetic progression. Prove that the line connecting the centroid and the center of the incircle is parallel to the side of the triangle with middle length.

2008 — 3

Record: mkd_2008_10b634. **Category:** contest. Source.

Two lines p and q intersect at C. The point C divides the line p into two half-lines that, together with one of the half-lines of q with starting point at C, form two angles pCq and qCp. On the bisector of the angle pCq a point M is chosen such that $MN \parallel p$. The segment MN intersects the line q at a point D. Prove that D is the midpoint of the segment MN.

2008 — 3

Record: mkd_2008_14b254. **Category:** contest. Source.

$ABCD$ is a convex quadrilateral with area P. Let the point A_1 be on the ray AB such that $\overline{AB} = \overline{BA_1}$, B_1 on the ray BC such that $\overline{BC} = \overline{CB_1}$, C_1 on the ray CD such that $\overline{CD} = \overline{DC_1}$ and D_1 on the ray DA such that $\overline{DA} = \overline{AD_1}$. Find the area of $A_1B_1C_1D_1$.

2008 — 2

Record: `mkd_2008_2363f9`. **Category:** contest. Source.

Prove that there is an angle x such that

$$sinx = \frac{\sin\beta \cdot \sin\gamma}{1 - \cos\alpha \cdot \cos\beta \cdot \cos\gamma},$$

for every angle α, and β and γ acute angles.

2008 — 3

Record: `mkd_2008_251f69`. **Category:** contest. Source.

The difference of two complementary angles α and β is $20^\circ 52'$. Determine α and β.

2008 — 1

Record: `mkd_2008_2708eb`. **Category:** contest. Source.

Calculate the length of the leg of an isosceles trapeze with bases $18\,extcm$ and $10\,extcm$, if it is known that its middle line is $\frac{2}{7}$ of its perimeter.

2008 — 3

Record: `mkd_2008_281dfd`. **Category:** contest. Source.

The base regular four-sided prism is a rhomb with area of $\frac{2}{3}k^2$, $k > 0$. The smaller diagonal plane intersection is a square with area of k^2. a) Express the area and volume of the prism using only k. b) For which value of k the area and the volume have equal measure numbers?

2008 — 4

Record: `mkd_2008_2d1896`. **Category:** contest. Source.

Let k is the circumscribed circle for the quadrangle $ABCD$. The angle at the vertex B is twice bigger than the angle at the vertex A and for 40° smaller than the angle at the vertex D. Calculate the angles of $ABCD$.

2008 — 3

Record: `mkd_2008_30218d`. **Category:** contest. Source.

$2n$ ($n > 1$) points are given in the plane. Line p lies in the plane and does not cross the given points. Prove that this line cuts no more than n^2 segments with end in the given points.

2008 — 4

Record: `mkd_2008_321a57`. **Category:** contest. Source.

Let H be the orthocenter of the triangle $\triangle ABC$ and A_1, B_1, C_1 be the feet of the altitudes from A, B, C respectively. Find $\frac{\overline{AH}}{HA_1} \cdot \frac{\overline{BH}}{HB_1} \cdot \frac{\overline{CH}}{HC_1}$ if $\frac{\overline{AH}}{HA_1} + \frac{\overline{BH}}{HB_1} + \frac{\overline{CH}}{HC_1} = 2008$.

2008 — 1

Record: `mkd_2008_3a0197`. **Category:** contest. Source.

An acute-angled scalene triangle ABC is given, with $AC > BC$. Let O be its circumcentre, H its orthocentre, and F the foot of the altitude from C. Let P be the point (other than A) on the line AB such that $AF = PF$, and M be the midpoint of AC. We denote the intersection of PH and BC by X, the intersection of OM and FX by Y, and the intersection of OF and AC by Z. Prove that the points F, M, Y and Z are concyclic.

2008 — 2

Record: `mkd_2008_3fa8e5`. **Category:** contest. Source.

In the triangle ABC, $\angle BAC = 120^\circ$. On the bisector of the angle $\angle BAC$, a point D is chosen such that $\overline{AD} = \overline{AB} + \overline{AC}$. Prove that $\triangle BCD$ is equilateral.

2008 — 2

Record: `mkd_2008_4d994d`. **Category:** contest. Source.

A square shaped orchard with side length 26m is fenced in with 3 rows of wire. Is it possible with the same wire to fence in a rectangular shaped orchard with side lengths 95m and 60m?

2008 — 5

Record: `mkd_2008_4fd525`. **Category:** contest. Source.

Calculate the angles in the triangle ABC, if the angle between the altitude from C and the bisector of the angle ACB is 90°, and the angle between the bisectors of the exterior angles at the vertices A and B is 61°.

2008 — 4

Record: `mkd_2008_5089cf`. **Category:** contest. Source.

Calculate the angles of the triangle if it is known that the sum of two of its angles is $\frac{5}{6}$ from the right angle and one of these angles is 20° bigger than the other.

2008 — 4

Record: `mkd_2008_56f717`. **Category:** contest. Source.

The height of isosceles trapeze equals to h and his area to h^2. What is the measure of the angle between the two diagonals?

2008 — 2

Record: `mkd_2008_84a61d`. **Category:** contest. Source.

In the triangle ABC on the side AC a point D is chosen such that $\angle BDC = \angle ABC$. If it is known that $\overline{AD} = 7$cm and $\overline{DC} = 9$cm, calculate the length of BC and the ratio $\overline{BD} : \overline{BA}$.

2008 — 3. (A)

Record: `mkd_2008_89e744`. **Category:** contest. Source.

Two sides of a triangular pyramid are equilateral triangles with length a cm. The planes of these triangles are normal to each other. Find the area and volume of the pyramid.

2008 — 2

Record: `mkd_2008_8a894d`. **Category:** contest. Source.

Let p be a line through the vertex D of a given parallelogram $ABCD$, such that the vertices A, B and C are on the same side of p. Let A', B' and C' be the bases of the altitudes from A, B and C to p correspondingly. Prove that $\overline{BB'} = \overline{AA'} + \overline{CC'}$.

2008 — 3

Record: `mkd_2008_8fbac0`. **Category:** contest. Source.

Let n be a positive integer. The rectangle $ABCD$ with side lengths $AB = 90n + 1$ and $BC = 90n + 5$ is partitioned into unit squares with sides parallel to the sides of $ABCD$. Let S be the set of all points which are vertices of these unit squares. Prove that the number of lines which pass through at least two points from S is divisible by 4.

2008 — 4

Record: `mkd_2008_94cb49`. **Category:** contest. Source.

Let p be a given line. A and B are two points such that $A \notin p$ and $B \notin p$. Construct a point C on the line p such that $\overline{AC} = \overline{BC}$.

2008 — 2. (AB)

Record: `mkd_2008_9d5968`. **Category:** contest. Source.

Let ABC be an isosceles triangle with $\overline{AB} = \overline{AC}$. Let D be the midpoint of BC, M the midpoint of AD and N the projection of D to BM. Prove that $\angle ANC = 90^\circ$.

2008 — 4. (A)

Record: `mkd_2008_a0992f`. **Category:** contest. Source.

Prove that $a^3 + b^3 + 3abc > c^3$, where a, b, c are the sides of a triangle.

2008 — 2. (AB)

Record: `mkd_2008_a15b49`. **Category:** contest. Source.

From a point inside the equilateral triangle ABC are drawn perpendicular lines to the sides AB, BC and CA. The lengths of these lines are m, n, p and the length of the side of the triangle ABC is equal to a. Find the ratio of the areas of the triangle ABC and the triangle formed with the points of the intersection of the perpendicular lines and sides.

2008 — 4

Record: `mkd_2008_a24f7e`. **Category:** contest. Source.

$ABCDE$ is pentagon where K, L, M, N are the midpoints of AB, BC, CD, DE respectively. Let P, Q, F be the midpoints of KM, LN, AD, respectively. Prove that PQ and AE are parallel and $\overline{AE} = 4\overline{PQ}$.

2008 — 3

Record: `mkd_2008_aa70ee`. **Category:** contest. Source.

A rhomb with length of one diagonal equal to $7.2\,extcm$ has area of $34.56\,extcm^2$. Calculate the perimeter of the rhomb and the radius of its inscribed circle.

2008 — 5

Record: `mkd_2008_af4ffe`. **Category:** contest. Source.

Let $A_1B_1C_1$ and $A_2B_2C_2$ are given triangles. Let T_1 and T_2 are their centers of mass correspondently. Prove that $3\overline{T_1T_2} = \overline{A_1A_2} + \overline{B_1B_2} + \overline{C_1C_2}$.

2008 — 4

Record: `mkd_2008_b54730`. **Category:** contest. Source.

Two lines intersect in one point and form four angles, two acute and two obtuse. The sum of the two acute angles is half of the one obtuse angle. Calculate these angles?

2008 — 1. (AB)

Record: `mkd_2008_c9697c`. **Category:** contest. Source.

Prove that $\frac{\operatorname{tg}\alpha}{\operatorname{tg}\beta} = \frac{a^2+c^2-b^2}{b^2+c^2-a^2}$ for every triangle ABC (a, b, c and α, β are the sides and the angles in the triangle, respectively).

2008 — 4

Record: `mkd_2008_d6418a`. **Category:** contest. Source.

In the isosceles triangle ABC, M is the middle point of the base AB. Let N be a point from the leg BC, such that $MN \perp BC$ and S be the middle point of the segment MN. Prove that AN is perpendicular with CS.

2008 — 3. (B)

Record: `mkd_2008_d79f9b`. **Category:** contest. Source.

In a square shaped area on the ground with side length 12 m a cylindrical hole is dug with diameter 8 m. The dug soil is equally allotted on the square shaped area, outside of the hole, and then is pressed as much as was before the digging. How deep should be the dug in order the hole to be deep?

2008 — 2

Record: `mkd_2008_de5dba`. **Category:** contest. Source.

Let B be an interior point of the segment AC. The equilateral triangles $\triangle ABM$ and $\triangle BCN$ are constructed in same half plane determined by the line AC. The lines AN and CM intersect in L. Find the angle $\angle CLN$.

2008 — 1

Record: `mkd_2008_e5b591`. **Category:** contest. Source.

One square meter of tin costs 35 denars. How much money is needed to buy tin for 50 opened pots (without roof) with polyhedral shape with dimensions: length 20 cm, width 2 dm and height 4 dm?

2008 — 4

Record: `mkd_2008_f2aa63`. **Category:** contest. Source.

The sides of the triangle are $a = 3$, $b = 4$ and $c = 5$. Find if there is a point inside the triangle so that the distance to his sides is less than 1.

2008 — 2

Record: `mkd_2008_fba2da`. **Category:** contest. Source.

In a given rectangle $ABCD$ the length of AB is two times bigger than the length of BC. On the side CD a point M is chosen such that the angle AMD is equal to the angle AMB.

a) Determine the measure of the angle AMD.

b) If $\overline{DM} = 1$, determine the area of the rectangle $ABCD$?

2009 — 3

Record: `bmo_2009_f956f2`. **Category:** contest. Source.

A 9×12 rectangle is divided into unit squares. The centers of all the unit squares, except the four corner squares and the eight squares adjacent (by side) to them, are colored red. Is it possible to numerate the red centers by $C_1, C_2, \ldots, C_{96}$ so that the following two conditions are fulfilled: 1° All segments $C_1C_2, C_2C_3, \ldots C_{95}C_{96}, C_{96}C_1$ have the length $\sqrt{13}$; 2° The poligonal line $C_1C_2 \ldots C_{96}C_1$ is centrally symmetric?

2009 — 2.

Record: `mkd_2009_311bb5`. **Category:** contest. Source.

Let O be the circumcentre of the triangle ABC. Let K and L be the intersection points of the circumcircles of the triangles BOC and AOC with the bisectors of the angles at A and B correspondently. Let P be the midpoint of $\overline{KL}$, M symmetrical to O relatively P and N symmetrical to O relatively KL. Prove that $KLMN$ cyclic.

2009 — 4A.

Record: `mkd_2009_5131ff`. **Category:** contest. Source.

Секоја точка од рамнината е обоена во една од две бои, сина или црвена. Да се докаже дека во таа рамнина постои рамностран триаголник чии темиња се обоени во една иста боја.

2009 — 3A.

Record: `mkd_2009_ec6791`. **Category:** contest. Source.

Нека a, b, c се страните а α, β, γ соодветните агли во триаголникот ABC со плоштина P. Докажи дека важи равенството

$$a^2(\sin 2\beta + \sin 2\gamma) + b^2(\sin 2\gamma + \sin 2\alpha) + c^2(\sin 2\alpha + \sin 2\beta) = 12P.$$

2010 — Problem 2.

Record: `mkd_2010_6b8e73`. **Category:** contest. Source.

Let ABC be an acute triangle with orthocenter H, and let M be the midpoint of AC. The point C_1 on AB is such that CC_1 is an altitude of the triangle ABC. Let H_1 be the reflection of H in AB. The orthogonal projections of C_1 onto the lines AH_1, AC and BC are P, Q and R, respectively. Let M_1 be the point such that the circumcentre of triangle PQR is the midpoint of the segment MM_1. Prove that M_1 lies on the segment BH_1.

2010 — Problem 3.

Record: `mkd_2010_b41ce7`. **Category:** contest. Source.

A strip of width w is the set of all points which lie on, or between, two parallel lines distance w apart. Let S be a set of n ($n \geq 3$) points on the plane such that any three different points of S can be covered by a strip of width 1. Prove that S can be covered by a strip of width 2.

2010 — Problem 3.

Record: `rou_2010_b512fc`. **Category:** contest. Source.

A strip of breadth ℓ is the set of those points in the plane, situated between or on two parallel lines at distance ℓ one from the other. Consider a finite set S of $n \geq 3$ (distinct) points in the plane, with the property that any three points of S can be covered by a strip of breadth 1. Prove the set S can be covered by a strip of breadth 2.

2011 — 4

Record: `mkd_2011_b4aec1`. **Category:** contest. Source.

Let $ABCDEF$ be a convex hexagon of area 1 whose opposite sides are parallel. The lines AB, CD and EF meet in pairs to determine the vertices of a triangle. Similarly, the lines BC, DA and FA meet in pairs to determine the vertices of another triangle. Show that the area of the least one of these two triangles is at least $\frac{3}{2}$.

2012 — 2

Record: `mkd_2012_857951`. **Category:** contest. Source.

$$\sum_{\text{cyc}} (x+y)\sqrt{(z+x)(z+y)} \geq 4(xy+yz+zx)$$

for all positive real numbers x, y, z.

2012 — 1

Record: `mkd_2012_d5746f`. **Category:** contest. Source.

Let A, B, C be points lying on a circle Γ with the center O and assume that $\angle ABC > 90^\circ$. Let D be the point of intersection of the line AB and the line perpendicular to AC at C. Let l be the line through D and perpendicular to AO. Let E be the point of intersection of l and the line AC, and F be the point of intersection of Γ and l that lies between D and E. Prove that the circumcircles of the triangles BFE and CFD are tangent at F.

2013 — Problem 1.

Record: `grc_2013_d207eb`. **Category:** contest. Source.

In a triangle ABC, the excircle ω_a opposite A touches AB at P and AC at Q, and the excircle ω_b opposite B touches BA at M and BC at N. Let K be the projection of C onto MN, and let L be the projection of C onto PQ. Show that the quadrilateral $MKLP$ is cyclic.

2014 — 4

Record: `mkd_2014_9ea0c9`. **Category:** contest. Source.

Let n be a positive integer. A regular hexagon with side length n is divided into equilateral triangles with side length 1 by lines parallel to its sides. Find the number of regular hexagons all of whose vertices are along the vertices of the equilateral triangles.

2018 — 1

Record: `mkd_2018_85c0d3`. **Category:** contest. Source.

A quadrilateral $ABCD$ is inscribed in a circle k, where $AB > CD$ and AB is not parallel to CD. Point M is the intersection of the diagonals AC and BD and the perpendicular from M to AB intersects the segment AB at the point E. If EM bisects the angle CED, prove that AB is a diameter of the circle k.

2018 — 2

Record: `mkd_2018_881ec5`. **Category:** contest. Source.

Let q be a positive rational number. Two ants are initially at the same point X in the plane. In the n-th minute ($n = 1, 2, 3, \ldots$) each of them chooses whether to walk due north, east, south or west and then walks the distance of q^n meter. After a whole number of minutes, they are at the same point in the plane (not necessarily X), but have not taken exactly the same route within that time. Determine all possible values of q.

BMO-SL source statements

2010 — G2

Record: `bmo_2010_14ebeb`. **Category:** shortlist. Source.

Consider a cyclic quadrilateral such that the midpoints of its sides form another cyclic quadrilateral. Prove that the area of the smaller circle is less than or equal to half the area of the bigger circle.

2010 — G3

Record: `bmo_2010_50813e`. **Category:** shortlist. Source.

The incircle of a triangle $A_0B_0C_0$ touches the sides B_0C_0, C_0A_0, A_0B_0 at the points A, B, C, respectively, and the incircle of the triangle ABC with incenter I touches the sides BC, CA, AB at the points A_1, B_1, C_1, respectively. Let $\sigma(ABC)$ and $\sigma(A_1B_1C)$ be the areas of the triangles ABC and A_1B_1C respectively. Show that if $\sigma(ABC) = 2\sigma(A_1B_1C)$, then the lines AA_0, BB_0, IC_1 pass through a common point.

2010 — G7

Record: `bmo_2010_a08034`. **Category:** shortlist. Source.

A triangle ABC is given. Let M be the midpoint of the side AC of the triangle and Z the image of point B along the line BM. The circle with center M and radius MB intersects the lines BA and BC at the points E and G respectively. Let H be the point of intersection of EG with the line AC, and K the point of intersection of HZ with the line EB. The perpendicular from point K to the line BH, intersects the lines BZ and BH at the points L and N, respectively. If P is the second point of intersection of the circumscribed circles of the triangles KZL and BLN, prove that the lines BZ, KN and HP intersect at a common point.

2010 — G8

Record: `bmo_2010_a8a66d`. **Category:** shortlist. Source.

Let $c(O, R)$ be a circle with diameter AB and C a point on it different than A and B such that $\angle AOC > 90^\circ$. On the radius OC we consider the point K and the circle (c_1) with center K and radius $KC = R_1$. We draw the tangents AD and AE from A to the circle (c_1). Prove that the straight lines AC, BK and DE are concurrent.

2010 — G4

Record: `bmo_2010_cfb901`. **Category:** shortlist. Source.

Let ABC be a given triangle and l be a line that meets the lines BC, CA and AB in A_1, B_1 and C_1 respectively. Let A' be the midpoint of the segment connecting the projections of A_1 onto the lines AB and AC. Construct analogously the points B' and C'.

a. Show that the points A', B' and C' are collinear on some line l'.

b. Show that if l contains the circumcenter of the triangle ABC, then l' contains the center of its Euler circle.

2010 — C3

Record: `bmo_2010_d8caea`. **Category:** shortlist. Source.

By a *strip* of breadth b we mean a closed part of the plane consisting of all points that lie between two parallel lines at distance b from each other. Let S be a finite set of n ($n \geq 4$) points in the plane, such that any three points from S can be covered by a strip of breadth 1. Prove that S can be covered by a strip of breadth 2.

2012 — A4

Record: `bmo_2012_0ec531`. **Category:** shortlist. Source.

Let $ABCD$ be a square in the plane P. Find the minimum and the maximum values of the function $f : P \to \mathbb{R}$ defined by

$$f(P) = \frac{PA + PB}{PC + PD}$$

where $\mathbb{R}$ is the set of all real numbers.

2012 — G6

Record: `bmo_2012_3d1c38`. **Category:** shortlist. Source.

Let P and Q be points inside a triangle ABC such that $\angle PAC = \angle QAB$ and $\angle PBC = \angle QBA$. Let D and E be the feet of the perpendiculars from P to the lines BC and AC, and F be the foot of perpendicular from Q to the line AB. Let M be the intersection of the lines DE and AB. Prove that $MP \perp CF$.

2012 — G3

Record: `bmo_2012_70f90f`. **Category:** shortlist. Source.

Let ABC be a triangle with circumcircle c and circumcenter O, and let D be a point on the side BC different from the vertices and the midpoint of BC. Let K be the point where the circumcircle c_1 of the triangle BOD intersects c for the second time and let Z be the point where c_1 meets the line AB. Let M be the point where the circumcircle c_2 of the triangle COD intersects c for the second time and let E be the point where c_2 meets the line AC. Finally let N be the point where the circumcircle c_3 of the triangle AEZ meets c again. Prove that the triangles ABC and NKM are congruent.

2012 — G2

Record: `bmo_2012_bfceb3`. **Category:** shortlist. Source.

Let ABC be a triangle, O its circumcenter and AD the bisector of the angle A where $D \in BC$. Let ℓ be the line passing through O and parallel to the bisector AD. Prove that ℓ passes through the orthocenter H of the triangle ABC if and only if ABC is isosceles or $\angle BAC = 120^\circ$.

2012 — G5

Record: bmo_2012_d294bd. **Category:** shortlist. Source.

The incircle of a triangle ABC touches its sides BC, CA, AB at the points A_1, B_1, C_1, respectively. Let the projections of the orthocenter H_1 of the triangle $A_1B_1C_1$ to the lines AA_1 and BC be P and Q, respectively. Show that the line PQ bisects the line segment B_1C_1.

2012 — G4

Record: bmo_2012_e25616. **Category:** shortlist. Source.

Let M be the point of intersection of the diagonals of a cyclic quadrilateral $ABCD$. Let I_1 and I_2 be the incenters of triangles AMD and BMC, respectively, and let L be the point of intersection of the lines DI_1 and CI_2. The foot of the perpendicular from the midpoint T of I_1I_2 to CL is N, and F is the midpoint of TN. Let G and J be the points of intersection of the line LF with I_1N and I_1I_2, respectively. Let O_1 be the circumcenter of triangle LI_1J, and let Γ_1 and Γ_2 be the circles with diameters O_1L and O_1J, respectively. Let V and S be the second points of intersection of I_1O_1 with Γ_1 and Γ_2, respectively. If K is the point where the circles Γ_1 and Γ_2 meet again, prove that K is the circumcenter of the triangle SVG.

2014 — G7

Record: bmo_2014_1d88b3. **Category:** shortlist. Source.

Let I be the incenter of $\triangle ABC$ and let H_a, H_b, and H_c be the orthocenters of $\triangle BIC$, $\triangle CIA$, and $\triangle AIB$, respectively. The line H_aH_b meets AB at X and the line H_aH_c meets AC at Y. If the midpoint T of the median AM of $\triangle ABC$ lies on XY, prove that the line H_aT is perpendicular to BC.

2014 — G4

Record: bmo_2014_2a985d. **Category:** shortlist. Source.

Let $A_0B_0C_0$ be a triangle with area equal to $\sqrt{2}$. We consider the excenters A_1, B_1 and C_1 then we consider the excenters, say A_2, B_2 and C_2, of the triangle $A_1B_1C_1$. By continuing this procedure, examine if it is possible to arrive to a triangle $A_nB_nC_n$ with all coordinates rational.

2014 — N3

Record: bmo_2014_35e42c. **Category:** shortlist. Source.

Prove that there exist infinitely many non isosceles triangles with rational side lengths, rational lengths of altitudes, and perimeter equal to 3.

2014 — G6

Record: bmo_2014_4c2639. **Category:** shortlist. Source.

In triangle ABC with $AB = AC$, M is the midpoint of BC, H is the projection of M onto AB and D is arbitrary point on the side AC. Let E be the intersection point of the parallel line through B to HD with the parallel line through C to AB. Prove that DM is the bisector of $\angle ADE$.

2014 — G3

Record: bmo_2014_4dd9ae. **Category:** shortlist. Source.

Let ABC be an isosceles triangle, ($AB = AC$). Let D and E be two points on the side BC such that $D \in BE$, $E \in DC$ and $2\angle DAE = \angle BAC$. Prove that we can construct a triangle XYZ such that $XY = BD$, $YZ = DE$ and $ZX = EC$. Find $\angle BAC + \angle YXZ$.

2014 — C2

Record: bmo_2014_5daa9c. **Category:** shortlist. Source.

Let $M = \{1, 2, \dots, 2013\}$ and let Γ be a circle. For every nonempty subset $\mathcal{A}$ of the set M, denote by $S(\mathcal{A})$ the sum of elements of the set $\mathcal{A}$, and define $S(\emptyset) = 0$ ($\emptyset$ is the empty set). Is it possible to join every subset $\mathcal{A}$ of M with some point A on the circle Γ so that following conditions are fulfilled: 1. Different subsets are joined with different points; 2. All joined points are vertices of a regular polygon; 3. If $A_1, A_2, \dots, A_k$ are some of the joined points, $k > 2$, such that $A_1A_2 \dots A_k$ is a regular k-gon, then 2014 divides $S(\mathcal{A}_1) + S(\mathcal{A}_2) + \cdots + S(\mathcal{A}_k)$?

2014 — C3

Record: bmo_2014_96bc5b. **Category:** shortlist. Source.

Consider a lattice of side length 1 equilateral triangles forming a regular hexagon of side length n. Show that the number of ways of simultaneously selecting six vertices of the lattice to form the vertices of a regular hexagon is a perfect square.

2014 — G2

Record: bmo_2014_dc76b3. **Category:** shortlist. Source.

Triangle ABC is said to be perpendicular to triangle DEF if the perpendiculars from A to EF, from B to FD and from C to DE are concurrent. Prove that if ABC is perpendicular to DEF then DEF is perpendicular to ABC.

2014 — G5

Record: bmo_2014_ed7234. **Category:** shortlist. Source.

Let $ABCD$ be a trapezium inscribed in a circle k with diameter AB. A circle with center B and radius BE, where E is the intersection point of the diagonals AC and BD meets k at points K and L. If the line, perpendicular to BD at E, intersects CD at M, prove that $KM \perp DL$.

2015 — G6.

Record: bmo_2015_0762b9. **Category:** shortlist. Source.

Let AB be a diameter of a circle (ω) with centre O. From an arbitrary point M on AB such that $MA < MB$ we draw the circles (ω_1) and (ω_2) with diameters AM and BM respectively. Let CD be an exterior common tangent of (ω_1), (ω_2) such that C belongs to (ω_1) and D belongs to (ω_2). The point E is diametrically opposite to C with respect to (ω_1) and the tangent to (ω_1) at the point E intersects (ω_2) at the points F, G. If the line of the common chord of the circumcircles of the

triangles CED and CFG intersects the circle (ω) at the points K, L and the circle (ω_2) at the point N (with N closer to L), then prove that $KC = NL$.

2015 — G2.

Record: `bmo_2015_2e45cc`. **Category:** shortlist. Source.

Let ABC be a triangle and C its circumcircle. The point D lies on the arc BC of C and is different from B, C and the midpoint of BC. The tangent line to C at D intersects the lines BC, CA, AB at A', B', C', respectively. The lines BB' and CC' intersect at E. The line AA' intersects again the circle C at F. Prove that the points D, E, F are collinear.

2015 — G7.

Record: `bmo_2015_58e034`. **Category:** shortlist. Source.

Let scalene triangle ABC have orthocentre H and circumcircle Γ. AH meets Γ at D distinct from A. BH and CH meet CA and AB at E and F respectively, and EF meets BC at P. The tangents to Γ at B and C meet at T. Show that AP and DT are concurrent on the circumcircle of AFE.

2015 — G1.

Record: `bmo_2015_98379f`. **Category:** shortlist. Source.

In the acute angled triangle ABC consider the altitudes BB' and CC'. The half-line $C'B'$ intersects the circumcircle of triangle at B'' and denote by α_A the angle $\widehat{ABB''}$. In a similar way define the angles α_B and α_C. Prove the inequality

$$\sin\alpha_A \sin\alpha_B \sin\alpha_C \leq \frac{3\sqrt{6}}{32}.$$

2015 — G4.

Record: `bmo_2015_abceba`. **Category:** shortlist. Source.

Let $\triangle ABC$ be a triangle with incenter I and circumcircle (ω). The lines AI, BI, CI intersect (ω) for second time at the points D, E, F respectively. The parallel lines through I to the sides BC, AC, AB intersect the lines EF, DF, DE at the points K, L, M respectively. Prove that the points K, L, M are collinear.

2015 — G5.

Record: `bmo_2015_b0bd6c`. **Category:** shortlist. Source.

A quadrilateral $ABCD$ is given with $AD \parallel BC$. The midpoints of AD and BC are denoted by M and N, respectively. The line MN intersects the diagonals AC and BD in points K and L, respectively. Prove that the circumcircles of the triangles AKM and BNL have a common point on the line AB.

2015 — G3.

Record: `bmo_2015_bbd6c6`. **Category:** shortlist. Source.

A set of points in the plane is called *obtuse* when it contains no three collinear points, and every triangle with its vertices in this set has one angle $> 91^\circ$. Is it true that every finite obtuse set can be extended to an infinite obtuse set?

2015 — A2.

Record: `bmo_2015_c975a8`. **Category:** shortlist. Source.

Let a, b and c be the sides of a triangle and r, R and s be the inradius, the circumradius and the semiperimeter of the triangle respectively. Prove that

$$\frac{1}{a+b}+\frac{1}{a+c}+\frac{1}{b+c}\leq\frac{r}{16Rs}+\frac{s}{16Rr}+\frac{11}{8s}$$

2015 — A3.

Record: `bmo_2015_e4e619`. **Category:** shortlist. Source.

Let a, b, c be the lengths of the sides of a given triangle and m_a, m_b, m_c be the lengths of the corresponding medians. Prove that:

$$m_a\left(\frac{b}{a}-1\right)\left(\frac{c}{a}-1\right)+m_b\left(\frac{a}{b}-1\right)\left(\frac{c}{b}-1\right)+m_c\left(\frac{a}{c}-1\right)\left(\frac{b}{c}-1\right)\geq 0.$$

2017 — G2

Record: `bmo_2017_1b3e01`. **Category:** shortlist. Source.

Let ABC be an acute triangle and D a variable point on side AC. Point E is on BD such that $BE=\frac{BC^2-CD\cdot CA}{BD}$. As D varies on side AC prove that the circumcircle of ADE passes through a fixed point other than A.

2017 — C4

Record: `bmo_2017_1f4aba`. **Category:** shortlist. Source.

For any set of points $A_1, A_2, ..., A_n$ on the plane, one defines $r(A_1, A_2, ..., A_n)$ as the radius of the smallest circle that contains all of these points. Prove that if $n \geq 3$, there are indices i, j, k such that

$$r(A_1, A_2, ..., A_n) = r(A_i, A_j, A_k).$$

2017 — C3

Record: `bmo_2017_3f4864`. **Category:** shortlist. Source.

Let $n \geq 4$ points in the plane, no three of them are collinear. Prove that the number of parallelograms of area 1, formed by these points, is at most $\frac{n^2-3n}{4}$.

2017 — G3

Record: bmo_2017_43e8f4. **Category:** shortlist. Source.

Let $\triangle ABC$ be a triangle with $AB < AC$ inscribed into a circle c. The tangent of c at the point C meets the parallel from B to AC at the point D. The tangent of c at the point B meets the parallel from C to AB at the point E and the tangent of c at the point C at the point L. Suppose that the circumcircle c_1 of the triangle BDC meets AC at the point T and the circumcircle c_2 of the triangle BEC meets AB at the point S. Prove that the lines ST, BC, AL are concurrent.

2017 — G8

Record: bmo_2017_61b522. **Category:** shortlist. Source.

Given an acute triangle $\triangle ABC$ ($AC \neq AB$) and let (C) be its circumcircle. The excircle C_1 corresponding to the vertex A, of center I_a, tangents to the side BC at the point D and to the extensions of the sides AB, AC at the points E, Z respectively. Let I and L be the intersection points of the circles (C) and (C_1), H the orthocenter of the triangle EDZ and N the midpoint of segment EZ. The parallel line through the point I_a to the line HL meets the line HI at the point G. Prove that the perpendicular line (e) through the point N to the line BC and the parallel line (δ) through the point G to the line IL meet each other on the line HI_a.

2017 — C6

Record: bmo_2017_79742c. **Category:** shortlist. Source.

What is the least positive integer k such that, in every convex 101-gon, the sum of any k diagonals is greater than or equal to the sum of the remaining diagonals?

2017 — G7

Record: bmo_2017_b6dffd. **Category:** shortlist. Source.

Let ABC be an acute triangle with $AB \neq AC$ and circumcircle Γ. The angle bisector of BAC intersects BC and Γ at D and E respectively. Circle with diameter DE intersects Γ again at $F \neq E$. Point P is on AF such that $PB = PC$ and X and Y are feet of perpendiculars from P to AB and AC respectively. Let H and H' be the ortocenters of ABC and AXY respectively. AH meets Γ again at Q. If AH' and HH' intersect the circle with diameter AH again at points S and T, respectively, prove that the lines AT, HS and FQ are concurrent.

2017 — G1

Record: bmo_2017_ba2286. **Category:** shortlist. Source.

Let $\triangle ABC$ be an acute triangle. Variable points E and F are on sides AC and AB respectively such that $BC^2 = BA \cdot BF + CE \cdot CA$. As E and F vary prove that the circumcircle of AEF passes through a fixed point other than A.

2017 — G6

Record: bmo_2017_c82538. **Category:** shortlist. Source.

Construct outside the acute-angled triangle ABC the isosceles triangles ABA_B, ABB_A, ACA_C, ACC_A, BCB_C and BCC_B, so that

$$AB = AB_A = BA_B, \quad AC = AC_A = CA_C, \quad BC = BC_B = CB_C$$

and

$$\angle BAB_A = \angle ABA_B = \angle CAC_A = \angle ACA_C = \angle BCB_C = \angle CBC_B = \alpha < 90^\circ.$$

Prove that the perpendiculars from A to BA_C, from B to A_BC_B and from C to AC_BC_C are concurrent.

2017 — A4

Record: `bmo_2017_f27d27`. **Category:** shortlist. Source.

Let $M = \{(a, b, c) \in \mathbb{R}^3 : 0 < a, b, c < \frac{1}{2} \text{ with } a + b + c = 1\}$ and $f : M \to \mathbb{R}$ given as

$$f(a, b, c) = 4\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) - \frac{1}{abc}$$

Find the best (real) bounds α and β such that

$$f(M) = \{f(a, b, c) : (a, b, c) \in M\} \subseteq [\alpha, \beta]$$

and determine whether any of them is achievable.

2017 — G4

Record: `bmo_2017_f5ba57`. **Category:** shortlist. Source.

The acute-angled triangle ABC with circumcenter O is given. The midpoints of the sides BC, CA and AB are D, E and F respectively. An arbitrary point M on the side BC, different from D, is chosen. The straight lines AM and EF intersect at the point N and the straight line ON cuts again the circumscribed circle of the triangle ODM at the point P. Prove that the reflection of the point M with respect to the midpoint of the segment DP belongs on the nine points circle of the triangle ABC.

2019 — G5

Record: `bmo_2019_01a377`. **Category:** shortlist. Source.

Let $\triangle ABC$ ($BC > AC$) be an acute triangle with circumcircle k centered at O. The tangent to k at C intersects the line AB at the point D. The circumcircles of triangles BCD, OCD and AOB intersect the ray CA (beyond A) at the points Q, P and K, respectively, such that $P \in (AK)$ and $K \in (PQ)$. The line PD intersects the circumcircle of triangle BKQ at the point T, so that P and T are in different halfplanes with respect to BQ. Prove that $TB = TQ$.

2019 — G3

Record: `bmo_2019_3dfdc9`. **Category:** shortlist. Source.

Let ABC be a scalene and acute triangle, with circumcentre O. Let ω be the circle with centre A, tangent to BC at D. Suppose there are two points F and G on ω such that $FG \perp AO$, $\angle BFD = \angle DGC$ and the couples of points (B, F) and (C, G) are in different halfplanes with respect to the line AD. Show that the tangents to ω at F and G meet on the circumcircle of ABC.

2019 — G6

Record: bmo_2019_a9a4d4. **Category:** shortlist. Source.

Let ABC be an acute triangle, and AX, AY two isogonal lines. Also, suppose that K, S are the feet of perpendiculars from B to AX, AY, and T, L are the feet of perpendiculars from C to AX, AY respectively. Prove that KL and ST intersect on BC.

2019 — G9

Record: bmo_2019_c69cb4. **Category:** shortlist. Source.

Given semicircle (c) with diameter AB and center O. On the (c) we take point C such that the tangent at the C intersects the line AB at the point E. The perpendicular line from C to AB intersects the diameter AB at the point D. On the (c) we get the points H, Z such that $CD = CH = CZ$. The line HZ intersects the lines CO, CD, AB at the points S, I, K respectively and the parallel line from I to the line AB intersects the lines CO, CK at the points L, M respectively. We consider the circumcircle (k) of the triangle LMD, which intersects again the lines AB, CK at the points P, U respectively. Let (e_1), (e_2), (e_3) be the tangents of the (k) at the points L, M, P respectively and $R = (e_1) \cap (e_2)$, $X = (e_2) \cap (e_3)$, $T = (e_1) \cap (e_3)$. Prove that if Q is the center of (k), then the lines RD, TU, XS pass through the same point, which lies in the line IQ.

2019 — G4

Record: bmo_2019_d00da9. **Category:** shortlist. Source.

Given an acute triangle ABC, let M be the midpoint of BC and H the orthocentre. Let Γ be the circle with diameter HM, and let X, Y be distinct points on Γ such that AX, AY are tangent to Γ. Prove that $BXYC$ is cyclic.

2019 — G1

Record: bmo_2019_e1aff2. **Category:** shortlist. Source.

Let $ABCD$ be a square of center O and let M be the symmetric of the point B with respect to the point A. Let E be the intersection of CM and BD, and let S be the intersection of MO and AE. Show that SO is the angle bisector of $\angle ESB$.

2019 — G7

Record: bmo_2019_e57988. **Category:** shortlist. Source.

Let AD, BE, and CF denote the altitudes of triangle $\triangle ABC$. Points E' and F' are the reflections of E and F over AD, respectively. The lines BF' and CE' intersect at X, while the lines BE' and CF' intersect at the point Y. Prove that if H is the orthocenter of $\triangle ABC$, then the lines AX, YH, and BC are concurrent.

2019 — G2

Record: bmo_2019_f013ef. **Category:** shortlist. Source.

Let be a triangle $\triangle ABC$ with $m(\angle ABC) = 75^\circ$ and $m(\angle ACB) = 45^\circ$. The angle bisector of $\angle CAB$ intersects CB at the point D. We consider the point $E \in (AB)$, such that $DE = DC$. Let P be the intersection of the lines AD and CE. Prove that P is the midpoint of the segment AD.

2021 — N7

Record: `bmo_2021_ac9fb9`. **Category:** shortlist. Source.

A *super-integer* triangle is defined to be a triangle whose lengths of all sides and at least one height are positive integers. We will deem certain positive integer numbers to be *good* with the condition that if the lengths of two sides of a super-integer triangle are two (not necessarily different) good numbers, then the length of the remaining side is also a good number. Let 5 be a good number. Prove that all integers larger than 2 are good numbers.

2022 — C2.

Record: `bmo_2022_2d08c3`. **Category:** shortlist. Source.

Alice is drawing a shape on a piece of paper. She starts by placing her pencil at the origin, and then draws line segments of length 1, alternating between vertical and horizontal segments. Eventually, her pencil returns to the origin, forming a closed, non-self-intersecting shape. Show that the area of this shape is even if and only if its perimeter is a multiple of eight.

2022 — C3.

Record: `bmo_2022_9fea53`. **Category:** shortlist. Source.

Find the largest positive integer k for which there exists a convex polyhedron P with the following properties: (a) P has exactly 2022 edges. (b) The degrees of the vertices of P don't differ by more than 1. (c) It is possible to colour the edges of P with k colours such that for every colour c, and every pair of vertices (v_1, v_2) of P, there is a monochromatic path between v_1 and v_2 in the colour c.

2023 — G1

Record: `bmo_2023_0b3edf`. **Category:** shortlist. Source.

Let $ABCD$ be a circumscribed quadrilateral (i.e. convex with sides that are all tangent to a single circle) and let X be the intersection point of its diagonals AC and BD. Let I_1, I_2, I_3, I_4 be the incenters of $\triangle DXC$, $\triangle BXC$, $\triangle AXB$, and $\triangle DXA$, respectively. The circumcircle of $\triangle CI_1I_2$ intersects the sides CB and CD at points P and Q, respectively. The circumcircle of $\triangle AI_3I_4$ intersects the sides AB and AD at points M and N, respectively. Prove that

$$AM + CQ = AN + CP.$$

2023 — G6

Record: `bmo_2023_0bb743`. **Category:** shortlist. Source.

Let ABC be an acute triangle ($AB < BC < AC$) with circumcircle Γ. Assume there exists $X \in AC$ satisfying $AB = BX$ and $AX = BC$. Points $D, E \in \Gamma$ are taken such that $\angle ADB < 90^\circ$, $DA = DB$ and $BC = CE$. Let P be the intersection point of AE with the tangent line to Γ at B, and let Q be the intersection point of AB with tangent line to Γ at C. Show that the projection of D onto PQ lies on the circumcircle of $\triangle PAB$.

2023 — G2

Record: bmo_2023_5e5234. **Category:** shortlist. Source.

Let $ABCD$ be a cyclic quadrilateral with circumcenter O lying in the interior. Let E and F be the midpoints of the segments BC and AD, respectively. Let X be the point lying on the same side of the line EF as the vertex C such that $\triangle EXF$ and $\triangle BOA$ are similar. Prove that $XC = XD$.

2023 — G3

Record: bmo_2023_83afda. **Category:** shortlist. Source.

In triangle ABC, the incircle touches sides BC, CA, AB at D, E, F respectively. Assume there exists a point X on the line EF such that

$$\angle XBC = \angle XCB = 45^\circ.$$

Let M be the midpoint of the arc BC on the circumcircle of ABC not containing A. Prove that MD passes through E or F.

2023 — G5

Record: bmo_2023_e7a0b1. **Category:** shortlist. Source.

Let ABC be a triangle with circumcenter O. Point X is the intersection of the parallel line from O to AB with the perpendicular line to AC from C. Let Y be the point where the external bisector of $\angle BXC$ intersects with AC. Let K be the projection of X onto BY. Prove that the lines AK, XO, BC have a common point.

2023 — G4

Record: bmo_2023_fea5bb. **Category:** shortlist. Source.

Let O and H be the circumcenter and orthocenter of a scalene triangle ABC, respectively. Let D be the intersection point of the lines AH and BC. Suppose the line OH meets the side BC at X. Let P and Q be the second intersection points of the circumcircles of $\triangle BDH$ and $\triangle CDH$ with the circumcircle of $\triangle ABC$, respectively. Show that the four points P, D, Q, and X lie on a circle.

2024 — G5.

Record: bmo_2024_16732a. **Category:** shortlist. Source.

Let ABC be a scalene acute triangle ABC, D be the orthogonal projection of A on BC, M and N are the midpoints of AB and AC respectively. Let P, Q are points on the minor arcs $\widehat{AB}$ and $\widehat{AC}$ of circumcircle of $\triangle ABC$ respectively such that $PQ \parallel BC$. Show that the circumcircles of $\triangle DPQ$ and $\triangle MND$ are tangent to each other if and only if PQ passes through M.

2024 — G2.

Record: bmo_2024_4d74e3. **Category:** shortlist. Source.

Let ABC be an acute triangle and P be a point inside the triangle such that $\angle APB = \angle BPC = \angle CPA$. Denote with S the area and with α, β, γ the angles of $\triangle ABC$. Prove that

$$\frac{1}{\sin\alpha} + \frac{1}{\sin\beta} + \frac{1}{\sin\gamma} \geq \frac{PA^2 + PB^2 + PC^2}{2S} + \frac{4}{\sqrt{3}}$$

When does the equality occur?

2024 — G4.

Record: `bmo_2024_5d4e8e`. **Category:** shortlist. Source.

Let ABC be an acute-angled triangle with $AB < AC$, orthocentre H, circumcircle Γ and circumcentre O. Let M be the midpoint of BC and let D be a point such that $ADOH$ is a parallelogram. Suppose that there exists a point X on Γ and on the opposite side of DH to A such that $\angle DXH + \angle DHA = 90^\circ$. Let Y be the midpoint of OX. Prove that if $MY = OA$ then $OA = 2OH$.

2024 — G6.

Record: `bmo_2024_cb70a8`. **Category:** shortlist. Source.

Let $\triangle ABC$ be a triangle and the points K and L on AB, M and N on BC and P and Q on CA are such that $AK = LB < \frac{1}{2}AB$, $BM = NC < \frac{1}{2}BC$ and $CP = QA < \frac{1}{2}CA$. The intersections of KN with MQ and LP are R and T respectively, and the intersections of NP with LM and KQ are D and E respectively. Prove that the lines DR, BE and CT pass through a common point.

2024 — G7.

Record: `bmo_2024_e13d37`. **Category:** shortlist. Source.

Let $f : \pi \to \mathbb{R}$ be a function from the Euclidean plane to the real numbers such that

$$f(A) + f(B) + f(C) = f(O) + f(G) + f(H)$$

for any acute triangle ABC with circumcenter O, centroid G, and orthocenter H. Prove that f is constant.

2024 — G3.

Record: `bmo_2024_eb762b`. **Category:** shortlist. Source.

Let A, B, C, D be fixed points on this order on a line. Let ω be a variable circle through C and D and suppose that it meets the perpendicular bisector of CD at the points X and Y. Let Z and T be the other points of intersection of AX and BY with ω. Prove that XY passes through a fixed point which is independent of the circle ω.

2025 — G3

Record: `bmo_2025_3a6895`. **Category:** shortlist. Source.

Let $\triangle ABC$ be an acute-angled triangle with orthocenter H and let D be an arbitrary point on side BC. Let E and Z be the points on the segments AB and AC respectively such that the

quadrilaterals $ABDZ$ and $ACDE$ are cyclic, and let segments BZ and CE intersect at P. Let L be the intersection point of line HA and the tangent to the circumcircle of triangle $\triangle PBC$ at point C. If lines BH and CP intersect at X, prove that D lies on the line LX.

2025 — G4

Record: `bmo_2025_5312cc`. **Category:** shortlist. Source.

Let I be the incenter of triangle $\triangle ABC$, and D be an arbitrary point on arc $\widehat{AB}$, not containing point C, of the circumcircle of $\triangle ABC$. Let I_A and I_B be the incenters of $\triangle BCD$ and $\triangle ACD$ respectively. Lines DI_A and II_B intersect at point X_A, and lines DI_B and II_A intersect at point X_B. Prove that the intersection of lines X_AX_B and I_AI_B lies on a fixed line, independent of the choice of D.

2025 — G1

Record: `bmo_2025_84d1b3`. **Category:** shortlist. Source.

Let $\triangle ABC$ be a triangle with $AB < AC$, and let D be a point on the side AC such that $AD = \frac{AC-AB}{2}$. Points X and Y are chosen on the line through A parallel to BC such that $BX = CY$ and line AC is tangent to the circumcircle of $\triangle XDY$. Prove that the tangents to the circumcircle of $\triangle XDY$ at points X and Y meet on line BC.

2025 — C3

Record: `bmo_2025_91a2a7`. **Category:** shortlist. Source.

A rabbit is at some point (x, y) in the Euclidean plane. There are some (possibly infinitely many) landmines, which are circles with any radius that do not intersect except possibly at one point (tangent). Every move, the rabbit can hop a distance of exactly 1, but cannot land in the interior of a landmine (but may possibly land on the edge). Suppose that neither the rabbit nor the origin is in a landmine. Find the minimum ε so that the rabbit can always reach a point with a distance of at most ε from the origin, regardless of the landmine configuration.

2025 — G6

Record: `bmo_2025_b1de93`. **Category:** shortlist. Source.

Let $\triangle ABC$ be an acute-angled triangle with $AB < AC$, which is inscribed in the circle ω_1 centered at O. Denote by H the orthocenter of $\triangle ABC$ and by M the midpoint of side BC. Let ω_2 be the circumcircle of triangle $\triangle BHC$, let the line HO meet ω_2 at $K \neq H$, and let the line MK meet ω_2 at $P \neq K$. Prove that the tangent line to ω_1 at A and the tangent line to ω_2 at P meet on the circumcircle of triangle $\triangle APK$.

2025 — G2

Record: `bmo_2025_e2cd93`. **Category:** shortlist. Source.

Let $\triangle ABC$ be a triangle with $\angle BAC = 2\angle CBA$ and $AB < AC$. A point P lies on segment AC such that $PC = AB + AP$. Let O be the circumcentre of triangle $\triangle ABP$, and let the line through O parallel to AB intersect BP at Q. Show that AQ passes through the midpoint of segment BC.

2025 — G5

Record: `bmo_2025_ef58f5`. **Category:** shortlist. Source.

Let $ABCD$ be a convex quadrilateral such that $AB^2 + BC^2 = AD^2 + CD^2$. Points X and Y are chosen such that $XD \perp CD$, $XB \perp AB$, $YB \perp BC$ and $YD \perp AD$. Let lines AC and XY meet at T and M be the midpoint of segment XY. Prove that points T, M, B, D lie on a circle.

CMO-China source statements

2022 — 3

Record: `chn_2022_aece1e`. **Category:** contest. Source.

Fix two positive integers m and n. Fix a way to color the vertices of a regular $(2m+2n)$-gon so that $2m$ of them are black and the other $2n$ are white. Define the coloring distance $d(B, C)$ between two black points B and C to be the lesser of the numbers of white points on either side of the line BC; similarly, define the coloring distance $d(W, X)$ between two white points W and X to be the lesser of the numbers of black points on either side of the line WX. A black pairing scheme $\mathcal{B}$ means to label the $2m$ black points as $B_1, \dots, B_m, C_1, \dots, C_m$, such that no two segments from $B_1C_1, \dots, B_mC_m$ intersect each other. For each such $\mathcal{B}$, put

$$P(\mathcal{B}) = \sum_{i=1}^{m} d(B_i, C_i).$$

A white pairing scheme $\mathcal{W}$ means to label the $2n$ white points as $W_1, \dots, W_n, X_1, \dots, X_n$, such that no two segments from $W_1X_1, \dots, W_nX_n$ intersect each other. For each such $\mathcal{W}$, put

$$P(\mathcal{W}) = \sum_{i=1}^{n} d(W_i, X_i).$$

Prove that, regardless of how the $2m + 2n$ vertices are colored, we always have

$$\max_{\mathcal{B}} P(\mathcal{B}) = \max_{\mathcal{W}} P(\mathcal{W}),$$

where the maxima are taken over all possible black pairing schemes and all possible white pairing schemes, respectively.

2023 — 5

Record: `chn_2023_ca3b99`. **Category:** contest. Source.

Let $\triangle ABC$ be an acute triangle, and let K be a point on the extension ray of BC (so that C lies on the segment BK). Let P be a point such that $BP = BK$ and $PK \parallel AB$, and let Q be a point such that $CQ = CK$ and $QK \parallel AC$. Assume that the circumcircle of $\triangle PQK$ and the line AK intersects at another point T.

(1) Prove that $\angle APB + \angle BTC = \angle CQA$.

(2) Prove that $AP \cdot BT \cdot CQ = AQ \cdot BP \cdot CT$.

China-TST source statements

2014 — 1

Record: `chn_2014_fcf4c1`. **Category:** selection. Source.

Let O be the circumcenter of triangle ABC. H_A is the projection of A onto BC. The extension of AO intersects the circumcircle of BOC at A'. The projections of A' onto AB, AC are D, E, and O_A is the circumcenter of triangle DH_AE. Define H_B, O_B, H_C, O_C similarly. Prove: H_AO_A, H_BO_B, H_CO_C are concurrent.

2021 — 3

Record: `chn_2021_0b68e6`. **Category:** selection. Source.

Find the largest constant $C > 0$, such that for any integer $n \geq 2$, one can find real numbers $x_1, x_2, \ldots, x_n \in [-1, 1]$ satisfying

$$\prod_{1 \leq i < j \leq n} (x_i - x_j) \geq C^{\frac{n(n-1)}{2}}.$$

2021 — 6

Record: `chn_2021_202359`. **Category:** selection. Source.

Find the least positive real a satisfying this condition: for any three points A, B, C on the unit circle, there exists an equilateral triangle PQR with side length a, such that A, B, C are all inside or on the boundary of triangle PQR.

2021 — 1

Record: `chn_2021_376391`. **Category:** selection. Source.

Given a convex n-gon ($n \geq 5$) $\Omega : P_1P_2 \ldots P_n$, of which no three diagonals are concurrent inside Ω. Prove that one can choose a point inside every quadrilateral $P_iP_jP_kP_l$ ($1 \leq i < j < k < l \leq n$) and not on any diagonal of Ω, such that the C_n^4 points obtained are distinct and the segment connecting any two of them intersects with at least one diagonal of Ω. (Contributed by Leng Fusheng)

2021 — 2

Record: `chn_2021_c29903`. **Category:** selection. Source.

As illustrated in Fig. 2.1, in the acute $\triangle ABC$, $AB < AC$, I is the incentre, and $\odot O$ is the circumcentre. Let M, N be the midpoints of $\overset{\frown}{BAC}$ and $\overset{\frown}{BC}$, respectively. Let D be a point on $\overset{\frown}{AC}$ such that $AD//BC$. The inscribed circle of $\triangle ABC$ against $\angle BAC$ touches BC at E. Let F be a point inside $\triangle ABC$, satisfying $IF//BC$ and $\angle BAF = \angle CAE$. The line NF meets $\odot O$ at R other than N, the lines AF and DI meet at K, and the lines AR and IF meet at L. Prove: $NK \perp ML$.

2021 — 6

Record: `chn_2021_cb4abc`. **Category:** selection. Source.

Prove that there exists a constant $\lambda > 0$, such that: for any positive integer m, in the rectangular coordinates system, if all vertices of $\triangle ABC$ are integral points, and there is a unique interior (not on the side) integral point whose x and y coordinates are multiples of m, then the area of $\triangle ABC$ is less than λm^3.

(Contributed by Wang Bin and Fu Yunhao)

2021 — 5

Record: `chn_2021_fd4121`. **Category:** selection. Source.

Find the smallest real number α, such that for any convex polygon P of area 1, there exists a point M in the plane, such that the area of the convex hull of $P \cup Q$ is at most α, where Q is the central-symmetric figure of P about M. (Contributed by Qu Zhenhua and Wu Yuchi)

2023 — 4

Record: `chn_2023_2e71eb`. **Category:** selection. Source.

Let three circles $\odot O$, $\odot O_1$, $\odot O_2$, each externally tangent to the other two, lie on the same side of a line ℓ. Let them be tangent to ℓ at points A, A_1, A_2 respectively, where point A lies on segment A_1A_2. Denote the points of tangency of $\odot O$ with $\odot O_1$ and $\odot O_2$ as B_1 and B_2 respectively, and the point of tangency of $\odot O_1$ with $\odot O_2$ as C. Let line A_1C intersect line A_2B_2 at point D_1, and line A_2C intersect line A_1B_1 at point D_2. Prove that line D_1D_2 is parallel to line ℓ.

2023 — 1

Record: `chn_2023_4ae399`. **Category:** selection. Source.

Given two fixed points A and B on the unit circle ω, satisfying $\sqrt{2} < AB < 2$. Let P be a moving point on ω such that $\triangle ABP$ is an acute-angled triangle and $AP > AB > BP$.

For this moving point P, let H be the orthocenter of $\triangle ABP$. Take a point S on the arc $\overset{\frown}{AP}$ such that $SH = AH$, and take a point T on the arc $\overset{\frown}{AB}$ such that $TB \parallel AP$. Let Q be the intersection of lines ST and BP.

Prove that there exists a fixed point in the plane such that the circle with diameter HQ passes through it.

2023 — 1

Record: `chn_2023_8e18d4`. **Category:** selection. Source.

Does there exist an irrational number x such that there are at most finitely many positive integers n satisfying

$$\{kx\} \geq \frac{1}{n+1}$$

for every $k \in \{1, \ldots, n\}$?

Note: Here, for a positive real number y, $\{y\}$ denotes the fractional part of y.

2025 — Problem 6

Record: `chn_2025_0db89f`. **Category:** selection. Source.

Given an odd prime p. Find the largest positive integer n such that there exist n integer-coordinate points $A_1, A_2, \ldots, A_n$ in the plane, with no three collinear, and for any $1 \le i < j < k \le n$, twice the area of triangle $A_iA_jA_k$ is **not** divisible by p.

2025 — Problem 9

Record: `chn_2025_53da42`. **Category:** selection. Source.

A finite point set $P = \{(x_i, y_i) \mid i = 1, 2, \ldots, n\}$ in the plane is called **sparse** if for any $1 \le i < j \le n$,

$$|x_i - x_j| \cdot |y_i - y_j| \ge 1.$$

(1) Let $\{Q_1, Q_2, Q_3\}$ be a sparse set where no Q_i lies inside the axis-aligned rectangle with the other two points as diagonal vertices. Prove that the area of triangle $Q_1Q_2Q_3$ is at least $\frac{\sqrt{5}}{2}$.

(2) Let P be a sparse set with no three collinear points, all lying inside an axis-aligned square of side length a. Prove that the number of points in P does not exceed

$$\frac{a^2}{\sqrt{5}} + 2a + 1.$$

2025 — Problem 8

Record: `chn_2025_a179df`. **Category:** selection. Source.

Let $A_1A_2A_3A_4$ be a convex quadrilateral that is **not** cyclic and whose opposite sides are **not** parallel. For $1 \le i \le 4$, let M_i be the midpoint of $A_{i-1}A_{i+1}$. Let B_i be a point on the tangent to the circumcircle of triangle $A_{i-1}A_iA_{i+1}$ at A_i, such that the reflection of M_i over the angle bisector of $\angle A_{i+1}A_{i+2}A_{i+3}$ lies on line B_iA_{i+2}. Let C_i be the unique intersection point of lines A_iA_{i+1} and B_iB_{i+1}. (All indices are modulo 4.) Prove that C_1, C_2, C_3, C_4 are collinear.

2025 — Problem 2

Record: `chn_2025_e11d17`. **Category:** selection. Source.

Let point P lie on the nine-point circle of triangle ABC. A line through P perpendicular to AP intersects BC at Q. A line through A perpendicular to AQ intersects PQ at X. Let H be the orthocenter of triangle ABC, and let D and M be the midpoints of segments BC and AQ, respectively. Prove that $XH \perp DM$.

EGMO source statements

2012 — Problem 7

Record: `egm_2012_d8922a`. **Category:** contest. Source.

Let ABC be an acute-angled triangle with circumcircle Γ and orthocentre H. Let K be a point of Γ on the other side of BC from A. Let L be the reflection of K in the line AB, and let M be the reflection of K in the line BC. Let E be the second point of intersection of Γ with the circumcircle of triangle BLM. Show that the lines KH, EM and BC are concurrent. (The orthocentre of a triangle is the point on all three of its altitudes.)

2013 — Problem 1.

Record: `egm_2013_197205`. **Category:** contest. Source.

The side BC of the triangle ABC is extended beyond C to D so that $CD = BC$. The side CA is extended beyond A to E so that $AE = 2CA$. Prove that if $AD = BE$, then the triangle ABC is right-angled.

2013 — Problem 2.

Record: `egm_2013_35d4e6`. **Category:** contest. Source.

Determine all integers m for which the $m \times m$ square can be dissected into five rectangles, the side lengths of which are the integers $1, 2, 3, \ldots, 10$ in some order.

2013 — Problem 5.

Record: `egm_2013_408e36`. **Category:** contest. Source.

Let Ω be the circumcircle of the triangle ABC. The circle ω is tangent to the sides AC and BC, and it is internally tangent to Ω at the point P. A line parallel to AB and intersecting the interior of triangle ABC is tangent to ω at Q. Prove that $\angle ACP = \angle QCB$.

2014 — 1.

Record: `egm_2014_971b39`. **Category:** contest. Source.

Determine all real constants t such that whenever a, b, c are the lengths of the sides of a triangle, then so are $a^2 + bct$, $b^2 + cat$, $c^2 + abt$.

2014 — 2.

Record: `egm_2014_c365d4`. **Category:** contest. Source.

Let D and E be two points on the sides AB and AC, respectively, of a triangle ABC, such that $DB = BC = CE$, and let F be the point of intersection of the lines CD and BE. Prove that the

incenter I of the triangle ABC, the orthocenter H of the triangle DEF and the midpoint M of the arc BAC of the circumcircle of the triangle ABC are collinear.

2014 — 2.

Record: `fra_2014_3b24d7`. **Category:** contest. Source.

Soient D et E des points appartenant respectivement aux intérieurs des côtés $[AB]$ et $[AC]$ d'un triangle ABC, tels que $DB = BC = CE$. Soient F le point d'intersection des droites CD et BE, I le centre du cercle inscrit au triangle ABC, H l'orthocentre du triangle DEF et M le milieu de l'arc BAC du cercle circonscrit au triangle ABC. Montrer que I, H et M sont alignés.

2014 — 1.

Record: `fra_2014_41466c`. **Category:** contest. Source.

Déterminer tous les nombres réels t tels que si a, b, c sont les longueurs des côtés d'un triangle, alors il en est de même pour $a^2 + bct, b^2 + cat, c^2 + abt$.

2016 — Problem 4.

Record: `rou_2016_6b1f90`. **Category:** contest. Source.

Two circles, ω_1 and ω_2, of equal radius intersect at different points X_1 and X_2. Consider a circle ω externally tangent to ω_1 at a point T_1, and internally tangent to ω_2 at a point T_2. Prove that lines X_1T_1 and X_2T_2 intersect at a point lying on ω. Luxembourg

2016 — Problem 2.

Record: `rou_2016_aeff43`. **Category:** contest. Source.

Let $ABCD$ be a cyclic quadrilateral, and let diagonals AC and BD intersect at X. Let C_1, D_1 and M be the midpoints of segments CX, DX and CD, respectively. Lines AD_1 and BC_1 intersect at Y, and line MY intersects diagonals AC and BD at different points E and F, respectively. Prove that line XY is tangent to the circle through E, F and X.

2020 — Problem 3

Record: `egm_2020_4c41d5`. **Category:** contest. Source.

Let $ABCDEF$ be a convex hexagon such that $\angle A = \angle C = \angle E$ and $\angle B = \angle D = \angle F$ and the (interior) angle bisectors of $\angle A, \angle C$, and $\angle E$ are concurrent. Prove that the (interior) angle bisectors of $\angle B, \angle D$, and $\angle F$ must also be concurrent.

Note that $\angle A = \angle FAB$. The other interior angles of the hexagon are similarly described.

2020 — Problem 5

Record: `egm_2020_9202c0`. **Category:** contest. Source.

Consider the triangle ABC with $\angle BCA > 90°$. The circumcircle Γ of ABC has radius R. There is a point P in the interior of the line segment AB such that $PB = PC$ and the length of PA is R. The perpendicular bisector of PB intersects Γ at the points D and E.

Prove that P is the incentre of triangle CDE.

2021 — Problem 3.

Record: `egm_2021_755fd4`. **Category:** contest. Source.

Let ABC be a triangle with an obtuse angle at A. Let E and F be the intersections of the external bisector of angle A with the altitudes of ABC through B and C respectively. Let M and N be the points on the segments EC and FB respectively such that $\angle EMA = \angle BCA$ and $\angle ANF = \angle ABC$. Prove that the points E, F, N, M lie on a circle.

2022 — Problem 1.

Record: `egm_2022_a8807e`. **Category:** contest. Source.

Let ABC be an acute-angled triangle in which $BC < AB$ and $BC < CA$. Let point P lie on segment AB and point Q lie on segment AC such that $P \neq B$, $Q \neq C$ and $BQ = BC = CP$. Let T be the circumcentre of triangle APQ, H the orthocentre of triangle ABC, and S the point of intersection of the lines BQ and CP. Prove that T, H and S are collinear.

2024 — Problem 2

Record: `egm_2024_82b391`. **Category:** contest. Source.

Let ABC be a triangle with $AC > AB$, and denote its circumcircle by Ω and incentre by I. Let its incircle meet sides BC, CA, AB at D, E, F respectively. Let X and Y be two points on minor arcs $\widehat{DF}$ and $\widehat{DE}$ of the incircle, respectively, such that $\angle BXD = \angle DYC$. Let line XY meet line BC at K. Let T be the point on Ω such that KT is tangent to Ω and T is on the same side of line BC as A. Prove that lines TD and AI meet on Ω.

IMO source statements

1983 — 2.

Record: `esp_1983_1fea65`. **Category:** contest. Source.

Sea A uno de los dos puntos de intersección distintos de dos círculos distintos C_1, C_2 de centros O_1, O_2, respectivamente. Una de las tangentes comunes a los dos círculos toca a C_1 en P_1 y a C_2 en P_2, mientras que la otra toca a C_1 en Q_1 y a C_2 en Q_2. Sea M_1 el punto medio de P_1Q_1 y M_2 el punto medio de P_2Q_2. Demostrar que $\widehat{O_1AO_2} = \widehat{M_1AM_2}$.

1983 — 6.

Record: `esp_1983_48b69e`. **Category:** contest. Source.

Sean a, b y c las longitudes de los lados de un triángulo. Demostrar que

$$a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geq 0$$

Determinar en qué casos se cumple la igualdad.

1983 — 4.

Record: `esp_1983_f5c138`. **Category:** contest. Source.

Sea ABC un triángulo equilátero, y $\mathcal{E}$ el conjunto de todos los puntos contenidos en los tres segmentos AB, BC y CA (con A, B y C incluidos). Determinar si es cierto que para cada partición de $\mathcal{E}$ en dos conjuntos disjuntos, por lo menos uno de los dos conjuntos contiene los vértices de un triángulo rectángulo. Justificar la respuesta.

1984 — 3.

Record: `esp_1984_25d462`. **Category:** contest. Source.

Tenemos en el plano dos puntos diferentes, A y O. Para cada punto X del plano distinto de O, denotamos por $\alpha(X)$ la medida del ángulo entre OA y OX, en radianes, y contado en sentido antihorario desde OA $(0 \leq \alpha(X) < 2\pi)$.

Sea $C(X)$ la circunferencia de centro O y radio de longitud $OX + \frac{\alpha(X)}{OX}$. Tenemos un número finito de colores y coloreamos cada uno de los puntos del plano con ellos.

Demostrar que existe un punto Y tal que $\alpha(Y) > 0$ y tal que su color aparece sobre la circunferencia de $C(Y)$.

1984 — 5.

Record: `esp_1984_28162f`. **Category:** contest. Source.

Sea d la suma de las longitudes de todas las diagonales de un polígono convexo plano de n vértices $(n > 3)$, y sea p su perímetro. Demostrar que

$$n - 3 < \frac{2d}{p} < \left[\frac{n}{2}\right]\left[\frac{n+1}{2}\right] - 2$$

siendo $[x]$ la parte entera de x.

1986 — 4.

Record: `esp_1986_151259`. **Category:** contest. Source.

Sean A, B vértices adyacentes de un n-ágono regular $(n \geq 5)$ del plano que tiene centro en O. Un triángulo XYZ que es congruente con OAB e inicialmente coincide con él, se mueve en el plano de forma que Y y Z describan la frontera del polígono, dejando X en el interior. Hallar el lugar geométrico de X.

1986 — 2.

Record: `esp_1986_62f468`. **Category:** contest. Source.

Tenemos en el plano un punto P_0 y un triángulo $A_1A_2A_3$. Definimos $A_s = A_{s-3}$ para todo $s \geq 4$. Construimos una sucesión de puntos $P_1, P_2, P_3, ...$, de forma que P_{k+1} es la imagen de P_k por la rotación de centro A_{k+1} y ángulo 120° en sentido horario, para $k = 0, 1, 2 ...$ Demostrar que si $P_{1986} = P_0$, entonces el triángulo $A_1A_2A_3$ es equilátero.

1987 — 2.

Record: `esp_1987_9c23fd`. **Category:** contest. Source.

En un triángulo acutángulo ABC la bisectriz interior del ángulo A corta a BC en L y corta la circunferencia circunscrita de ABC de nuevo en N. Trazamos perpendiculares desde L a AB y AC, con pies K y M, respectivamente. Demostrar que el cuadrilátero $AKNM$ y el triángulo ABC tienen la misma área.

2006 — 6

Record: `usa_2006_72bde5`. **Category:** contest. Source.

Assign to each side b of a convex polygon $\mathcal{P}$ the maximum area of a triangle that has b as a side and is contained in $\mathcal{P}$. Show that the sum of the areas assigned to the sides of $\mathcal{P}$ is at least twice the area of $\mathcal{P}$.

2006 — 2

Record: `usa_2006_d819ea`. **Category:** contest. Source.

Let $\mathcal{P}$ be a regular 2006-gon. A diagonal of $\mathcal{P}$ is called *good segment* if its endpoints divide the boundary of $\mathcal{P}$ into two parts, each composed of an odd number of sides of $\mathcal{P}$. The sides of $\mathcal{P}$ are also called *good segment*. Suppose $\mathcal{P}$ has been dissected into triangles by 2003 diagonals, no two of which have a common point in the interior of $\mathcal{P}$. Find the maximum number of isosceles triangles having two *good segments* that could appear in such a configuration.

2009 — 5

Record: `usa_2009_ed888c`. **Category:** contest. Source.

Determine all functions f from the set of positive integers to the set of positive integers such that, for all positive integers a and b, there exists a non-degenerate triangle with sides of lengths a, $f(b)$ and $f(b + f(a) - 1)$.

2013 — 2

Record: `esp_2013_93e99b`. **Category:** contest. Source.

Una configuración de 4027 puntos del plano, de los cuales 2013 son rojos y 2014 azules, y no hay tres de ellos que sean colineales, se llama colombiana. Trazando algunas rectas, el plano queda dividido en varias regiones. Una colección de rectas es buena para una configuración colombiana si se cumplen las dos siguientes condiciones: * ninguna recta pasa por ninguno de los puntos de la configuración; * ninguna región contiene puntos de ambos colores.

Hallar el menor valor de k tal que para cualquier configuración colombiana de 4027 puntos hay una colección buena de k rectas.

2014 — 3

Record: `esp_2014_8835c6`. **Category:** contest. Source.

En el cuadrilátero convexo $ABCD$, se tiene $\angle ABC = \angle CDA = 90^\circ$. La perpendicular a BD desde A corta a BD en el punto H. Los puntos S y T están en los lados AB y AD, respectivamente, y son tales que H está dentro del triángulo SCT y

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Demostrar que la recta BD es tangente a la circunferencia circunscrita del triángulo TSH.

2014 — 4

Record: `esp_2014_888344`. **Category:** contest. Source.

Los puntos P y Q están en el lado BC del triángulo acutángulo ABC de modo que $\angle PAB = \angle BCA$ y $\angle CAQ = \angle ABC$. Los puntos M y N están en las rectas AP y AQ, respectivamente, de modo que P es el punto medio de AM, y Q es el punto medio de AN. Demostrar que las rectas BM y CN se cortan en la circunferencia circunscrita del triángulo ABC.

IMO-SL source statements

2006 — G9

Record: `imo_2006_40d146`. **Category:** shortlist. Source.

Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC, respectively. The circumcircles of triangles AB_1C_1, BC_1A_1, CA_1B_1 intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2, respectively ($A_2 \neq A$, $B_2 \neq B$, $C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar. (Russia)

2006 — C7

Record: `imo_2006_47470d`. **Category:** shortlist. Source.

Consider a convex polyhedron without parallel edges and without an edge parallel to any face other than the two faces adjacent to it. Call a pair of points of the polyhedron antipodal if there exist two parallel planes passing through these points and such that the polyhedron is contained between these planes. Let A be the number of antipodal pairs of vertices, and let B be the number of antipodal pairs of midpoints of edges. Determine the difference $A - B$ in terms of the numbers of vertices, edges and faces.

2006 — G10

Record: `imo_2006_4bd52b`. **Category:** shortlist. Source.

To each side a of a convex polygon we assign the maximum area of a triangle contained in the polygon and having a as one of its sides. Show that the sum of the areas assigned to all sides of the polygon is not less than twice the area of the polygon. (Serbia)

2006 — G6

Record: `imo_2006_78aa45`. **Category:** shortlist. Source.

Circles ω_1 and ω_2 with centres O_1 and O_2 are externally tangent at point D and internally tangent to a circle ω at points E and F, respectively. Line t is the common tangent of ω_1 and ω_2 at D. Let AB be the diameter of ω perpendicular to t, so that A, E and O_1 are on the same side of t. Prove that lines AO_1, BO_2, EF and t are concurrent.

(Brasil)

2006 — C2

Record: `imo_2006_816c88`. **Category:** shortlist. Source.

A diagonal of a regular 2006-gon is called odd if its endpoints divide the boundary into two parts, each composed of an odd number of sides. Sides are also regarded as odd diagonals. Suppose the 2006-gon has been dissected into triangles by 2003 nonintersecting diagonals. Find the maximum possible number of isosceles triangles with two odd sides. (Serbia)

2006 — A5

Record: `imo_2006_b08bab`. **Category:** shortlist. Source.

Let a, b, c be the sides of a triangle. Prove that

$$\frac{\sqrt{b+c-a}}{\sqrt{b}+\sqrt{c}-\sqrt{a}}+\frac{\sqrt{c+a-b}}{\sqrt{c}+\sqrt{a}-\sqrt{b}}+\frac{\sqrt{a+b-c}}{\sqrt{a}+\sqrt{b}-\sqrt{c}} \leq 3.$$

2006 — G5

Record: `imo_2006_b62599`. **Category:** shortlist. Source.

In triangle ABC, let J be the centre of the excircle tangent to side BC at A_1 and to the extensions of sides AC and AB at B_1 and C_1, respectively. Suppose that the lines A_1B_1 and AB are perpendicular and intersect at D. Let E be the foot of the perpendicular from C_1 to line DJ. Determine the angles $\angle BEA_1$ and $\angle AEB_1$. (Greece)

2006 — G2

Record: `imo_2006_c0d21b`. **Category:** shortlist. Source.

Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD, respectively, so that $AK/KB = DL/LC$. Suppose that there are points P and Q on the line segment KL satisfying

$$\angle APB = \angle BCD \quad \text{and} \quad \angle CQD = \angle ABC.$$

Prove that the points P, Q, B and C are concyclic.

2006 — G8

Record: `imo_2006_db3585`. **Category:** shortlist. Source.

Let $ABCD$ be a convex quadrilateral. A circle passing through the points A and D and a circle passing through the points B and C are externally tangent at a point P inside the quadrilateral. Suppose that

$$\angle PAB + \angle PDC \leq 90^\circ \quad \text{and} \quad \angle PBA + \angle PCD \leq 90^\circ.$$

Prove that $AB + CD \geq BC + AD$.

2006 — G1

Record: `imo_2006_e64448`. **Category:** shortlist. Source.

Let ABC be a triangle with incentre I. A point P in the interior of the triangle satisfies

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

Show that $AP \geq AI$ and that equality holds if and only if P coincides with I.

2006 — G7

Record: `imo_2006_e9b24c`. **Category:** shortlist. Source.

In a triangle ABC, let M_a, M_b, M_c be respectively the midpoints of the sides BC, CA, AB and T_a, T_b, T_c be the midpoints of the arcs BC, CA, AB of the circumcircle of ABC, not containing the opposite vertices. For $i \in \{a, b, c\}$, let ω_i be the circle with M_iT_i as diameter. Let p_i be the common external tangent to $\omega_j, \omega_k (\{i, j, k\} = \{a, b, c\})$ such that ω_i lies on the opposite side of p_i than ω_j, ω_k do. Prove that the lines p_a, p_b, p_c form a triangle similar to ABC and find the ratio of similitude.

(Slovakia)

2006 — G4

Record: `imo_2006_ee6585`. **Category:** shortlist. Source.

A point D is chosen on the side AC of a triangle ABC with $\angle C < \angle A < 90^\circ$ in such a way that $BD = BA$. The incircle of ABC is tangent to AB and AC at points K and L, respectively. Let J be the incentre of triangle BCD. Prove that the line KL intersects the line segment AJ at its midpoint. (Russia)

2006 — C3

Record: `imo_2006_f50971`. **Category:** shortlist. Source.

Let S be a finite set of points in the plane such that no three of them are on a line. For each convex polygon P whose vertices are in S, let $a(P)$ be the number of vertices of P, and let $b(P)$ be the number of points of S which are outside P. Prove that for every real number x

$$\sum_P x^{a(P)}(1-x)^{b(P)} = 1,$$

where the sum is taken over all convex polygons with vertices in S.

NB. A line segment, a point and the empty set are considered as convex polygons of 2, 1 and 0 vertices, respectively.

2007 — C2

Record: `imo_2007_58d0fc`. **Category:** shortlist. Source.

A unit square is dissected into $n > 1$ rectangles such that their sides are parallel to the sides of the square. Any line, parallel to a side of the square and intersecting its interior, also intersects the interior of some rectangle. Prove that in this dissection, there exists a rectangle having no point on the boundary of the square.

2007 — C5

Record: `imo_2007_be294f`. **Category:** shortlist. Source.

In the Cartesian coordinate plane define the strip $S_n = \{(x, y) \mid n \leq x < n+1\}$ for every integer n. Assume that each strip S_n is colored either red or blue, and let a and b be two distinct positive integers. Prove that there exists a rectangle with side lengths a and b such that its vertices have the same color.

2007 — C8

Record: `imo_2007_c1c48b`. **Category:** shortlist. Source.

Given a convex n-gon P in the plane. For every three vertices of P, consider the triangle determined by them. Call such a triangle good if all its sides are of unit length. Prove that there are not more than $\frac{2}{3}n$ good triangles.

2008 — G5

Record: `imo_2008_384044`. **Category:** shortlist. Source.

Let k and n be integers with $0 \leq k \leq n-2$. Consider a set L of n lines in the plane such that no two of them are parallel and no three have a common point. Denote by I the set of intersection points of lines in L. Let O be a point in the plane not lying on any line of L. A point $X \in I$ is colored red if the open line segment OX intersects at most k lines in L. Prove that I contains at least $\frac{1}{2}(k+1)(k+2)$ red points.

2008 — G3

Record: `imo_2008_b56254`. **Category:** shortlist. Source.

Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

2008 — C3

Record: `imo_2008_bdc772`. **Category:** shortlist. Source.

In the coordinate plane consider the set S of all points with integer coordinates. For a positive integer k, two distinct points $A, B \in S$ will be called k-friends if there is a point $C \in S$ such that the area of the triangle ABC is equal to k. A set $T \subset S$ will be called a k-clique if every two points in T are k-friends. Find the least positive integer k for which there exists a k-clique with more than 200 elements.

2008 — G2

Record: `imo_2008_c61d18`. **Category:** shortlist. Source.

Given trapezoid $ABCD$ with parallel sides AB and CD, assume that there exist points E on line BC outside segment BC, and F inside segment AD, such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF, and by J the point of intersection of AB and EF. Let K be the midpoint of segment EF; assume it does not lie on line AB. Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ.

2008 — C1

Record: `imo_2008_edd8bc`. **Category:** shortlist. Source.

In the plane we consider rectangles whose sides are parallel to the coordinate axes and have positive length. Such a rectangle will be called a box. Two boxes intersect if they have a common

point in their interior or on their boundary. Find the largest n for which there exist n boxes $B_1, \ldots, B_n$ such that B_i and B_j intersect if and only if $i \not\equiv j \pm 1 (\mathrm{mod} n)$.

2008 — G6

Record: `imo_2008_f38735`. **Category:** shortlist. Source.

There is given a convex quadrilateral $ABCD$. Prove that there exists a point P inside the quadrilateral such that

$$\angle PAB + \angle PDC = \angle PBC + \angle PAD = \angle PCD + \angle PBA = \angle PDA + \angle PCB = 90^\circ \quad (1)$$

2009 — C4

Record: `imo_2009_c91ba6`. **Category:** shortlist. Source.

For an integer $m \geq 1$, we consider partitions of a $2^m \times 2^m$ chessboard into rectangles consisting of cells of the chessboard, in which each of the 2^m cells along one diagonal forms a separate rectangle of side length 1. Determine the smallest possible sum of rectangle perimeters in such a partition.

2010 — G6

Record: `imo_2010_625cd8`. **Category:** shortlist. Source.

The vertices X, Y, Z of an equilateral triangle XYZ lie respectively on the sides BC, CA, AB of an acute-angled triangle ABC. Prove that the incenter of triangle ABC lies inside triangle XYZ.

The vertices X, Y, Z of an equilateral triangle XYZ lie respectively on the sides BC, CA, AB of a triangle ABC. Prove that if the incenter of triangle ABC lies outside triangle XYZ, then one of the angles of triangle ABC is greater than 120°.

2010 — G3

Record: `imo_2010_da3f72`. **Category:** shortlist. Source.

Let $A_1A_2 \ldots A_n$ be a convex polygon. Point P inside this polygon is chosen so that its projections $P_1, \ldots, P_n$ onto lines $A_1A_2, \ldots, A_nA_1$ respectively lie on the sides of the polygon. Prove that for arbitrary points $X_1, \ldots, X_n$ on sides $A_1A_2, \ldots, A_nA_1$ respectively,

$$\max\left\{\frac{X_1X_2}{P_1P_2}, \ldots, \frac{X_nX_1}{P_nP_1}\right\} \geq 1$$

2011 — G7

Record: `imo_2011_7f236e`. **Category:** shortlist. Source.

Let $ABCDEF$ be a convex hexagon all of whose sides are tangent to a circle ω with center O. Suppose that the circumcircle of triangle ACE is concentric with ω. Let J be the foot of the perpendicular from B to CD. Suppose that the perpendicular from B to DF intersects the line EO at a point K. Let L be the foot of the perpendicular from K to DE. Prove that $DJ = DL$.

2011 — G8

Record: `imo_2011_b03c26`. **Category:** shortlist. Source.

Let ABC be an acute triangle with circumcircle ω. Let t be a tangent line to ω. Let t_a, t_b, and t_c be the lines obtained by reflecting t in the lines BC, CA, and AB, respectively. Show that the circumcircle of the triangle determined by the lines t_a, t_b, and t_c is tangent to the circle ω.

2011 — C5

Record: `imo_2011_bc8c85`. **Category:** shortlist. Source.

Let m be a positive integer and consider a checkerboard consisting of m by m unit squares. At the midpoints of some of these unit squares there is an ant. At time 0, each ant starts moving with speed 1 parallel to some edge of the checkerboard. When two ants moving in opposite directions meet, they both turn 90° clockwise and continue moving with speed 1. When more than two ants meet, or when two ants moving in perpendicular directions meet, the ants continue moving in the same direction as before they met. When an ant reaches one of the edges of the checkerboard, it falls off and will not re-appear.

Considering all possible starting positions, determine the latest possible moment at which the last ant falls off the checkerboard or prove that such a moment does not necessarily exist.

2011 — G4

Record: `imo_2011_e40585`. **Category:** shortlist. Source.

Let ABC be an acute triangle with circumcircle Ω. Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB. Let D be the foot of the altitude from A, and let G be the centroid of the triangle ABC. Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G, and X are collinear.

2011 — G6

Record: `imo_2011_e661f3`. **Category:** shortlist. Source.

Let ABC be a triangle with $AB = AC$, and let D be the midpoint of AC. The angle bisector of $\angle BAC$ intersects the circle through D, B, and C in a point E inside the triangle ABC. The line BD intersects the circle through A, E, and B in two points B and F. The lines AF and BE meet at a point I, and the lines CI and BD meet at a point K. Show that I is the incenter of triangle KAB.

2012 — G2

Record: `imo_2012_758467`. **Category:** shortlist. Source.

Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E. The extensions of the sides AD and BC beyond A and B meet at F. Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD. Prove that D, H, F, G are concyclic.

2013 — G2

Record: `imo_2013_22bd4a`. **Category:** shortlist. Source.

Source issue: One proof is truncated; G05 supplies a complete coordinate argument and intersection branch check.

Let ω be the circumcircle of a triangle ABC. Denote by M and N the midpoints of the sides AB and AC, respectively, and denote by T the midpoint of the arc BC of ω not containing A. The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y, respectively; assume that X and Y lie inside the triangle ABC. The lines MN and XY intersect at K. Prove that $KA = KT$.

2013 — C7

Record: `imo_2013_5a8513`. **Category:** shortlist. Source.

Let $n \geqslant 2$ be an integer. Consider all circular arrangements of the numbers $0, 1, \ldots, n$; the $n+1$ rotations of an arrangement are considered to be equal. A circular arrangement is called beautiful if, for any four distinct numbers $0 \leqslant a, b, c, d \leqslant n$ with $a + c = b + d$, the chord joining numbers a and c does not intersect the chord joining numbers b and d. Let M be the number of beautiful arrangements of $0, 1, \ldots, n$. Let N be the number of pairs (x, y) of positive integers such that $x + y \leqslant n$ and $\gcd(x, y) = 1$. Prove that

$$M = N + 1.$$

2013 — G3

Record: `imo_2013_6cfbd8`. **Category:** shortlist. Source.

In a triangle ABC, let D and E be the feet of the angle bisectors of angles A and B, respectively. A rhombus is inscribed into the quadrilateral $AEDB$ (all vertices of the rhombus lie on different sides of $AEDB$). Let φ be the non-obtuse angle of the rhombus. Prove that $\varphi \leqslant \max\{\angle BAC, \angle ABC\}$.

2013 — G5

Record: `imo_2013_d2b8b1`. **Category:** shortlist. Source.

Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD, BE, and CF are concurrent.

2013 — G4

Record: `imo_2013_d4a0bb`. **Category:** shortlist. Source.

Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C. Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.

2013 — C2

Record: `imo_2013_eb8b9e`. **Category:** shortlist. Source.

In the plane, 2013 red points and 2014 blue points are marked so that no three of the marked points are collinear. One needs to draw k lines not passing through the marked points and dividing the plane into several regions. The goal is to do it in such a way that no region contains points of both colors. Find the minimal value of k such that the goal is attainable for every possible configuration of 4027 points.

2014 — G2

Record: `imo_2014_32e67b`. **Category:** shortlist. Source.

Let ABC be a triangle. The points K, L, and M lie on the segments BC, CA, and AB, respectively, such that the lines AK, BL, and CM intersect in a common point. Prove that it is possible to choose two of the triangles ALM, BMK, and CKL whose inradii sum up to at least the inradius of the triangle ABC.

2014 — G1

Record: `imo_2014_a0b822`. **Category:** shortlist. Source.

The points P and Q are chosen on the side BC of an acute-angled triangle ABC so that $\angle PAB = \angle ACB$ and $\angle QAC = \angle CBA$. The points M and N are taken on the rays AP and AQ, respectively, so that $AP = PM$ and $AQ = QN$. Prove that the lines BM and CN intersect on the circumcircle of the triangle ABC.

2014 — C5

Record: `imo_2014_d42980`. **Category:** shortlist. Source.

Consider $n \geqslant 3$ lines in the plane such that no two lines are parallel and no three have a common point. These lines divide the plane into polygonal regions; let $\mathcal{F}$ be the set of regions having finite area. Prove that it is possible to colour $\lceil \sqrt{n/2} \rceil$ of the lines blue in such a way that no region in $\mathcal{F}$ has a completely blue boundary. (For a real number x, $\lceil x \rceil$ denotes the least integer which is not smaller than x.)

2014 — C7

Record: `imo_2014_fb1712`. **Category:** shortlist. Source.

Let M be a set of $n \geqslant 4$ points in the plane, no three of which are collinear. Initially these points are connected with n segments so that each point in M is the endpoint of exactly two segments. Then, at each step, one may choose two segments AB and CD sharing a common interior point and replace them by the segments AC and BD if none of them is present at this moment. Prove that it is impossible to perform $n^3/4$ or more such moves.

2015 — C2

Record: `imo_2015_ae4598`. **Category:** shortlist. Source.

Let $\mathcal{V}$ be a finite set of points in the plane. We say that $\mathcal{V}$ is balanced if for any two distinct points $A, B \in \mathcal{V}$, there exists a point $C \in \mathcal{V}$ such that $AC = BC$. We say that $\mathcal{V}$ is center-free if for any distinct points $A, B, C \in \mathcal{V}$, there does not exist a point $P \in \mathcal{V}$ such that $PA = PB = PC$.

a. Show that for all $n \geqslant 3$, there exists a balanced set consisting of n points.

b. For which $n \geqslant 3$ does there exist a balanced, center-free set consisting of n points?

2015 — G4

Record: `imo_2015_b11e64`. **Category:** shortlist. Source.

Let ABC be an acute triangle, and let M be the midpoint of AC. A circle ω passing through B and M meets the sides AB and BC again at P and Q, respectively. Let T be the point such that the quadrilateral $BPTQ$ is a parallelogram. Suppose that T lies on the circumcircle of the triangle ABC. Determine all possible values of BT/BM.

2015 — G6

Record: `imo_2015_d499b1`. **Category:** shortlist. Source.

Source issue: Recovered proof omits definitions and preliminary collinearity. Reconstructed in G14.

Let ABC be an acute triangle with $AB > AC$, and let Γ be its circumcircle. Let H, M, and F be the orthocenter of the triangle, the midpoint of BC, and the foot of the altitude from A, respectively. Let Q and K be the two points on Γ that satisfy $\angle AQH = 90^\circ$ and $\angle QKH = 90^\circ$. Prove that the circumcircles of the triangles KQH and KFM are tangent to each other.

2016 — G2

Record: `imo_2016_022acb`. **Category:** shortlist. Source.

Let ABC be a triangle with circumcircle Γ and incentre I. Let M be the midpoint of side BC. Denote by D the foot of perpendicular from I to side BC. The line through I perpendicular to AI meets sides AB and AC at F and E respectively. Suppose the circumcircle of triangle AEF intersects Γ at a point X other than A. Prove that lines XD and AM meet on Γ.

2016 — G5

Record: `imo_2016_02a08b`. **Category:** shortlist. Source.

Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC. A circle ω with centre S passes through A and D, and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC, and let M be the midpoint of BC. Prove that the circumcentre of triangle XSY is equidistant from P and M.

2016 — G7

Record: `imo_2016_622aa2`. **Category:** shortlist. Source.

Let I be the incentre of a non-equilateral triangle ABC, I_A be the A-excentre, I'_A be the reflection of I_A in BC, and l_A be the reflection of line AI'_A in AI. Define points I_B, I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B.

a. Prove that P lies on line OI where O is the circumcentre of triangle ABC.

b. Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y. Show that $\angle XIY = 120^\circ$.

2016 — G1

Record: `imo_2016_a6910c`. **Category:** shortlist. Source.

In a convex pentagon $ABCDE$, let F be a point on AC such that $\angle FBC = 90^\circ$. Suppose triangles ABF, ACD and ADE are similar isosceles triangles with

$$\angle FAB = \angle FBA = \angle DAC = \angle DCA = \angle EAD = \angle EDA.$$

Let M be the midpoint of CF. Point X is chosen such that $AMXE$ is a parallelogram. Show that BD, EM and FX are concurrent.

2016 — N7

Record: `imo_2016_c1bb34`. **Category:** shortlist. Source.

Let n be an odd positive integer. In the Cartesian plane, a cyclic polygon P with area S is chosen. All its vertices have integral coordinates, and the squares of its side lengths are all divisible by n. Prove that $2S$ is an integer divisible by n.

2016 — G8

Record: `imo_2016_d6cbea`. **Category:** shortlist. Source.

Let A_1, B_1 and C_1 be points on sides BC, CA and AB of an acute triangle ABC respectively, such that AA_1, BB_1 and CC_1 are the internal angle bisectors of triangle ABC. Let I be the incentre of triangle ABC, and H be the orthocentre of triangle $A_1B_1C_1$. Show that

$$AH + BH + CH \geqslant AI + BI + CI.$$

2016 — C7

Record: `imo_2016_f01bd2`. **Category:** shortlist. Source.

Let $n \geqslant 2$ be an integer. In the plane, there are n segments given in such a way that any two segments have an intersection point in the interior, and no three segments intersect at a single point. Jeff places a snail at one of the endpoints of each of the segments and claps his hands $n-1$ times. Each time when he claps his hands, all the snails move along their own segments and stay at the next intersection points until the next clap. Since there are $n-1$ intersection points on each segment, all snails will reach the furthest intersection points from their starting points after $n-1$ claps.

a) Prove that if n is odd then Jeff can always place the snails so that no two of them ever occupy the same intersection point.

b) Prove that if n is even then there must be a moment when some two snails occupy the same intersection point no matter how Jeff places the snails.

2017 — G8

Record: `imo_2017_a6d9fd`. **Category:** shortlist. Source.

There are 2017 mutually external circles drawn on a blackboard, such that no two are tangent and no three share a common tangent. A tangent segment is a line segment that is a common tangent to two circles, starting at one tangent point and ending at the other one. Luciano is drawing tangent segments on the blackboard, one at a time, so that no tangent segment intersects any

other circles or previously drawn tangent segments. Luciano keeps drawing tangent segments until no more can be drawn. Find all possible numbers of tangent segments when he stops drawing.

2017 — C8

Record: `imo_2017_abff99`. **Category:** shortlist. Source.

Let n be a given positive integer. In the Cartesian plane, each lattice point with nonnegative coordinates initially contains a butterfly, and there are no other butterflies. The neighborhood of a lattice point c consists of all lattice points within the axis-aligned $(2n+1) \times (2n+1)$ square centered at c, apart from c itself. We call a butterfly lonely, crowded, or comfortable, depending on whether the number of butterflies in its neighborhood N is respectively less than, greater than, or equal to half of the number of lattice points in N.

Every minute, all lonely butterflies fly away simultaneously. This process goes on for as long as there are any lonely butterflies. Assuming that the process eventually stops, determine the number of comfortable butterflies at the final state.

(Bulgaria)

2017 — G2

Record: `imo_2017_bb6ddb`. **Category:** shortlist. Source.

Let R and S be distinct points on circle Ω, and let t denote the tangent line to Ω at R. Point R' is the reflection of R with respect to S. A point I is chosen on the smaller arc RS of Ω so that the circumcircle Γ of triangle ISR' intersects t at two different points. Denote by A the common point of Γ and t that is closest to R. Line AI meets Ω again at J. Show that JR' is tangent to Γ.

2017 — G1

Record: `imo_2017_e0103e`. **Category:** shortlist. Source.

Source issue: A displayed pentagon angle sum has a plus/minus transcription error. Corrected in G09.

Let $ABCDE$ be a convex pentagon such that $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that the perpendicular line from E to BC and the line segments AC and BD are concurrent.

2018 — G7

Record: `imo_2018_fa8234`. **Category:** shortlist. Source.

Let O be the circumcentre, and Ω be the circumcircle of an acute-angled triangle ABC. Let P be an arbitrary point on Ω, distinct from A, B, C, and their antipodes in Ω. Denote the circumcentres of the triangles AOP, BOP, and COP by O_A, O_B, and O_C, respectively. The lines ℓ_A, ℓ_B, and ℓ_C perpendicular to BC, CA, and AB pass through O_A, O_B, and O_C, respectively. Prove that the circumcircle of the triangle formed by ℓ_A, ℓ_B, and ℓ_C is tangent to the line OP.

2019 — C4

Record: `imo_2019_335dd1`. **Category:** shortlist. Source.

On a flat plane in Camelot, King Arthur builds a labyrinth $\mathfrak{L}$ consisting of n walls, each of which is an infinite straight line. No two walls are parallel, and no three walls have a common point. Merlin then paints one side of each wall entirely red and the other side entirely blue. At the intersection of two walls there are four corners: two diagonally opposite corners where a red side and a blue side meet, one corner where two red sides meet, and one corner where two blue sides meet. At each such intersection, there is a two-way door connecting the two diagonally opposite corners at which sides of different colours meet. After Merlin paints the walls, Morgana then places some knights in the labyrinth. The knights can walk through doors, but cannot walk through walls. Let $k(\mathfrak{L})$ be the largest number k such that, no matter how Merlin paints the labyrinth $\mathfrak{L}$, Morgana can always place at least k knights such that no two of them can ever meet. For each n, what are all possible values for $k(\mathfrak{L})$, where $\mathfrak{L}$ is a labyrinth with n walls?

2019 — C6

Record: `imo_2019_5db64d`. **Category:** shortlist. Source.

Let $n > 1$ be an integer. Suppose we are given $2n$ points in a plane such that no three of them are collinear. The points are to be labelled $A_1, A_2, \ldots, A_{2n}$ in some order. We then consider the $2n$ angles $\angle A_1A_2A_3, \angle A_2A_3A_4, \ldots, \angle A_{2n-2}A_{2n-1}A_{2n}, \angle A_{2n-1}A_{2n}A_1, \angle A_{2n}A_1A_2$. We measure each angle in the way that gives the smallest positive value (i.e. between 0° and 180°). Prove that there exists an ordering of the given points such that the resulting $2n$ angles can be separated into two groups with the sum of one group of angles equal to the sum of the other group.

2019 — G8

Record: `imo_2019_7bd8a9`. **Category:** shortlist. Source.

Let $\mathcal{L}$ be the set of all lines in the plane and let f be a function that assigns to each line $\ell \in \mathcal{L}$ a point $f(\ell)$ on ℓ. Suppose that for any point X, and for any three lines ℓ_1, ℓ_2, ℓ_3 passing through X, the points $f(\ell_1), f(\ell_2), f(\ell_3)$ and X lie on a circle. Prove that there is a unique point P such that $f(\ell) = P$ for any line ℓ passing through P.

2019 — G3

Record: `imo_2019_7f17b7`. **Category:** shortlist. Source.

In triangle ABC, let A_1 and B_1 be two points on sides BC and AC, and let P and Q be two points on segments AA_1 and BB_1, respectively, so that line PQ is parallel to AB. On ray PB_1, beyond B_1, let P_1 be a point so that $\angle PP_1C = \angle BAC$. Similarly, on ray QA_1, beyond A_1, let Q_1 be a point so that $\angle CQ_1Q = \angle CBA$. Show that points P, Q, P_1, and Q_1 are concyclic. (Ukraine)

2019 — G4

Record: `imo_2019_8ac837`. **Category:** shortlist. Source.

Let P be a point inside triangle ABC. Let AP meet BC at A_1, let BP meet CA at B_1, and let CP meet AB at C_1. Let A_2 be the point such that A_1 is the midpoint of PA_2, let B_2 be the point such that B_1 is the midpoint of PB_2, and let C_2 be the point such that C_1 is the midpoint of PC_2. Prove that points A_2, B_2, and C_2 cannot all lie strictly inside the circumcircle of triangle ABC.

2019 — G2

Record: `imo_2019_8e3c8d`. **Category:** shortlist. Source.

Let ABC be an acute-angled triangle and let D, E, and F be the feet of altitudes from A, B, and C to sides BC, CA, and AB, respectively. Denote by ω_B and ω_C the incircles of triangles BDF and CDE, and let these circles be tangent to segments DF and DE at M and N, respectively. Let line MN meet circles ω_B and ω_C again at $P \neq M$ and $Q \neq N$, respectively. Prove that $MP = NQ$. (Vietnam)

2019 — G5

Record: `imo_2019_96377f`. **Category:** shortlist. Source.

Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2 \cdot \angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $\text{area}(BCD) + \text{area}(ADE) = \text{area}(ABD) + \text{area}(ABP)$.

2019 — G7

Record: `imo_2019_bb6505`. **Category:** shortlist. Source.

The incircle ω of acute-angled scalene triangle ABC has centre I and meets sides BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω again at R. Line AR meets ω again at P. The circumcircles of triangles PCE and PBF meet again at $Q \neq P$. Prove that lines DI and PQ meet on the external bisector of angle BAC.

2019 — G1

Record: `imo_2019_be46cd`. **Category:** shortlist. Source.

Let ABC be a triangle. Circle Γ passes through A, meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G. The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T. Suppose that points A and T are distinct. Prove that line AT is parallel to BC.

2019 — G6

Record: `imo_2019_eee1dd`. **Category:** shortlist. Source.

Let I be the incentre of acute-angled triangle ABC. Let the incircle meet BC, CA, and AB at D, E, and F, respectively. Let line EF intersect the circumcircle of the triangle at P and Q, such that F lies between E and P. Prove that $\angle DPA + \angle AQD = \angle QIP$.

(Slovakia)

2021 — G1

Record: `imo_2021_18d688`. **Category:** shortlist. Source.

Let $ABCD$ be a parallelogram such that $AC = BC$. A point P is chosen on the extension of the segment AB beyond B. The circumcircle of the triangle ACD meets the segment PD again at Q, and the circumcircle of the triangle APQ meets the segment PC again at R. Prove that the lines CD, AQ, and BR are concurrent.

2021 — G8

Record: `imo_2021_75d3e7`. **Category:** shortlist. Source.

Let ω be the circumcircle of a triangle ABC, and let Ω_A be its excircle which is tangent to the segment BC. Let X and Y be the intersection points of ω and Ω_A. Let P and Q be the projections of A onto the tangent lines to Ω_A at X and Y, respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R. Prove that $AR \perp BC$.

2021 — G3

Record: `imo_2021_96bcae`. **Category:** shortlist. Source.

Version 1. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|S| = 4n^2$). Assume that $\mathcal{F}$ is a set consisting of n^2 quadrilaterals such that all their vertices lie in S, and each point in S is a vertex of exactly one of the quadrilaterals in $\mathcal{F}$. Determine the largest possible sum of areas of all n^2 quadrilaterals in $\mathcal{F}$.

Version 2. Let n be a fixed positive integer, and let S be the set of points (x, y) on the Cartesian plane such that both coordinates x and y are nonnegative integers smaller than $2n$ (thus $|S| = 4n^2$). Assume that $\mathcal{F}$ is a set of polygons such that all vertices of polygons in $\mathcal{F}$ lie in S, and each point in S is a vertex of exactly one of the polygons in $\mathcal{F}$. Determine the largest possible sum of areas of all polygons in $\mathcal{F}$.

2021 — G6

Record: `imo_2021_a375f0`. **Category:** shortlist. Source.

Determine all integers $n \geqslant 3$ satisfying the following property: every convex n-gon whose sides all have length 1 contains an equilateral triangle of side length 1. (Every polygon is assumed to contain its boundary.)

2021 — G5

Record: `imo_2021_e17daf`. **Category:** shortlist. Source.

Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcentre of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at B_1 and D_1, respectively. Let O_B be the centre of the circle which passes through B and is tangent to AC at D_1. Similarly, let O_D be the centre of the circle which passes through D and is tangent to AC at B_1. Assume that $BD_1 \parallel DB_1$. Prove that O lies on the line O_BO_D.

2023 — G3.

Record: `imo_2023_13b85f`. **Category:** shortlist. Source.

Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A. Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$. Prove that lines AD, PM, BC are concurrent.

2023 — G4.

Record: `imo_2023_1ae32a`. **Category:** shortlist. Source.

Let ABC be an acute-angled triangle with $AB < AC$. Denote its circumcircle by Ω and denote the midpoint of arc CAB by S. Let the perpendicular from A to BC meet BS and Ω at D and $E \neq A$ respectively. Let the line through D parallel to BC meet line BE at L and denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P, and line BS intersect on the internal bisector of $\angle BAC$.

2023 — G8.

Record: `imo_2023_1fa49d`. **Category:** shortlist. Source.

Let ABC be an equilateral triangle. Points A_1, B_1, C_1 lie inside triangle ABC such that triangle $A_1B_1C_1$ is scalene, $BA_1 = A_1C$, $CB_1 = B_1A$, $AC_1 = C_1B$ and

$$\angle BA_1C + \angle CB_1A + \angle AC_1B = 480^\circ.$$

Lines BC_1 and CB_1 intersect at A_2; lines CA_1 and AC_1 intersect at B_2; and lines AB_1 and BA_1 intersect at C_2. Prove that the circumcircles of triangles AA_1A_2, BB_1B_2, CC_1C_2 have two common points.

2023 — G2.

Record: `imo_2023_da43cf`. **Category:** shortlist. Source.

Let ABC be a triangle with $AC > BC$. Let ω be the circumcircle of triangle ABC and let r be the radius of ω. Point P lies on segment AC such that $BC = CP$ and point S is the foot of the perpendicular from P to line AB. Let ray BP intersect ω again at D and let Q lie on line SP such that $PQ = r$ and S, P, Q lie on the line in that order. Finally, let the line perpendicular to CQ from A intersect the line perpendicular to DQ from B at E. Prove that E lies on ω.

2023 — A7.

Record: `imo_2023_f7b402`. **Category:** shortlist. Source.

Let N be a positive integer. Prove that there exist three permutations $a_1, a_2, \ldots, a_N$; $b_1, b_2, \ldots, b_N$; and $c_1, c_2, \ldots, c_N$ of $1, 2, \ldots, N$ such that

$$\left|\sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} - 2\sqrt{N}\right| < 2023$$

for every $k = 1, 2, \ldots, N$.

2024 — G2

Record: `imo_2024_7db46b`. **Category:** shortlist. Source.

Let ABC be a triangle with $AB < AC < BC$, incentre I and incircle ω. Let X be the point in the interior of side BC such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point in the interior of side BC such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC again at $P \neq A$. Let K and L be the midpoints of AB and AC, respectively. Prove that $\angle KIL + \angle YPX = 180^\circ$. (Poland)

2024 — G1

Record: `imo_2024_a60b6c`. **Category:** shortlist. Source.

Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90^\circ$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD, and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD, and $BD = AF$. Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

IMO-TST source statements

2001 — 4

Record: `usa_2001_02fd39`. **Category:** selection. Source.

Let P be a point in the plane of triangle ABC such that the segments PA, PB, and PC are the sides of an obtuse triangle. Assume that in this triangle the obtuse angle opposes the side congruent to PA. Prove that $\angle BAC$ is acute.

2001 — 7

Record: `usa_2001_154ed9`. **Category:** selection. Source.

Let $ABCD$ be a convex quadrilateral such that $\angle ABC = \angle ADC = 135^\circ$ and

$$AC^2 \cdot BD^2 = 2AB \cdot BC \cdot CD \cdot DA.$$

Prove that the diagonals of quadrilateral $ABCD$ are perpendicular.

2001 — 5

Record: `usa_2001_23c27a`. **Category:** selection. Source.

In triangle ABC, $\angle B = 2\angle C$. Let P and Q be points on the perpendicular bisector of segment BC such that rays AP and AQ trisect $\angle A$. Prove that $PQ < AB$ if and only if $\angle B$ is obtuse.

2001 — 2

Record: `usa_2001_51d215`. **Category:** selection. Source.

Let $\triangle ABC$ be a triangle and let ω be its incircle. Denote by D_1 and E_1 the points where ω is tangent to sides BC and AC, respectively. Denote by D_2 and E_2 the points on sides BC and AC, respectively, such that $CD_2 = BD_1$ and $CE_2 = AE_1$, and denote by P the point of intersection of segments AD_2 and BE_2. Circle ω intersects segment AD_2 at two points, the closer of which to the vertex A is denoted by Q. Prove that $AQ = D_2P$.

2001 — 5

Record: `usa_2001_713702`. **Category:** selection. Source.

In a triangle ABC, let segment AP bisect $\angle BAC$, with P on side BC, and let segment BQ bisect $\angle ABC$, with Q on side CA. It is known that $\angle BAC = 60^\circ$ and that $AB + BP = AQ + QB$. What are the possible angles of triangle ABC?

2001 — 1

Record: `usa_2001_77684e`. **Category:** selection. Source.

Let $\triangle ABC$ be an acute-angled triangle with O as its circumcenter. Let P on line BC be the foot of the altitude from A. Assume that $\angle BCA \geq \angle ABC + 30^\circ$. Prove that $\angle CAB + \angle COP < 90^\circ$.

2001 — 3

Record: `usa_2001_8e6431`. **Category:** selection. Source.

Let a, b, and c be nonnegative real numbers such that

$$a^2 + b^2 + c^2 + abc = 4.$$

Prove that

$$0 \leq ab + bc + ca - abc \leq 2.$$

2001 — 6

Record: `usa_2001_c629af`. **Category:** selection. Source.

Let $a > b > c > d$ be positive integers and suppose

$$ac + bd = (b + d + a - c)(b + d - a + c).$$

Prove that $ab + cd$ is not prime.

2001 — 2

Record: `vnm_2001_57564f`. **Category:** selection. Source.

In plane, let be given two circles intersecting at A and B and let PQ be a common tangent (P and Q are touching points). Let S be the point of intersection of the tangents at P and at Q of the circumcircle of triangle APQ and let H be the reflection of B in the line PQ. Prove that the points A, S, H are collinear.

2003 — 4

Record: `vnm_2003_8e14da`. **Category:** selection. Source.

On the sides of triangle ABC take the points M_1, N_1, P_1 such that each line MM_1, NN_1, PP_1 divides the perimeter of ABC into two equal parts (M, N, P are respectively the midpoints of sides BC, CA, AB).

1/ Prove that the lines MM_1, NN_1, PP_1 are concurrent at a point K.

2/ Prove that among the ratios $\frac{KA}{BC}$, $\frac{KB}{CA}$, $\frac{KC}{AB}$ there exists at least a ratio which is not less than $\frac{1}{\sqrt{3}}$.

2003 — 2

Record: `vnm_2003_de0c30`. **Category:** selection. Source.

A triangle ABC inscribed in a circle with center O has three altitudes AH, BK, CL. Let A_0, B_0, C_0 respectively be the midpoints of AH, BK, CL. The incircle with center I of triangle ABC touches the sides BC, CA, AB respectively at D, E, F.

Prove that the four lines A_0D, B_0E, C_0F and OI are concurrent. (When O coincides with I, we consider the line OI as an arbitrary line passing through O).

2004 — 4

Record: `usa_2004_17c52d`. **Category:** selection. Source.

Let $\triangle ABC$ be a triangle and let D be a point in its interior. Construct a circle ω_1 passing through B and D and a circle ω_2 passing through C and D such that the point of intersection of ω_1 and ω_2 other than D lies on line AD. Denote by E and F the points where ω_1 and ω_2 intersect side BC, respectively, and by X and Y the intersections of lines DF, AB and DE, AC, respectively. Prove that $XY \parallel BC$.

2004 — 1

Record: `usa_2004_2b56a1`. **Category:** selection. Source.

Let $ABCD$ be a quadrilateral circumscribed about a circle, whose interior and exterior angles are at least 60°. Prove that

$$\frac{1}{3}|AB^3 - AD^3| \leq |BC^3 - CD^3| \leq 3|AB^3 - AD^3|.$$

When does equality hold?

2004 — 5

Record: `usa_2004_33bab6`. **Category:** selection. Source.

Let $A = (0, 0, 0)$ be the origin in the three dimensional coordinate space. The *weight* of a point is the sum of the absolute values of its coordinates. A point is a *primitive lattice point* if all its coordinates are integers with their greatest common divisor equal to 1. A square $ABCD$ is called a *unbalanced primitive integer square* if it has integer side length and the points B and D are primitive lattice points with different weights. Show that there are infinitely many unbalanced primitive integer squares $AB_iC_iD_i$ such that the plane containing the squares are not parallel to each other.

2004 — 3

Record: `usa_2004_6ca0a8`. **Category:** selection. Source.

For what real values of $k > 0$ is it possible to dissect a $1 \times k$ rectangle into two similar, but noncongruent, polygons?

2004 — 1

Record: `usa_2004_92e118`. **Category:** selection. Source.

Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be real numbers such that

$$(a_1^2 + a_2^2 + \cdots + a_n^2 - 1)(b_1^2 + b_2^2 + \cdots + b_n^2 - 1) > (a_1b_1 + a_2b_2 + \cdots + a_nb_n - 1)^2.$$

Show that $a_1^2 + a_2^2 + \cdots + a_n^2 > 1$ and $b_1^2 + b_2^2 + \cdots + b_n^2 > 1$.

2004 — 3

Record: `usa_2004_954202`. **Category:** selection. Source.

A 2004×2004 array of points is drawn. Find the largest integer n such that it is possible to draw a convex n-sided polygon whose vertices lie on the points of the array.

2006 — 6

Record: `usa_2006_4d3687`. **Category:** selection. Source.

Let ABC be a triangle. Triangles PAB and QAC are constructed outside of triangle ABC such that $AP = AB$ and $AQ = AC$ and $\angle BAP = \angle CAQ$. Segments BQ and CP meet at R. Let O be the circumcenter of triangle BCR. Prove that $AO \perp PQ$.

2006 — 2

Record: `usa_2006_642c16`. **Category:** selection. Source.

In acute triangle ABC, segments AD, BE, and CF are its altitudes, and H is its orthocenter. Circle ω, centered at O, passes through A and H and intersects sides AB and AC again at Q and P (other than A), respectively. The circumcircle of triangle OPQ is tangent to segment BC at R. Prove that $CR/BR = ED/FD$.

2008 — 3

Record: `usa_2008_de1133`. **Category:** selection. Source.

For a pair $A = (x_1, y_1)$ and $B = (x_2, y_2)$ of points on the coordinate plane, let $d(A, B) = |x_1 - x_2| + |y_1 - y_2|$. We call a pair (A, B) of (unordered) points *harmonic* if $1 < d(A, B) \leq 2$. Determine the maximum number of harmonic pairs among 100 points on the plane.

(This problem was suggested by Zuming Feng and Oleg Golberg.)

2008 — 2

Record: `usa_2008_e46816`. **Category:** selection. Source.

Let P, Q, and R be the points on sides BC, CA, and AB of an acute triangle ABC such that triangle PQR is equilateral and has minimal area among all such equilateral triangles. Prove that the perpendiculars from A to line QR, from B to line RP, and from C to line PQ are concurrent.

2010 — 4

Record: `usa_2010_25637d`. **Category:** selection. Source.

Let ABC be a triangle. Point M and N lie on sides AC and BC respectively such that $MN \parallel AB$. Points P and Q lie on sides AB and CB respectively such that $PQ \parallel AC$. The incircle of triangle CMN touches segment AC at E. The incircle of triangle BPQ touches segment AB at F. Line EN and AB meet at R, and lines FQ and AC meet at S. Given that $AE = AF$, prove that the incenter of triangle AEF lies on the incircle of triangle ARS.

2010 — 3

Record: `usa_2010_51ef6c`. **Category:** selection. Source.

In acute triangle ABC, denote by h_a, h_b, h_c the lengths of the altitudes to bases BC, CA, AB, respectively. Point P lies inside the triangle. Prove that

$$\frac{PA}{h_b+h_c}+\frac{PB}{h_c+h_a}+\frac{PC}{h_a+h_b}\geq 1.$$

2010 — 7

Record: `usa_2010_d62a38`. **Category:** selection. Source.

Points P and Q lie inside acute triangle ABC such that $\angle PAB=\angle QAC$ and $\angle PBA=\angle QBC$. Point D lies on side BC. Prove that

$$\angle DPC+\angle APB=180^\circ \quad \text{if and only if} \quad \angle DQB+\angle AQC=180^\circ.$$

2010 — Problem 1

Record: `vnm_2010_38d622`. **Category:** selection. Source.

Given a triangle ABC, not right-angled in A. Let M be the midpoint of BC. Consider a point D moving on the line AM, in such a way that D is not coincident with M. Denote by (O_1) the circle passing through D and touching BC in B; by (O_2) the circle passing through D and touching BC in C. The line AB intersects (O_1) in the second point P and the line AC intersects (O_2) in the second point Q. Show that the tangents at P to (O_1) and the tangents at Q to (O_2) intersect each other and the intersection point lies in a fixed line.

2011 — 4

Record: `ind_2011_9fc5bb`. **Category:** selection. Source.

Let $(a_0,a_1,a_2,...)$ and $(b_0,b_1,b_2,...)$ be two infinite sequences of integers such that

$$(a_n-a_{n-1})(a_n-a_{n-2})+(b_n-b_{n-1})(b_n-b_{n-2})=0,$$

for all integers $n\geq 2$. Prove that there exists a positive integer K such that

$$a_{K+2011}=a_{K+(2011)^{2011}}.$$

2011 — 2

Record: `ind_2011_a86186`. **Category:** selection. Source.

Let ABC be a triangle each of whose angles is greater than 30°. Suppose a circle, with centre T, cuts the segments BC in P,Q; CA in K,L; and AB in M,N, such that P,Q,K,L,M,N are on the circle in counter-clockwise direction in that order. Suppose further that the triangles TQK, TLM and TNP are all equilateral. Prove that: (i) the radius of the circle is $\frac{2abc}{a^2+b^2+c^2+4\sqrt{3}\Delta}$; (ii) $a\cdot AT=b\cdot BT=c\cdot CT$.

2011 — 3

Record: `ind_2011_d8fd7b`. **Category:** selection. Source.

Let $ABCDE$ be a convex pentagon in which $BC\parallel AE$, $AB=BC+AE$ and $\angle ABC=\angle CDE$. Let M be the mid-point of CE and let O be the circumcentre of triangle BCD. Suppose $\angle DMO=90^\circ$. Prove that $2\angle BDA=\angle CDE$.

2011 — 2

Record: `vnm_2011_6c5400`. **Category:** selection. Source.

On the plane given a circle (O) and a point A lying outside the circle. Through A, draw the tangents to (O); let B and C be the tangent points. Consider a point P moving on the opposite ray to ray BA and a point Q moving on the opposite ray to ray CA, such that the line PQ is tangent to (O). The line BC intersects the line passing through P, parallel with AC, in E, and intersects the line passing through Q, parallel to AB, in F. Show that 1/ The line EQ passes through a fixed point, say, M; the line FP passes through a fixed point, say, N. 2/ The product $|PM| \cdot |QN|$ is constant.

2011 — 1

Record: `vnm_2011_f6e2ec`. **Category:** selection. Source.

1. At the point $(1, 1)$ of the coordinate plane (Oxy), there is a cào cào. From that point the cào cào can only jump to other integral points by the rule: from the positive integral point A, the cào cào jumps to the positive integral point B if the triangle OAB has area equal to $\frac{1}{2}$.

1/ Find all points (m, n) to which the cào cào can jump to after a finite number of jumps, starting from the point $(1, 1)$.

2/ Let (m, n) be a positive integral point with property mentioned in 1/. Show that there exists a way, in which the cào cào can jump from point $(1, 1)$ to point (m, n) with not more than $|m - n|$ jumps.

2012 — Bài 1

Record: `vnm_2012_7554dc`. **Category:** selection. Source.

Bài 1. (7 điểm). Cho đường tròn (O) và hai điểm B, C cố định trên (O) sao cho BC không là đường kính của (O). Điểm A thay đổi trên (O) ($A \neq B, C$). Gọi D, K, J tương ứng là trung điểm các cạnh BC, CA, AB; E, M, N tương ứng là hình chiếu vuông góc của A, B, C trên BC, DJ, DK. Các tiếp tuyến tại M và N của đường tròn ngoại tiếp tam giác EMN cắt nhau tại T. Chứng minh rằng T là điểm cố định.

2013 — 7.

Record: `usa_2013_88fa26`. **Category:** selection. Source.

In a table with n rows and $2n$ columns where n is a fixed positive integer, we write either zero or one into each cell so that each row has n zeros and n ones. For $1 \leq k \leq n$ and $1 \leq i \leq n$, we define $a_{k,i}$ so that the ith zero in the kth row is the $a_{k,i}^{th}$ column. Let $\mathcal{F}$ be the set of such tables with $a_{1,i} \geq a_{2,i} \geq ... \geq a_{n,i}$ for every i with $1 \leq i \leq n$. We associate another $n \times 2n$ table $f(C)$ from $C \in \mathcal{F}$ as follows: for the kth row of $f(C)$, we write n ones in the columns $a_{n,k} - k + 1, a_{n-1,k} - k + 2, ..., a_{1,k} - k + n$ (and we write zeros in the other cells in the row).

a. Show that $f(C) \in \mathcal{F}$.

b. Show that $f(f(f(f(f(C))))) = C$ for any $C \in \mathcal{F}$.

2014 — 5.

Record: `usa_2014_160d98`. **Category:** selection. Source.

Let $ABCD$ be a cyclic quadrilateral, and let E, F, G, and H be the midpoints of AB, BC, CD, and DA, respectively. Let W, X, Y, and Z be the orthocenters of triangles AHE, BEF, CFG, and DGH, respectively. Prove that quadrilaterals $ABCD$ and $WXYZ$ have the same area. (This problem was suggested by Zhonghao Ye)

2014 — 2.

Record: `usa_2014_402d1a`. **Category:** selection. Source.

A finite sequence of integers $a_1, a_2, \ldots, a_n$ is called *regular* if there exists a real number x satisfying

$$\lfloor kx \rfloor = a_k \quad \text{for } 1 \leq k \leq n.$$

Given a regular sequence $a_1, a_2, \ldots, a_n$, for $1 \leq k \leq n$ we say the term a_k is *forced* if the following condition is satisfied: the sequence

$$a_1, a_2, \ldots, a_{k-1}, b$$

is regular if and only if $b = a_k$. Find the maximum possible number of forced terms in a regular sequence with 1000 terms.

2014 — 1.

Record: `usa_2014_61740b`. **Category:** selection. Source.

Let $\triangle ABC$ be an acute triangle, and let X be a variable interior point on the minor arc $\widehat{BC}$ of its circumcircle. Let P and Q be the feet of the perpendiculars from X to lines CA and CB, respectively. Let R be the intersection of line PQ and the perpendicular from B to AC. Let ℓ be the line through P parallel to XR. Prove that as X varies along minor arc $\widehat{BC}$, the line ℓ always passes through a fixed point. (Specifically: prove that there is a point F, determined by triangle ABC, such that no matter where X is on arc $\widehat{BC}$, line ℓ passes through F.)

2014 — 1.

Record: `usa_2014_97160c`. **Category:** selection. Source.

Let ABC be a triangle and D, E, F be the midpoints of arcs BC, CA, AB on the circumcircle. Line l_a passes through the feet of the perpendiculars from A to DB and DC. Line m_a passes through the feet of the perpendiculars from D to AB and AC. Let A_1 denote the intersection of lines l_a and m_a. Define points B_1 and C_1 similarly. Prove that triangles DEF and $A_1B_1C_1$ are similar to each other.

2014 — 3.

Record: `usa_2014_d4dc03`. **Category:** selection. Source.

Divide the plane into an infinite square grid by drawing all the lines $x = m$ and $y = n$ for integers m and n. Next, if a square's upper-right corner has both coordinates even, color it black; otherwise, color it white (in this way, exactly $1/4$ of the squares are black and no two black squares are adjacent). Let r and s be odd integers, and let (x, y) be a point in the interior of any white square such that $rx - sy$ is irrational. Shoot a laser out of this point with slope r/s; lasers pass through white squares and reflect off black squares. Prove that the path of this laser will form a closed loop.

2014 — 9.

Record: `usa_2014_ef9290`. **Category:** selection. Source.

Let r be a rational number in the interval $[-1, 1]$, and let $\theta = \cos^{-1} r$. Call a subset S of the plane good if S is unchanged upon rotation by θ around any point of S (in both clockwise and counterclockwise directions). Determine all values of r satisfying the following property: The midpoint of any two points in a good set also lies in the set.

(This problem was suggested by Ricky Liu.)

2015 — 5

Record: `vnm_2015_33a9bc`. **Category:** selection. Source.

Given an acute, non-isosceles triangle ABC, and a point P inside the triangle such that $\angle APB = \angle APC = \alpha$ with $\alpha > 80^\circ - \angle BAC$. The circle (APB) intersects the line AC at E, the circle (APC) intersects the line AB at F. Let Q be the inside point of the triangle AEF such that $\angle AQE = \angle AQF = \alpha$. Let D be the symmetric point of Q through EF. The bisector of $\angle EDF$ intersects AP at $T.

a) Prove that $\angle DET = \angle ABC$, $\angle DFT = \angle ACB$.

b) The line AP intersects the line DE, DF respectively at M, N. Let I and J be centers of incircles of PEM and PFN respectively. Let K be the center of (DIJ). The line DT intersects (K) at H. Prove that HK goes through the incircle center of the triangle DMN.

2015 — 2

Record: `vnm_2015_4700bf`. **Category:** selection. Source.

Given a circle (O) with a fixed chord BC (BC is not a diameter of the circle). Let A move on the bigger arc BC such that ABC is an acute triangle and $AB < AC$. Let I and H respectively be the midpoint of BC and the orthocenter of the triangle ABC. The ray IH intersects the circle (O) again at K, the line AH intersects the line BC at D and the line KD intersects the circle (O) again at M. From M, we draw a perpendicular line to BC, intersecting AI at N.

a) Prove that point N belongs to a fixed circle when A moves along the bigger arc BC.

b) The circle goes through and touching AK at A intersects AB, AC at P, Q, respectively. Let J be the midpoint of PQ. Prove that the line AJ always goes through a fixed point.

2017 — Problem 1

Record: `vnm_2017_71cd4a`. **Category:** selection. Source.

There are 44 distinct holes on a line and 2017 ants. Each ant crawled up from a hole, then moved to another hole and crawled down. Denote T as the set of time points that the ants crawled up or crawled down from some hole. Suppose that the speed of the ants are pairwise distinct and they did not change their own speed. Prove that if $|T| \leq 45$ then there exist two ants did not meet.

Note that two ants met when there exists a time point such that they were at the same location on the line, including the holes.

2018 — TSTST 5

Record: `usa_2018_4f1aa5`. **Category:** selection. Source.

Let ABC be an acute triangle with circumcircle ω, and let H be the foot of the altitude from A to $\overline{BC}$. Let P and Q be the points on ω with $PA = PH$ and $QA = QH$. The tangent to ω at P intersects lines AC and AB at E_1 and F_1 respectively; the tangent to ω at Q intersects lines AC and AB at E_2 and F_2 respectively. Show that the circumcircles of $\triangle AE_1F_1$ and $\triangle AE_2F_2$ are congruent, and the line through their centers is parallel to the tangent to ω at A.

2018 — TSTST 9

Record: `usa_2018_7ab8a8`. **Category:** selection. Source.

Show that there is an absolute constant $c < 1$ with the following property: whenever $\mathcal{P}$ is a polygon with area 1 in the plane, one can translate it by a distance of $\frac{1}{100}$ in some direction to obtain a polygon $\mathcal{Q}$, for which the intersection of the interiors of $\mathcal{P}$ and $\mathcal{Q}$ has total area at most c.

2018 — TSTST 3

Record: `usa_2018_9dcf2b`. **Category:** selection. Source.

Let ABC be an acute triangle with incenter I, circumcenter O, and circumcircle Γ. Let M be the midpoint of $\overline{AB}$. Ray AI meets $\overline{BC}$ at D. Denote by ω and γ the circumcircles of $\triangle BIC$ and $\triangle BAD$, respectively. Line MO meets ω at X and Y, while line CO meets ω at C and Q. Assume that Q lies inside $\triangle ABC$ and $\angle AQM = \angle ACB$. Consider the tangents to ω at X and Y and the tangents to γ at A and D. Given that $\angle BAC \neq 60^\circ$, prove that these four lines are concurrent on Γ.

2019 — TSTST 9

Record: `usa_2019_176024`. **Category:** selection. Source.

Let ABC be a triangle with incenter I. Points K and L are chosen on segment BC such that the incircles of $\triangle ABK$ and $\triangle ABL$ are tangent at P, and the incircles of $\triangle ACK$ and $\triangle ACL$ are tangent at Q. Prove that $IP = IQ$.

2019 — TST 6

Record: `usa_2019_77ff97`. **Category:** selection. Source.

Let ABC be a triangle with incenter I, and let D be a point on line BC satisfying $\angle AID = 90^\circ$. Let the excircle of triangle ABC opposite the vertex A be tangent to $\overline{BC}$ at point A_1. Define points B_1 on $\overline{CA}$ and C_1 on $\overline{AB}$ analogously, using the excircles opposite B and C, respectively. Prove that if quadrilateral $AB_1A_1C_1$ is cyclic, then $\overline{AD}$ is tangent to the circumcircle of $\triangle DB_1C_1$.

2019 — TSTST 5

Record: `usa_2019_84fcd5`. **Category:** selection. Source.

Let $\triangle ABC$ be an acute triangle with orthocenter H and circumcircle Γ. A line through H intersects segments AB and AC at E and F, respectively. Let K be the circumcenter of $\triangle AEF$, and suppose line AK intersects Γ again at a point D. Prove that line HK and the line through D perpendicular to $\overline{BC}$ meet on Γ.

2019 — TSTST 8

Record: `usa_2019_d4cda2`. **Category:** selection. Source.

Let S be a set of 16 points in the plane, no three collinear. Let $\chi(S)$ denote the number of ways to draw 8 line segments with endpoints in S, such that no two drawn segments intersect, even at endpoints. Find the smallest possible value of $\chi(S)$ across all such S.

2019 — TST 1

Record: `usa_2019_d5cc11`. **Category:** selection. Source.

Let ABC be a triangle and let M and N denote the midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Let X be a point such that $\overline{AX}$ is tangent to the circumcircle of triangle ABC. Denote by ω_B the circle through M and B tangent to $\overline{MX}$, and by ω_C the circle through N and C tangent to $\overline{NX}$. Show that ω_B and ω_C intersect on line BC.

2019 — TSTST 2

Record: `usa_2019_f2002c`. **Category:** selection. Source.

Let ABC be an acute triangle with circumcircle Ω and orthocenter H. Points D and E lie on segments AB and AC respectively, such that $AD = AE$. The lines through B and C parallel to DE intersect Ω again at P and Q, respectively. Denote by ω the circumcircle of $\triangle ADE$.

a. Show that lines PE and QD meet on ω.

b. Prove that if ω passes through H, then lines PD and QE meet on ω as well.

2020 — TST 6

Record: `usa_2020_55b457`. **Category:** selection. Source.

Let $P_1P_2\cdots P_{100}$ be a cyclic 100-gon, and let $P_i = P_{i+100}$ for all i. Define Q_i as the intersection of diagonals $\overline{P_{i-2}P_{i+1}}$ and $\overline{P_{i-1}P_{i+2}}$ for all integers i. Suppose there exists a point P satisfying $\overline{PP_i} \perp \overline{P_{i-1}P_{i+1}}$ for all integers i. Prove that the points $Q_1, Q_2, \ldots, Q_{100}$ are concyclic.

2020 — TST 3

Record: `usa_2020_8ea9dc`. **Category:** selection. Source.

Let $\alpha \geq 1$ be a real number. Hephaestus and Poseidon play a turn-based game on an infinite grid of unit squares. Before the game starts, Poseidon chooses a finite number of cells to be *flooded*. Hephaestus is building a *levee*, which is a subset of unit edges of the grid (called *walls*) forming a connected, non-self-intersecting path or loop. The game then begins with Hephaestus moving first. On each of Hephaestus's turns, he adds one or more walls to the levee, as long as the total length of the levee is at most αn after his nth turn. On each of Poseidon's turns, every cell which is adjacent to an already flooded cell and with no wall between them becomes flooded as well. Hephaestus wins if the levee forms a closed loop such that all flooded cells are contained in the interior of the loop — hence stopping the flood and saving the world. For which α can Hephaestus guarantee victory in a finite number of turns no matter how Poseidon chooses the initial cells to flood?

2020 — TSTST 6

Record: `usa_2020_93f219`. **Category:** selection. Source.

Let A, B, C, D be four points such that no three are collinear and D is not the orthocenter of triangle ABC. Let P, Q, R be the orthocenters of $\triangle BCD$, $\triangle CAD$, $\triangle ABD$, respectively. Suppose that lines AP, BQ, CR are pairwise distinct and are concurrent. Show that the four points A, B, C, D lie on a circle.

2020 — TST 2

Record: `usa_2020_95873f`. **Category:** selection. Source.

Two circles Γ_1 and Γ_2 have common external tangents l_1 and l_2 meeting at T. Suppose l_1 touches Γ_1 at A and l_2 touches Γ_2 at B. A circle Ω through A and B intersects Γ_1 again at C and Γ_2 again at D, such that quadrilateral $ABCD$ is convex. Suppose lines AC and BD meet at point X, while lines AD and BC meet at point Y. Show that T, X, Y are collinear.

2020 — TSTST 2

Record: `usa_2020_975da9`. **Category:** selection. Source.

Let ABC be a scalene triangle with incenter I. The incircle of ABC touches BC, CA, AB at points D, E, F, respectively. Let P be the foot of the altitude from D to EF, and let M be the midpoint of BC. The rays AP and IP intersect the circumcircle of triangle ABC again at points G and Q, respectively. Show that the incenter of triangle GQM coincides with D.

2020 — TSTST 3

Record: `usa_2020_992de5`. **Category:** selection. Source.

We say a nondegenerate triangle whose angles have measures $\theta_1, \theta_2, \theta_3$ is *quirky* if there exists integers r_1, r_2, r_3, not all zero, such that

$$r_1\theta_1 + r_2\theta_2 + r_3\theta_3 = 0.$$

Find all integers $n \geq 3$ for which a triangle with side lengths $n-1, n, n+1$ is quirky.

2020 — TSTST 9

Record: `usa_2020_fe77d1`. **Category:** selection. Source.

Ten million fireflies are glowing in $\mathbb{R}^3$ at midnight. Some of the fireflies are friends, and friendship is always mutual. Every second, one firefly moves to a new position so that its distance from each one of its friends is the same as it was before moving. This is the only way that the fireflies ever change their positions. No two fireflies may ever occupy the same point.

Initially, no two fireflies, friends or not, are more than a meter away. Following some finite number of seconds, all fireflies find themselves at least ten million meters away from their original positions. Given this information, find the greatest possible number of friendships between the fireflies.

In general, we show that when $n \geq 70$, the answer is $f(n) = \lfloor \frac{n^2}{3} \rfloor$.

2021 — 5

Record: `vnm_2021_6fbf99`. **Category:** selection. Source.

Given a fixed circle (O) and two fixed points B, C on that circle, let A be a moving point on (O) such that triangle ABC is acute and scalene. Let I be the midpoint of BC and let AD, BE and CF be the altitudes of triangle ABC. On two rays $\overrightarrow{FA}$, $\overrightarrow{EA}$, take M, N such that $FM = CE$, $EN = BF$. Let L be the intersection of MN and EF, and let $G \neq L$ be the second intersection of the circumcircles of triangles LEN and LFM.

a) Show that the circumcircle of triangle MNG always goes through a fixed point.

b) Let AD intersects (O) at K which is differs from A. On the tangent line through D of circumcircle of DKI, take P, Q such that GP is parallel to AB and GQ is parallel to AC. Let T be the circumcenter of triangle GPQ. Show that GT always goes through a fixed point.

2021 — 3

Record: `vnm_2021_e27739`. **Category:** selection. Source.

Let ABC be a triangle and N be a point that differs from A, B and C. Let A_b be the reflection of A through NB, and B_a be the reflection of B through NA. We define B_c, C_b, A_c and C_a similarly. Let m_a be the line passing through N and perpendicular to B_cC_b. Define m_b, m_c similarly.

a) Assume that N is the orthocenter of triangle ABC, show that the respective reflections of the lines m_a, m_b and m_c through each bisector of angles $\angle BNC$, $\angle CNA$ and $\angle ANB$ are coincident.

b) Assume that N is the nine-point center of triangle ABC, show that the respective reflections of the lines m_a, m_b and m_c through the lines BC, CA and AB concur.

2022 — 4

Record: `chn_2022_9e340b`. **Category:** selection. Source.

Find all positive integers k such that there are finitely many triangles on the Descartes coordinate plane such that (1) the center of mass of each triangle is an integral point; (2) the intersection of any two triangles is either the empty set, a common vertex, or an edge joining two common vertices; and (3) the union of these triangles is a square with side length k. (The vertices of the squares are not required to be integral points, and the edges are not required to be parallel to the coordinate axes.)

2022 — Problem 4

Record: `vnm_2022_98e5bf`. **Category:** selection. Source.

Given an acute, scalene triangle ABC with circumcircle (O). The line passing through O and midpoint I of BC intersects AB, AC at E, F. Let D, G be the reflections of A over O and the circumcenter of triangle AEF. Let K be the reflection of O over the circumcenter of triangle OBC.

a) Prove that D, G and K are collinear.

b) Take M on KB and N on KC such that IM is perpendicular to AC and IN is perpendicular to AB. The perpendicular bisector of IK intersects MN at H. Suppose that IH meets AB, AC at P, Q respectively. Prove that the circumcircle of triangle APQ intersects (O) again at a point on AI.

2023 — 2

Record: `chn_2023_b592f6`. **Category:** selection. Source.

In acute triangle ABC which is not isosceles, AP, BQ, CR are three altitudes, H is the orthocenter. The parallel line to BC passing through A intersects line RQ at point D. Let A_1 be the midpoint of BC, and let K be the intersection of RQ and AA_1. The line passing through the midpoint of AH and point K intersects line DA_1 at point A_2. Similarly define points B_2 and C_2.

Suppose that the circumcircle of non-degenerate triangle $A_2B_2C_2$ is ω. Prove: There exist three circles $\odot A'$, $\odot B'$ and $\odot C'$ inside ω that are tangent to ω and satisfy the following conditions: (1) $\odot A'$ is tangent to sides AB, AC, $\odot B'$ is tangent to sides BA, BC, and $\odot C'$ is tangent to sides CA, CB; (2) The centers of the three circles, A', B', C', are distinct and collinear.

2023 — 5

Record: `chn_2023_b8f301`. **Category:** selection. Source.

Suppose $P_1, \ldots, P_n$ are n points inside triangle ABC, such that any three points among $P_1, \ldots, P_n, A, B, C$ are not collinear. Prove: It is possible to divide $\triangle ABC$ into a union of $2n+1$ smaller triangles such that each smaller triangle has its vertices among $P_1, \ldots, P_n, A, B, C$, and the number of smaller triangles with vertices including at least one point among A, B, C is no less than $n + \sqrt{n} + 1$.

2023 — 1

Record: `chn_2023_c81d0b`. **Category:** selection. Source.

Suppose $n \geq 2$ is a positive integer, and $A_1A_2 \cdots A_{2n}$ is a convex $2n$-gon inscribed in a circle. It is known that there exists a point P inside this $2n$-gon such that

$$\angle PA_1A_2 = \angle PA_2A_3 = \cdots = \angle PA_{2n-1}A_{2n} = \angle PA_{2n}A_1.$$

Prove that the following equation holds

$$\prod_{i=1}^{n} |A_{2i-1}A_{2i}| = \prod_{i=1}^{n} |A_{2i}A_{2i+1}|,$$

where $A_{2n+1} = A_1$.

2023 — Problem 2

Record: `ind_2023_26115d`. **Category:** selection. Source.

In triangle ABC, let D be the foot of the perpendicular from A to line BC. Point K lies inside triangle ABC such that $\angle KAB = \angle KCA$ and $\angle KAC = \angle KBA$. The line through K perpendicular to line DK meets the circle with diameter BC at points X and Y. Prove that $AX \cdot DY = DX \cdot AY$.

2023 — Problem 1

Record: `ind_2023_6c27f2`. **Category:** selection. Source.

Suppose an acute scalene triangle ABC has incentre I and incircle touching BC at D. Let Z be the antipode of A in the circumcircle of ABC. Point L is chosen on the internal angle bisector of

$\angle BZC$ such that $AL = LI$. Let M be midpoint of arc BZC, and let V be the midpoint of ID. Prove that $\angle IML = \angle DVM$.

2023 — USA TST 2023/1

Record: `usa_2023_942b5a`. **Category:** selection. Source.

There are 2022 equally spaced points on a circular track γ of circumference 2022. The points are labeled $A_1, A_2, \ldots, A_{2022}$ in some order, each label used once. Initially, Bunbun the Bunny begins at A_1. She hops along γ from A_1 to A_2, then from A_2 to A_3, until she reaches A_{2022}, after which she hops back to A_1. When hopping from P to Q, she always hops along the shorter of the two arcs $\widehat{PQ}$ of γ; if $\widehat{PQ}$ is a diameter of γ, she moves along either semicircle. Determine the maximal possible sum of the lengths of the 2022 arcs which Bunbun traveled, over all possible labellings of the 2022 points.

2023 — 5

Record: `vnm_2023_030c5c`. **Category:** selection. Source.

Let $ABCD$ be a convex quadrilateral with $\angle B < \angle A < 90^\circ$. Let I be the midpoint of AB and S the intersection of AD and BC. Let R be a variable point inside the triangle SAB such that $\angle ASR = \angle BSR$. On the lines AR, BR, take the points E, F, respectively so that BE, AF are parallel to RS. Suppose that EF intersects the circumcircle of triangle SAB at points H, K. On the segment AB, take points M, N such that $\angle AHM = \angle BHI, \angle BKN = \angle AKI$.

a) Prove that the center J of the circumcircle of triangle SMN lies on a fixed line.

b) On BE, AF, take the points P, Q respectively so that CP is parallel to SE and DQ is parallel to SF. The lines SE, SF intersect the circumcircle of SAB, respectively, at U, V. Let G be the intersection of AU and BV. Prove that the median of vertex G of the triangle GPQ always passes through a fixed point.

2023 — 3

Record: `vnm_2023_29c942`. **Category:** selection. Source.

Let ABC be an acute, non-isosceles triangle with circumcircle (O). BE, CF are the heights of $\triangle ABC$, and BE, CF intersect at H. Let M be the midpoint of AH, and K be the point on EF such that HK is perpendicular to EF. A line not going through A and parallel to BC intersects the minor arcs AB and AC of (O) at P, Q, respectively. Show that the tangent line of the circumcircle of CQE at E, the tangent line of the circumcircle of BPF at F, and MK concur.

2024 — 3

Record: `ind_2024_1b53f9`. **Category:** selection. Source.

Let ABC be an acute-angled triangle with $AB < AC$, and let O, H be its circumcentre and orthocentre respectively. Points Z, Y lie on segments AB, AC respectively, such that

$$\angle ZOB = \angle YOC = 90^\circ.$$

The perpendicular line from H to line YZ meets lines BO and CO at Q, R respectively. Let the tangents to the circumcircle of $\triangle AYZ$ at points Y and Z meet at point T. Prove that Q, R, O, T are concyclic.

2024 — 2.

Record: `ind_2024_759b6c`. **Category:** selection. Source.

Let $\triangle ABC$ be an acute angled triangle with $AC > AB$ and incircle ω. Let ω touch the sides BC, CA, and AB at D, E, and F respectively. Let X and Y be points outside $\triangle ABC$ satisfying

$$\angle BDX = \angle XEA = \angle YDC = \angle AFY = 45^\circ.$$

Prove that the circumcircles of $\triangle AXY$, $\triangle AEF$ and $\triangle ABC$ meet at a point $Z \neq A$.

2024 — 1

Record: `ind_2024_804c67`. **Category:** selection. Source.

A sleeping rabbit lies in the interior of a convex 2024-gon. A hunter picks three vertices of the polygon and he lays a trap which covers the interior and the boundary of the triangular region determined by them. Determine the minimum number of times he needs to do this to guarantee that the rabbit will be trapped.

2024 — 1

Record: `ind_2024_8851c8`. **Category:** selection. Source.

Let ABC be an acute-angled triangle with $AB < AC$, incenter I, and let M be the midpoint of major arc BAC. Suppose the perpendicular line from A to segment BC meets lines BI, CI, and MI at points P, Q, and K respectively. Prove that the A-median line in $\triangle AIK$ passes through the circumcentre of $\triangle PIQ$.

2025 — Problem 4.

Record: `chn_2025_48f370`. **Category:** selection. Source.

In convex quadrilateral $ABCD$, $AB \perp AD$ and $AD = DC$. Let point E lie inside segment BC, and point F lie on the extension of DE such that $\angle ABF = \angle DEC > 90^\circ$. Let O be the circumcenter of triangle CDE. Let P be a point on the extension of FO such that $FP = FB$. Let segment BP intersect AC at point Q. Prove: $\angle AQB = \angle DPF$.

2025 — Problem 9.

Record: `chn_2025_843721`. **Category:** selection. Source.

Given a circle ω and two points A, B outside ω. A quadrilateral inscribed in ω is called "good" if one pair of opposite sides intersects at A and the other pair intersects at B. Assume that at least one good quadrilateral exists. Prove that there exists a good quadrilateral Γ such that every other good quadrilateral has area smaller than Γ.

2025 — Problem 5.

Record: `chn_2025_f1f576`. **Category:** selection. Source.

Prove that there exist integers $x_1, x_2, \dots, x_{10}, y_1, y_2, \dots, y_{10}$ satisfying: (1) For $i = 1, 2, \dots, 10$, $|x_i| \le 10^{10}$ and $|y_i| \le 10^{10}$; (2) The point set in the plane

$$X = \left\{ \left(\sum_{i=1}^{10} a_i x_i, \sum_{i=1}^{10} a_i y_i \right) \middle| a_1, a_2, \dots, a_{10} \in \{0, 1\} \right\}$$

contains exactly 1024 distinct points; (3) For any two parallel lines in the plane at distance 1, the strip region between them (including the lines) contains at most two points from X.

2025 — 4

Record: `usa_2025_08c332`. **Category:** selection. Source.

Let ABC be a triangle, and let X, Y, and Z be collinear points such that $AY = AZ$, $BZ = BX$, and $CX = CY$. Points X', Y', and Z' are the reflections of X, Y, and Z over BC, CA, and AB, respectively. Prove that if $X'Y'Z'$ is a nondegenerate triangle, then its circumcenter lies on the circumcircle of ABC.

2025 — 3

Record: `usa_2025_a27f25`. **Category:** selection. Source.

Let $A_1A_2 \cdots A_{2025}$ be a convex 2025-gon, and let $A_i = A_{i+2025}$ for all integers i. Distinct points P and Q lie in its interior such that $\angle A_{i-1}A_iP = \angle QA_iA_{i+1}$ for all i. Define points P_i^j and Q_i^j for integers i and positive integers j as follows: * For all i, $P_i^1 = Q_i^1 = A_i$. * For all i and j, P_i^{j+1} and Q_i^{j+1} are the circumcenters of $PP_i^jP_{i+1}^j$ and $QQ_i^jQ_{i+1}^j$, respectively. Let $\mathcal{P}$ and $\mathcal{Q}$ be the polygons $P_1^{2025}P_2^{2025} \dots P_{2025}^{2025}$ and $Q_1^{2025}Q_2^{2025} \cdots Q_{2025}^{2025}$, respectively. (a) Prove that $\mathcal{P}$ and $\mathcal{Q}$ are cyclic. (b) Let O_P and O_Q be the circumcenters of $\mathcal{P}$ and $\mathcal{Q}$, respectively. Assuming that $O_P \neq O_Q$, show that O_PO_Q is parallel to PQ.

2025 — 6

Record: `usa_2025_abaaab`. **Category:** selection. Source.

Prove that there exists a real number $\varepsilon > 0$ such that there are infinitely many sequences of integers $0 < a_1 < a_2 < \cdots < a_{2025}$ satisfying

$$\gcd(a_1^2 + 1, a_2^2 + 1, \dots, a_{2025}^2 + 1) > a_{2025}^{1+\varepsilon}.$$

2025 — Problem 5

Record: `usa_2025_ed5827`. **Category:** selection. Source.

A tetrahedron $ABCD$ is said to be *angelic* if it has nonzero volume and satisfies

$$\angle BAC + \angle CAD + \angle DAB = \angle ABC + \angle CBD + \angle DBA,$$
$$\angle ACB + \angle BCD + \angle DCA = \angle ADB + \angle BDC + \angle CDA.$$

Across all angelic tetrahedrons, what's the maximum number of distinct lengths that could appear in the set $\{AB, AC, AD, BC, BD, CD\}$?

2025 — Problem 9

Record: `usa_2025_fde337`. **Category:** selection. Source.

Let triangle ABC have orthocenter H. Let B_1, C_1, B_2 and C_2 be collinear points which lie on lines AB, AC, BH, and CH, respectively. Let ω_B and ω_C be the circumcircles of triangles BB_1B_2 and CC_1C_2, respectively. Prove that the radical axis of ω_B and ω_C intersects the line through their centers on the nine point circle of triangle ABC.

IMOTC source statements

2023 — Problem 1

Record: `ind_2023_ec5cac`. **Category:** pipeline supplement. Source.

Let ABC be a triangle, and let D be the foot of the A-altitude. Points P, Q are chosen on BC such that $DP = DQ = DA$. Suppose AP and AQ intersect the circumcircle of ABC again at X and Y. Prove that the perpendicular bisectors of the lines PX, QY, and BC intersect.

IranMO source statements

2015 — 9

Record: `irn_2015_5e81d0`. **Category:** contest. Source.

Find the maximum number of rectangles with sides equal to 1 and 2 and parallel to the coordinate axes such that each two have an area equal to 1 in common.

JMO source statements

2006 — 1

Record: `jpn_2006_25bba3`. **Category:** pipeline supplement. Source.

Five distinct points A, M, B, C and D are on a circle O in this order with $MA = MB$. Let the lines AC and MD intersect at P, the lines BD and MC at Q. Let the line PQ meet the circle O at X and Y. Prove that $MX = MY$.

2006 — 11

Record: `jpn_2006_a56a34`. **Category:** pipeline supplement. Source.

For $0 \leq x, y \leq 1$, let

$$f(x, y) = xy^2\sqrt{1 - x^2} - x^2y\sqrt{1 - y^2}$$

Find the minimum constant c for which the following condition holds. For any integer $n > 1$ and any real numbers $a_1, a_2, \ldots, a_n$ such that $0 \leq a_1 < a_2 < \cdots < a_n \leq 1$,

$$f(a_1, a_2) + f(a_2, a_3) + \cdots + f(a_{n-1}, a_n) < c$$

2006 — 2

Record: `jpn_2006_edaff8`. **Category:** pipeline supplement. Source.

P is a point inside an equilateral triangle. The length of the perpendiculars from P towards three sides are 1, 2, and 3. Answer the length of one edge of this triangle.

2006 — 9

Record: `jpn_2006_fa0abe`. **Category:** pipeline supplement. Source.

The triangle ABC satisfies $BC = 5$, $CA = 7$ and $AB = 8$. Let O be a point inside the triangle ABC such that the triangles OBC, OCA and OAB have the same circumradius. Find their common circumradius.

2006 — 10

Record: `jpn_2006_ff34be`. **Category:** pipeline supplement. Source.

A dodecahedron and a vertex X of the dodecahedron are given. An ant started from X, walked along the edges of the dodecahedron, passing each of the vertices of the dodecahedron except X just once, and returned to X. How many such routes exist? We consider a route and its reversal to be different.

2007 — 3

Record: `jpn_2007_20976d`. **Category:** pipeline supplement. Source.

Let Γ be the circumcircle of triangle ABC. Let Γ_A be the circle which is tangent to AB, AC and tangent to Γ internally, Γ_B be the circle which is tangent to AB, BC and tangent to Γ internally, and Γ_C be the circle which is tangent to AC, BC and tangent to Γ internally. Denote $\Gamma_A, \Gamma_B, \Gamma_C$ are tangent to Γ at P, Q, R. Prove that the lines AP, BQ, CR are concurrent.

2007 — 4

Record: `jpn_2007_2549e4`. **Category:** pipeline supplement. Source.

On a plane, the band with width d is the set of all points whose distance from a line is less than or equal to $\frac{d}{2}$. There are four points A, B, C, D on the plane. If you chose any three points among them, there exists a band with width 1 containing them. Prove that there exists a band with width $\sqrt{2}$ containing all four points.

2007 — 8

Record: `jpn_2007_44d09d`. **Category:** pipeline supplement. Source.

How many ways are there to cut a cube S into tetrahedron $\{T_1, \ldots, T_k\}$ with following properties? (1) Every vertex of $T_1, \ldots, T_k$ is one of the vertices of S. (2) For every $i \neq j$, the intersection of T_i and T_j is a common face of them, a common edge of them, a common vertex of them or empty.

2007 — 2

Record: `jpn_2007_5dc580`. **Category:** pipeline supplement. Source.

A quadrilateral $ABCD$ that satisfies $AB = 5$, $BC = 7$, $CD = 6$ is given. And AC and BD are perpendicular to each other. Find the length of DA.

2007 — 1

Record: `jpn_2007_7175c5`. **Category:** pipeline supplement. Source.

Let $ABCD$ be a convex quadrilateral with $AB = 3$, $BC = 4$, $CD = 5$, $DA = 6$ and $\angle ABC = 90^\circ$. Find the area of $ABCD$.

2007 — 3

Record: `jpn_2007_7fb20e`. **Category:** pipeline supplement. Source.

AB is a segment on a plane with length 7, and P is a point such that the distance between P and line AB is 3. Find the smallest possible value of $AP \times BP$.

2007 — 5

Record: `jpn_2007_ed31dc`. **Category:** pipeline supplement. Source.

3 rectangles are placed on a plane, and any 2 rectangles have their edges that are parallel to each other. They divide the plane into several parts. Determine the maximum possible number of such parts. (We consider the area that are contained in no rectangles as one part. For example, if we have only 1 rectangle, the plane is divided into 2 parts.)

2008 — 11

Record: `jpn_2008_94af0d`. **Category:** pipeline supplement. Source.

For a convex quadrilateral $ABCD$, the conditions $AB = BC = 2$, $CD = 2\sqrt{3}$, $DA = 2\sqrt{5}$ are satisfied. Also, if we let M, N be the midpoint of AC and BD, respectively, then $MN = \sqrt{2}$. Determine the area of the quadrilateral $ABCD$. Here, by XY we are representing the length of the line segment XY.

2008 — 2

Record: `jpn_2008_d6c6e8`. **Category:** pipeline supplement. Source.

Let $ABCD$ be a square of side length 1. Let O be the circle having the side AD of the square as its diameter, and pick a point E on the side AB of the square in such a way that the line CE becomes a tangent line to the circle O. Determine the area of the triangle CBE.

2008 — 3

Record: `jpn_2008_fcdcf8`. **Category:** pipeline supplement. Source.

Let O be the circumcenter of an acute triangle ABC. A circle passing through points A, O intersects lines AB and AC at points P, Q distinct from A, respectively. If the lengths of the line segments PQ and BC are equal, what is the magnitude of the angle formed by the lines PQ and BC and less than 90°?

2009 — 12

Record: `jpn_2009_0a1555`. **Category:** pipeline supplement. Source.

A plane lying in the 3-dimensional space splits the space into 2 parts. We call one such part (excluding the plane) a **half space**. Let S be a set consisting of 10 points in the space, no 4 among them lying on a same plane. Determine the number of subsets of S which can be obtained as an intersection of S with some half space.

2009 — 4

Record: `jpn_2009_0cf9c5`. **Category:** pipeline supplement. Source.

Let M be the mid-point of the side BC of a triangle ABC. Suppose $AB = 4$ and $AM = 1$, where by XY we denote the length of the line segment XY. Determine the smallest possible value the angle $\angle BAC$ can take.

2009 — 4

Record: `jpn_2009_c834cd`. **Category:** pipeline supplement. Source.

Let Γ be the circum-circle of a triangle $\triangle ABC$. Suppose a circle with its center at a point O is tangent to the line segment BC at a point P, and is tangent to the arc BC (not containing A) at a point Q. If $\angle BAO = \angle CAO$, show that $\angle PAO = \angle QAO$ must be satisfied.

2009 — 6

Record: `jpn_2009_f65d67`. **Category:** pipeline supplement. Source.

Suppose for a tetrahedron $OABC$, $OA = 3$, $OB = 4$, $OC = 5$, where by XY we denote the length of the line segment XY. Suppose also that we have $\angle AOB = \angle AOC = 45^\circ$, and $\angle BOC = 60^\circ$. Determine the volume of the tetrahedron $OABC$.

2010 — 5

Record: `jpn_2010_04151c`. **Category:** pipeline supplement. Source.

Suppose a regular 2010-gon is given. How many combinations of 3 distinct vertices A, B, C of this polygon are there for which all the inner angles of the triangle ABC have integer-valued degrees (i.e., integral multiples of 1°)? Regard 2 triples as representing the same combination if one is a permutation of the other.

2010 — 5

Record: `jpn_2010_3dd59d`. **Category:** pipeline supplement. Source.

Suppose for a convex 2010-gon, any 3 diagonals do not share a common point except for vertices. Let us consider a closed zig-zag line segments (abbr. by Z.Z.L.S.) which goes through each of the vertices of the 2010-gon once and only once. Determine the maximum possible number of the self-intersection points for such a closed Z.Z.L.S. Here, by a closed Z.Z.L.S. we mean contiguous line segments $P_1P_2 \ldots P_nP_{n+1}$ for which $P_1 = P_{n+1}$ is satisfied.

2012 — 7

Record: `jpn_2012_51ed62`. **Category:** pipeline supplement. Source.

Let O be the circum-center of a triangle ABC. When points D and E were chosen on the line segments AB and AC, respectively, the mid-point of the line segment DE coincided with the point O. If $AD = 8$, $BD = 3$ and $AO = 7$, determine the value of CE. Here for a line segment XY we denote also by XY its length.

2012 — 4

Record: `jpn_2012_b90bde`. **Category:** pipeline supplement. Source.

Triangles PAB and PCD are placed on a plane. Suppose that $PA = PB$, $PC = PD$ are satisfied and that the 3 points P, A, C lie on a straight line in this order and the same is true for the 3 points B, P, D. Suppose further that a circle S_1 going through A, C and the circle S_2 going through B, D intersect at 2 distinct points X and Y. Prove that the orthocenter of the triangle PXY coincides with the mid-point of the line segment connecting the centers of the circles S_1 and S_2. Here, we denote for a line segment ZW its length also by ZW.

2012 — 1

Record: `jpn_2012_c71ac2`. **Category:** pipeline supplement. Source.

For a triangle ABC the line tangent to its circum-circle at A and the line BC intersects at a point P. Let Q and R be the points which are symmetrically located from P with respect to the lines AB and AC, respectively. Prove that the lines BC and QR intersect perpendicularly.

2012 — 4

Record: `jpn_2012_f8f195`. **Category:** pipeline supplement. Source.

Given two triangles PAB and PCD such that $PA = PB$, $PC = PD$, P, A, C and B, P, D are collinear in this order respectively. The circle S_1 passing through A, C intersects with the circle S_2 passing through B, D at distinct points X, Y. Prove that the circumcenter of the triangle PXY is the midpoint of the centers of S_1, S_2.

2013 — 5

Record: `jpn_2013_348c1d`. **Category:** pipeline supplement. Source.

Let n be a positive integer. Points $P_1, P_2, \ldots, P_{4n}$ are placed in a plane in such a way that no 3 points among them lie on any straight line. Furthermore, for each $i = 1, 2, \ldots, 4n$ if we rotate the half-line P_iP_{i-1} starting at P_i around the point P_i by 90° clockwise, then the half line falls onto the half-line P_iP_{i+1} starting at P_i. Determine the maximum possible number of the pairs (i, j) for which the line segments P_iP_{i+1} and P_jP_{j+1} intersect at a point different from the end points of the line segments. Here we let $P_0 = P_{4n}, P_{4n+1} = P_1$ and assume that $1 \leq i < j \leq 4n$.

2013 — 5

Record: `jpn_2013_6a6bc7`. **Category:** pipeline supplement. Source.

Let n be a positive integer. Points $P_1, P_2, \ldots, P_{4n}$ are placed in a plane in such a way that no 3 points among them lie on any straight line. Furthermore, for each $i = 1, 2, \ldots, 4n$ if we rotate the half-line P_iP_{i-1} starting at P_i around the point P_i by 90° clockwise, then the half line falls onto the half-line P_iP_{i+1} starting at P_i. Determine the maximum possible number of the pairs (i, j) for which the line segments P_iP_{i+1} and P_jP_{j+1} intersect at a point different from the end points of the line segments. Here we let $P_0 = P_{4n}, P_{4n+1} = P_1$ and assume that $1 \leq i < j \leq 4n$.

2013 — 5

Record: `jpn_2013_812fcd`. **Category:** pipeline supplement. Source.

Two circles O_1 and O_2 intersect at two distinct points P and Q. The tangent line to the circle O_1 at the point P intersects the circle O_2 at R, different from P, and the tangent line to the circle O_2 at the point Q intersects the circle O_1 at S, different from Q. Let X be the point of the intersection of the two lines PR and QS. If $XR = 9$ and $XS = 2$, what is the value of the ratio $\frac{r_1}{r_2}$, where r_1 and r_2 are the radii of the circles O_1, O_2, respectively? Here we denote by YZ the length of the line segment YZ.

2013 — 8

Record: `jpn_2013_b52f08`. **Category:** pipeline supplement. Source.

Let $ABCD$ be a convex quadrilateral for which the line segments AC and BD intersect perpendicularly at a point X. Suppose $AX = 5$, $BX = 6$ and $CX = 20$ are satisfied. Let

us denote by O_1, O_2, O_3 and O_4, respectively, the circle with center at A and radius AX, the circle with center at B and radius BX, the circle with center at C and radius CX and the circle with center at D and radius DX. Determine the value of DX if there exists a circle which is tangent to all of the circles O_1, O_2, O_3 and O_4. Here we denote by YZ the length of the line segment YZ.

2013 — 4

Record: `jpn_2013_d03b98`. **Category:** pipeline supplement. Source.

Let ABC be an acute triangle and let H be its orthocenter. Suppose that a circle going through the points B, C and the circle having the line segment AH as its diameter intersect at two distinct points X and Y. Let D be the foot of the perpendicular line drawn from A to the line BC, and K be the foot of the perpendicular line drawn from D to the line XY. Prove that $\angle BKD = \angle CKD$ must hold.

2014 — 9

Record: `jpn_2014_1b8093`. **Category:** pipeline supplement. Source.

Let O be the point of intersection of two diagonals of a square $ABCD$. Points P, Q, R, S lie on the line segments OA, OB, OC, OD, respectively, and satisfy $OP = 3$, $OQ = 5$, $OR = 4$. Here we denote for a line segment XY its length also by XY. If the point of intersection of lines AB and PQ, the point of intersection of lines BC and QR, and the point of intersection of lines CD and RS are collinear, what is the value of OS?

2014 — 4

Record: `jpn_2014_45d731`. **Category:** pipeline supplement. Source.

On the circumference of a circle, 6 points A, B, C, D, E, F are placed in this order counter-clockwise, and three lines AD, BE and CF intersect at a single point. If

$$AB = 1,\ BC = 2,\ CD = 3,\ DE = 4,\ EF = 5,$$

find the value of FA. Here we denote the length of a line segment XY also by XY.

2014 — 1

Record: `jpn_2014_51a9ec`. **Category:** pipeline supplement. Source.

Circle C_1 is internally tangent to circle C_2 at point A. Let O be the center of circle C_2. A tangent line to circle C_1 at point P on C_1 goes through point O. Let Q be the point of intersection of half-line OP and circle C_2, and let R be the point of intersection of the line tangent to circle C_1 at point A and the line OP. Suppose that the radius of circle C_2 is 9 and that $PQ = QR$ is satisfied. Here we denote the length of the line segment XY also by XY. Determine the value of OP.

2014 — 4

Record: `jpn_2014_f9e49e`. **Category:** pipeline supplement. Source.

Let Γ be the circumcircle of a triangle ABC, and ℓ be the line tangent to Γ at point A. Let D, E be interior points of the sides AB, AC, respectively, satisfying the condition $BD : DA =$

$AE : EC$. Let F, G be the two points of intersection of line DE and circle Γ, H be the point of intersection of the line ℓ and the line parallel to AC and going through point D, and I be the point of intersection of the line ℓ and the line parallel to AB and going through point E. Prove that four points F, G, H, I lie on the circumference of a circle which is tangent to line BC. Here, for a line segment XY we denote its length also by XY.

2015 — 7

Record: `jpn_2015_bd227a`. **Category:** pipeline supplement. Source.

Let l be a straight line lying on the xy-plane. For given 20×15 points $(m, n) : (m = 1, 2, \dots, 20, n = 1, 2, \dots, 15)$ on the xy-plane, there are 222 straight lines parallel to the line l (l itself may be considered as one of those parallel lines) and going through at least one of these points. How many straight lines are there which go through at least one of these points and are perpendicular to l?

2019 — 6

Record: `jpn_2019_13b27b`. **Category:** pipeline supplement. Source.

Some of the vertices of a regular 120-gon are labeled by $*$. What is the maximal possible number of vertices labeled with $*$ if the following condition must be satisfied?

There exists no isosceles triangle with the angle at the top vertex of 18° formed by $*$-labeled vertices of the 120-gon.

2019 — 4

Record: `jpn_2019_94e54f`. **Category:** pipeline supplement. Source.

Let I be the in-center and Ω be the inscribed circle of a triangle ABC, and let M be the mid-point of the side BC. Let K be the point of intersection of the line, going through A and perpendicular to line BC, and the line, going through M and perpendicular to the line AI. Prove that the circle having the line segment AK as a diameter is tangent to the circle Ω.

2019 — 10

Record: `jpn_2019_a9b93a`. **Category:** pipeline supplement. Source.

Suppose D, E are points on the sides AB, AC, respectively, of a triangle ABC, and the following are known: $AB = 6$, $AC = 9$, $AD = 4$, $AE = 6$. Suppose, furthermore, the circum-circle of the triangle ADE intersects the side BC at two points F, G and the points B, F, G, C are lined up in this order. If the point of intersection of the lines DF and EG lies on the circum-circle of the triangle ABC, find the value of $\frac{FG}{BC}$. Here XY denotes the length of the line segment XY as well.

2019 — 8

Record: `jpn_2019_d8e74b`. **Category:** pipeline supplement. Source.

Let I be the in-center of a triangle ABC satisfying $AB > AC$, and let D, E be points on the sides AB, AC, respectively, dividing the sides into two segments with $1 : 8$ ratio. If the triangle DIE becomes the regular triangle with side length 1, what is the length of AB? Here by XY we denote the length of the line segment XY as well.

2022 — 7

Record: `jpn_2022_183337`. **Category:** pipeline supplement. Source.

In isosceles right triangle ABC with $\angle BAC = 90^\circ$ and $AB = AC = 7$, let D, E and F be points on sides BC, CA and AB respectively. Given that $\angle EDF = 90^\circ$, $DE = 5$, $DF = 3$, find the length of BD.

2022 — 3

Record: `jpn_2022_3901aa`. **Category:** pipeline supplement. Source.

In isosceles triangle ABC with $AB = AC$, point O is in its interior (not including circumference) and the circle ω centered O and passing through C intersects the sides (excluding end points) BC and AC at D and E, respectively. Let Γ be the circumcircle of triangle AEO and intersect with ω again at $F \neq E$. Prove that the circumcenter of the triangle BDF lies on Γ. In the above, denote by XY the length of line segment XY.

2022 — 4

Record: `jpn_2022_88f911`. **Category:** pipeline supplement. Source.

There is a convex quadrilateral $ABCD$ and a point P inside such that the lines AP and AD are orthogonal and the lines BP and CD are orthogonal. When $AB = 7$, $AP = 3$, $BP = 6$, $AD = 5$, $CD = 10$, find the area of triangle ABC.

2022 — 12

Record: `jpn_2022_9e8b52`. **Category:** pipeline supplement. Source.

Let ABC be an acute triangle with $AB = 11$, $AC = 10$, and denote the orthocenter by H and the midpoint of BC by M. Point P in triangle ABC lies on the circumcircle of triangle BHC and satisfies $\angle ABP = \angle CPM$, $PM = 3$. Find the length of BC.

2024 — 3

Record: `jpn_2024_5e3413`. **Category:** pipeline supplement. Source.

Let us call a point in the xy-plane a good point if each of its coordinates is an integer from 1 to 2000. Let us also call polyline $ABCD$ a Z-shaped polyline if four points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, $D(x_4, y_4)$ satisfy all the following conditions: * A, B, C, D are good points. * $x_1 < x_2$, $y_1 = y_2$. * $x_2 > x_3$, $y_2 - x_2 = y_3 - x_3$. * $x_3 < x_4$, $y_3 = y_4$. Determine the smallest possible positive integer n such that, there exist Z-shaped polylines $Z_1, Z_2, \ldots, Z_n$ which satisfy the following condition: Any good point P lies on Z_i for some $1 \leq i \leq n$. Note that polyline $ABCD$ is the union sets of line segments (including both endpoints) AB, BC and CD.

2024 — 9

Record: `jpn_2024_957af4`. **Category:** pipeline supplement. Source.

A quadrilateral $ABCD$ is inscribed in a circle, with $AB = 7$ and $BC = 18$. The bisector of $\angle CDA$ intersects side BC at point E, and a point F on segment DE satisfies $\angle AED = \angle FCD$. When $BE = 5$ and $EF = 3$, find the length of segment DF.

2024 — 4

Record: `jpn_2024_b8af00`. **Category:** pipeline supplement. Source.

Let ABC be an acute triangle with $AB < AC$, O be its circumcenter, and M be the midpoint of arc BC of triangle ABC's circumcircle which does not include A. There is a point D on the extension of side AB beyond B satisfying $BD = BM$, and there is a point E on side AC (except for end points) satisfying $CE = CM$. If the circumcircles of triangle ABE and ACD intersect at the point X other than A, prove that the perpendicular bisector of segment DE is tangent to the circumcircle of triangle AOX.

2024 — 6

Record: `jpn_2024_c08f01`. **Category:** pipeline supplement. Source.

Let ABC be an isosceles triangle with $AB = AC = 5$. Let D be a point on side AB satisfying $AD = 3$, and let E be a point on side BC (excluding the endpoints B and C). Let ω be the circle passing through E and tangent to line AB at B. Suppose that ω is tangent to the circumcircle of triangle ADE. Let F be the intersection point of ω and line AE, other than E. When $CF = 10$, find the length of side BC.

2025 — 12

Record: `jpn_2025_1e02a4`. **Category:** pipeline supplement. Source.

A pentagon **ABCDE** is inscribed in a circle Ω, and it satisfies $AC = AD$ and $BC \parallel DE$. Take a point P on the arc CD of Ω which does not contain A. Let P_1, P_2, P_3, P_4, and P_5 be the points symmetric to P with respect to the lines AB, BC, CD, DE, and EA, respectively. When $PC : PD = P_1P_2 : P_4P_5 = 2 : 3$ and $CD : P_2P_4 = 4\sqrt{2} : 11$, determine the value of $\frac{P_1P_3}{P_1P_5}$.

2025 — 9

Record: `jpn_2025_95e290`. **Category:** pipeline supplement. Source.

Let ABC be an acute triangle with circumcenter O and let D be the foot of the perpendicular from A to BC. Assume $\angle AOD = 90^\circ$ and $OD = 4\sqrt{7}$ hold. Let E and F be the feet of perpendiculars from D to AB and AC respectively, and let the lines AO and EF meet at P. If $AP = 11$, find the length of EF.

2025 — 5

Record: `jpn_2025_96fe64`. **Category:** pipeline supplement. Source.

In the interior of an acute triangle ABC that is not isosceles, there are three distinct points A_1, B_1, C_1 such that $AB_1 : CB_1 = AB : CB$ and $AC_1 : BC_1 = AC : BC$. Let A_2 be the point symmetric to A_1 with respect to the line BC, let B_2 be the point symmetric to B_1 with respect to the line AC, and let C_2 be the point symmetric to C_1 with respect to the line AB. Then, all of the following conditions are satisfied. • The four points A, A_2, B, C_2 are concyclic. • The four points A, A_2, B_2, C are concyclic. • The four points B, B_2, C, C_2 are concyclic. • None of the three points A_2, B_2, C_2 lie on the circumcircle of the triangle ABC. Prove that the triangle $A_1B_1C_1$ and the triangle $A_2B_2C_2$ are similar.

2025 — 2

Record: `jpn_2025_e2076e`. **Category:** pipeline supplement. Source.

Let $\triangle ABC$ be an acute triangle with circumcenter O. Let O_1 and O_2 be the circumcenters of triangles ABO and ACO, respectively. Suppose that the circumcircle of triangle AO_1O_2 intersects line segment BC at two distinct points P and Q (excluding the endpoints), with the four points B, P, Q, C appearing in this order along the segment. Let O_3 be the circumcenter of triangle OPQ. Prove that the three points A, O, and O_3 lie on a straight line.

2025 — 5

Record: `jpn_2025_f58513`. **Category:** pipeline supplement. Source.

Let $ABCD$ be a cyclic quadrilateral with an incircle of radius 6. Let the extensions of sides AB and DC beyond B and C, respectively, meet at P, and let the extensions of sides AD and BC beyond D and C, respectively, meet at Q. The inradii of triangle PBC and QCD are 5 and 3, respectively. Find $\frac{BC}{CD}$.

KMO source statements

2006 — 5

Record: `kor_2006_db5ca5`. **Category:** pipeline supplement. Source.

For a convex hexagon $ABCDEFG$, three triangles ABC, CDE, EFA are similar. That is,

$$\angle BAC = \angle DCE = \angle FEA$$

$$\angle BCA = \angle DEC = \angle FAE.$$

Find conditions on these three triangles under which $\triangle ACE$ is an equilateral triangle if and only if $\triangle BDF$ is an equilateral triangle.

2006 — 1

Record: `kor_2006_fe3b44`. **Category:** pipeline supplement. Source.

Let $\triangle ABC$ be a triangle with $\angle B \neq \angle C$. The incircle I of a triangle ABC touches the sides BC, CA, AB at the points D, E, F, respectively. Let P be the intersection of AD and the incircle I, which is different from D. Let Q be the intersection of the line EF and the line passing P and perpendicular to AD, and let X, Y be intersections of the line AQ and DE, DF, respectively. Show that the point A is the midpoint of XY.

2008 — 1

Record: `kor_2008_08565a`. **Category:** pipeline supplement. Source.

Given a cyclic hexagon $ABCDEF$, let BD and CF meet at G, AC and BE meet at H, AD and CE meet at I. Suppose that BD is perpendicular to CF and $AI = CI$. Show that $CH = AH + DE$ if and only if $GH \cdot BD = BC \cdot DE$.

2008 — 5

Record: `kor_2008_7f3c09`. **Category:** pipeline supplement. Source.

Let a quadrilateral $ABCD$ be inscribed in a circle O with $\angle B$ and $\angle C$ obtuse. Let E be the point of intersection of the lines AB and CD. Let P and R be the feet of the perpendicular lines from E to the lines BC and AD, respectively. Let Q be the point of intersection of the lines EP and AD, and S be the point of intersection of the lines ER and BC. Let K be the midpoint of the line segment QS. Prove that the three points E, K, O are collinear.

2009 — 4

Record: `kor_2009_0baca7`. **Category:** pipeline supplement. Source.

Let $\triangle ABC$ be a triangle with $\angle B < \angle C$. Let O be the circle tangent to the line AC at the point C and passing through the point B. The lines AB and CO meet the circle O at the points $D(\neq B)$ and $P(\neq C)$, respectively. Let E be the intersection of AC and the line passing through P and

parallel to AO. The line EB and the circle O meet at the point $L(\neq B)$. Let F be the intersection of the bisector of the line segment BD and the line AC, and K be the intersection of LF and CD. Show that EK and CL are parallel.

2009 — 1

Record: `kor_2009_bab5c4`. **Category:** pipeline supplement. Source.

Let a, b and c be the sides of a triangle, and we set

$$A = \frac{a^2+bc}{b+c} + \frac{b^2+ca}{c+a} + \frac{c^2+ab}{a+b}$$

$$B = \frac{1}{\sqrt{(a+b-c)(b+c-a)}} + \frac{1}{\sqrt{(b+c-a)(c+a-b)}} + \frac{1}{\sqrt{(c+a-b)(a+b-c)}}$$

Prove that $AB \geq 9$.

2011 — 5

Record: `kor_2011_0a123e`. **Category:** pipeline supplement. Source.

Let ABC be a triangle with $AC < AB < BC$. Choose a point D on the side AB. Suppose the circumcircle of ABC meets the internal angle bisector of $\angle A$ and CD at E $(\neq A)$ and F $(\neq C)$, respectively. Let K be the intersection point of BC and DE. Prove that $CK = AC$ if and only if $DK \cdot EF = AC \cdot DF$.

2011 — 2

Record: `kor_2011_35d000`. **Category:** pipeline supplement. Source.

Let P be a point on the side BC of an acute triangle ABC. Let H be the orthocenter of $\triangle ABC$ and D be the foot of the perpendicular from H to AP. Let Γ_1, Γ_2 be the circumcircles of $\triangle ABD, \triangle ACD$, respectively. Let l be the line parallel to BC and passing through D. l intersects Γ_1 and Γ_2 at X and Y, different from D, respectively. l intersects AB and AC at E and F, respectively. Two lines XB and YC meet at Z. Show that $BP = CP$ if and only if $ZE = ZF$.

2013 — 4

Record: `kor_2013_1fbbe3`. **Category:** pipeline supplement. Source.

For a triangle ABC let B_1, C_1 be the excenters corresponding to B, C, respectively. The line B_1C_1 intersects the circumcircle of ABC at D $(\neq A)$. Let E be the intersection of the perpendiculars from B_1 to CA and from C_1 to AB. Let ω be the circumcircle of the triangle ADE. The tangent to ω at D intersects the line AE at F. The perpendicular from D to AE intersects AE at G and this line intersects ω at H $(\neq D)$. The circumcircle of the triangle HGF intersects ω at I $(\neq H)$. Let J be the perpendicular foot at AH from D. Show that AI passes through the midpoint of DJ.

2013 — 1

Record: `kor_2013_406ec8`. **Category:** pipeline supplement. Source.

Let ABC be a triangle satisfying $\angle B > \angle C$ and let D be the point on the side AC such that $\angle ADC = \angle C$. Let I be the incenter of ABC and E be the intersection of the circumcircle of CDI and the line AI which is not I. Let P be the intersection of the line BD and the line which is parallel to AB and passing E. Let J be the incenter of ABD and A' be the reflection of A with respect to I. Suppose that two lines JP and $A'C$ meet at the point Q. Show that $QJ = QA'$.

2014 — Problem 4

Record: `kor_2014_170404`. **Category:** pipeline supplement. Source.

Let ABC be an isosceles triangle with $AC = BC$. Let D be a point on a line BA such that A lies between B and D. Let O_1 be the circumcircle of triangle DAC. O_1 meets BC at point E. Let F be the point on the line BC such that FD is tangent to circle O_1, and let O_2 be the circumcircle of triangle DBF. Two circles O_1, O_2 meet at point G ($G \neq D$). Let O be the circumcenter of triangle BEG. Prove that the line FG is tangent to circle O if and only if DG is perpendicular to FO.

2014 — Problem 2

Record: `kor_2014_8107b0`. **Category:** pipeline supplement. Source.

Given a triangle ABC with $AC = BC > AB$, let E and F be the midpoints of AC and AB, respectively. The perpendicular bisector ℓ of AC meets AB at K and the line parallel to KC and passing through the point B intersects AC at a point L. For a point P on the line segment BF, let H be the orthocenter of the triangle ACP. The line segments BH and CP meet at a point J and the lines FJ and ℓ meet at a point M. Let W be the intersection point of FL and ℓ. Show that $AW = BW$ if and only if the points B, E, F, M are concyclic.

2015 — 4

Record: `kor_2015_28b822`. **Category:** pipeline supplement. Source.

Let H be the orthocenter of an acute triangle ABC, and a circle ω that passes through all of the points H, A, B meet the line segment BC at the point $D(\neq B)$. Let P be the intersecting points of the line DH and the line segment AC, and Q be the circumcenter of the triangle ADP. Show that the center of the circle ω lies on the circumscribed circle of the triangle BDQ.

2015 — 2

Record: `kor_2015_c69f98`. **Category:** pipeline supplement. Source.

Let I be the incenter of a triangle ABC whose incircle is tangent to the sides BC, AC, AB at D, E, F, respectively. Suppose the circumcircle of the triangle ABC intersects the line EF at P and Q. If O_1 and O_2 are the circumcenters of the triangles IAB and IAC, respectively, show that the circumcenter of the triangle DPQ lies on the line O_1O_2.

2016 — Problem 2

Record: `kor_2016_3ddc3f`. **Category:** pipeline supplement. Source.

Let n and k be integers satisfying $n \geq 2$ and $k \geq \frac{5}{2}n - 1$. Prove that in every choice of k distinct points among all integer points (x, y) with $1 \leq x, y \leq n$, there exists a circle going through at least four distinct chosen points.

2016 — Problem 5

Record: `kor_2016_82ba56`. **Category:** pipeline supplement. Source.

The incircle of a triangle ABC is tangent to BC, AC, AB at the points D, E, F, respectively. Suppose the line EF intersects the lines BI, CI, BC, DI at the points K, L, M, Q, respectively, where the incenter of $\triangle ABC$ is I. If the line passing through both the midpoint of CL and M intersects CK at a point P, show that

$$PQ = \frac{AB \cdot KQ}{BI}.$$

The incircle of a triangle *ABC* is tangent to *BC*, *AC*, *AB* at the points *D*, *E*, *F*, respectively. Suppose the line *EF* intersects the lines *BI*, *CI*, *BC*, *DI* at the points *K*, *L*, *M*, *Q*, respectively, where the incenter of *ΔABC* is *I*. If the line passing through both the midpoint of *CL* and *M* intersects *CK* at a point *P*, show that

$$PQ = \frac{AB \cdot KQ}{BI}$$

2023 — 6

Record: `kor_2023_57d25e`. **Category:** pipeline supplement. Source.

For an integer $n \geq 3$ and real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$, show the following inequality.

$$\sum_{i=1}^{n} a_i(b_i - b_{i+3}) \leq \frac{3n}{8} \sum_{i=1}^{n} \left[(a_i - a_{i+1})^2 + (b_i - b_{i+1})^2\right] (a_{n+1} = a_1 \text{ and } b_{n+1} = b_1 \text{ for } i = 1, 2, 3)$$

2023 — 1

Record: `kor_2023_7a2a3c`. **Category:** pipeline supplement. Source.

Let ABC be an acute triangle with $AB < AC$. Points D, E lie on the interiors of AB, AC, respectively. Let P be a point satisfying $PB = PD, PC = PE$. Let X be a point on the interior of arc AC of the circumcircle of ABC which does not include the point B. The line XA meets again the circumcircle of ADE at Y. Show that $PX = PY$.

RMM source statements

2015 — Problem 6

Record: `rmm_2015_4db6ce`. **Category:** contest. Source.

Given a positive integer n, determine the largest real number μ satisfying the following condition: for every $4n$-point configuration C in an open unit square U, there exists an open rectangle in U, whose sides are parallel to those of U, which contains exactly one point of C, and has an area greater than or equal to μ.

2015 — Problem 6

Record: `rou_2015_eb480f`. **Category:** contest. Source.

Given a positive integer n, determine the largest real number μ satisfying the following condition: for every $4n$-point configuration C in an open unit square U, there exists an open rectangle in U, whose sides are parallel to those of U, which contains exactly one point of C, and has an area greater than or equal to μ.

2016 — Problem 5.

Record: `rou_2016_093c05`. **Category:** contest. Source.

A convex hexagon $A_1B_1A_2B_2A_3B_3$ is inscribed in a circle Ω of radius R. The diagonals A_1B_2, A_2B_3, and A_3B_1 concur at X. For $i = 1, 2, 3$, let ω_i be the circle tangent to the segments XA_i and XB_i, and to the arc A_iB_i of Ω not containing other vertices of the hexagon; let r_i be the radius of ω_i.

a) Prove that $R \geq r_1 + r_2 + r_3$.

b) If $R = r_1 + r_2 + r_3$, prove that the six points where the circles ω_i touch the diagonals A_1B_2, A_2B_3, A_3B_1 are concyclic.

2016 — Problem 6.

Record: `rou_2016_185273`. **Category:** contest. Source.

A set of n points in Euclidean 3-dimensional space, no four of which are coplanar, is partitioned into two subsets A and B. An $\mathcal{AB}$-tree is a configuration of $n-1$ segments, each of which has an endpoint in A and the other in B, and such that no segments form a closed polyline. An $\mathcal{AB}$-tree is transformed into another as follows: choose three distinct segments A_1B_1, B_1A_2 and A_2B_2 in the $\mathcal{AB}$-tree such that A_1 is in A and $A_1B_1 + A_2B_2 > A_1B_2 + A_2B_1$, and remove the segment A_1B_1 to replace it by the segment A_1B_2. Given any $\mathcal{AB}$-tree, prove that every sequence of successive transformations comes to an end (no further transformation is possible) after finitely many steps. Russia

2016 — Problem 1.

Record: `rou_2016_eef1e4`. **Category:** contest. Source.

Let ABC be a triangle and let D be a point on the segment BC, $D \neq B$ and $D \neq C$. The circle ABD meets the segment AC again at an interior point E. The circle ACD meets the segment AB again at an interior point F. Let A' be the reflection of A in the line BC. The lines $A'C$ and DE meet at P, and the lines $A'B$ and DF meet at Q. Prove that the lines AD, BP and CQ are concurrent (or all parallel).

2017 — Problem 6

Record: `rou_2017_4320f5`. **Category:** contest. Source.

Let $ABCD$ be any convex quadrangle and let P, Q, R, S be points on the segments AB, BC, CD, and DA, respectively. It is given that the segments PR and QS dissect $ABCD$ into four quadrangles, each of which has perpendicular diagonals. Show that the points P, Q, R, S are concyclic.

Nikolai Beluhov, Bulgaria

2017 — Problem 4

Record: `rou_2017_6a40eb`. **Category:** contest. Source.

In the Cartesian plane, let γ_1 and γ_2 be the graphs of the quadratic functions $f_1(x) = p_1x^2 + q_1x + r_1$ and $f_2(x) = p_2x^2 + q_2x + r_2$, where $p_1 > 0 > p_2$. The graphs γ_1 and γ_2 cross at distinct points A and B. The four tangents to γ_1 and γ_2 at A and B form a convex quadrangle which has an inscribed circle. Prove that the graphs γ_1 and γ_2 have the same axis of symmetry.

2019 — Problem 4

Record: `rmm_2019_80fa3b`. **Category:** contest. Source.

Prove that for every positive integer n there exists a (not necessarily convex) polygon with no three collinear vertices, which admits exactly n different triangulations. (A triangulation is a dissection of the polygon into triangles by interior diagonals which have no common interior points with each other nor with the sides of the polygon.)

2019 — Problem 2

Record: `rmm_2019_f4d767`. **Category:** contest. Source.

Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC. Denote by ω and Ω the circumcircles of the triangles ABE and CDE, respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D. Prove that PE is tangent to Ω.

2023 — 2

Record: `rou_2023_9c5b23`. **Category:** contest. Source.

Fix an integer $n \geq 3$. Let S be a set of n points in the plane, no three of which are collinear. Given pairwise distinct points A, B, C in S, the triangle ABC is suitable for AB if area $ABC \leq$ area ABX for all X in S different from A and B. (Note that for a segment AB there could be several suitable triangles.) A triangle is adequate if its vertices are all in S and it is suitable for at least two of its sides. Prove that there are at least $\frac{1}{2}(n-1)$ adequate triangles. Bulgaria, Alexadar Ivanov

2024 — Problem 5.

Record: rou_2024_05cf8c. **Category:** contest. Source.

Let BC be a fixed segment in the plane, and let A be a variable point in the plane not on the line BC. Distinct points X and Y are chosen on the rays $\vec{CA}$ and $\vec{BA}$, respectively, such that $\angle CBX = \angle YCB = \angle BAC$. Assume that the tangents to the circumcircle of ABC at B and C meet line XY at P and Q, respectively, such that the points X, P, Y, and Q are pairwise distinct and lie on the same side of BC. Let Ω_1 be the circle through X and P centred on BC. Similarly, let Ω_2 be the circle through Y and Q centred on BC. Prove that Ω_1 and Ω_2 intersect at two fixed points as A varies. Denmark, Daniel Pham Nguyen

2024 — Problem 3.

Record: rou_2024_c3cc94. **Category:** contest. Source.

Given a positive integer n, a set S is *n-admissible* if: • each element of S is an unordered triple of distinct integers in $\{1, 2, \ldots, n\}$, • $|S| = n - 2$, and • for each $1 \leq k \leq n - 2$ and each choice of k distinct $A_1, A_2, \ldots, A_k \in S$,

$$|A_1 \cup A_2 \cup \cdots \cup A_k| \geq k + 2.$$

Is it true that, for all $n > 3$ and for each n-admissible set S, there exist pairwise distinct points $P_1, \ldots, P_n$ in the plane such that the angles of the triangle $P_iP_jP_k$ are all less than 61° for any triple $\{i, j, k\}$ in S?

USAMO source statements

2006 — 6

Record: `usa_2006_688c96`. **Category:** contest. Source.

Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC, respectively, such that $AE/ED = BF/FC$. Ray FE meets rays BA and CD at S and T, respectively. Prove that the circumcircles of triangles SAE, SBF, TCF, and TDE pass through a common point.

2008 — 4

Record: `usa_2008_2c2856`. **Category:** contest. Source.

Let $\mathcal{P}$ be a convex polygon with n sides, $n \geq 3$. Any set of $n-3$ diagonals of $\mathcal{P}$ that do not intersect in the interior of the polygon determine a *triangulation* of $\mathcal{P}$ into $n-2$ triangles. If $\mathcal{P}$ is regular and there is a triangulation of $\mathcal{P}$ consisting of only isosceles triangles, find all the possible values of n.

2008 — 2

Record: `usa_2008_9389ce`. **Category:** contest. Source.

Let ABC be an acute, scalene triangle, and let M, N, and P be the midpoints of $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Let the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$ intersect ray AM in points D and E respectively, and let lines BD and CE intersect in point F, inside of triangle ABC. Prove that points A, N, F, and P all lie on one circle.

2010 — 4

Record: `usa_2010_60d5ea`. **Category:** contest. Source.

Let ABC be a triangle with $\angle A = 90^\circ$. Points D and E lie on sides AC and AB, respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I. Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer lengths.

2010 — 1

Record: `usa_2010_7b28d0`. **Category:** contest. Source.

Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB. Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ, respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB.

2012 — 1

Record: `usa_2012_25ac44`. **Category:** contest. Source.

Find all integers $n \geq 3$ such that among any n positive real numbers $a_1, a_2, \ldots, a_n$ with

$$\max(a_1, a_2, \ldots, a_n) \leq n \cdot \min(a_1, a_2, \ldots, a_n),$$

there exist three that are the side lengths of an acute triangle.

2012 — 2

Record: `usa_2012_9b4f1a`. **Category:** contest. Source.

A circle is divided into 432 congruent arcs by 432 points. The points are colored in four colors such that some 108 points are colored Red, some 108 points are colored Green, some 108 points are colored Blue, and the remaining 108 points are colored Yellow. Prove that one can choose three points of each color in such a way that the four triangles formed by the chosen points of the same color are congruent.

2014 — 5.

Record: `usa_2014_59e29d`. **Category:** contest. Source.

Let ABC be a triangle with orthocenter H and let P be the second intersection of the circumcircle of triangle AHC with the internal bisector of the angle $\angle BAC$. Let X be the circumcenter of triangle APB and Y the orthocenter of triangle APC. Prove that the length of segment XY is equal to the circumradius of triangle ABC.

(This problem was suggested by Titu Andreescu and Cosmin Pohoata.)

2014 — 3.

Record: `usa_2014_eef9b1`. **Category:** contest. Source.

Prove that there exists an infinite set of points

$$\ldots, P_{-3}, P_{-2}, P_{-1}, P_0, P_1, P_2, P_3, \ldots$$

in the plane with the following property: For any three distinct integers a, b and c, points P_a, P_b and P_c are collinear if and only if $a + b + c = 2014$.

2025 — USAMO 3

Record: `usa_2025_a5e142`. **Category:** contest. Source.

Alice the architect and Bob the builder play a game. First, Alice chooses two points P and Q in the plane and a subset S of the plane, which are announced to Bob. Next, Bob marks infinitely many points in the plane, designating each a city. He may not place two cities within distance at most one unit of each other, and no three cities he places may be collinear. Finally, roads are constructed between the cities as follows: for each pair A, B of cities, they are connected with a road along the line segment AB if and only if the following condition holds: For every city C distinct from A and B, there exists $R \in S$ such that $\triangle PQR$ is directly similar to either $\triangle ABC$ or $\triangle BAC$.

Alice wins the game if (i) the resulting roads allow for travel between any pair of cities via a finite sequence of roads and (ii) no two roads cross. Otherwise, Bob wins. Determine, with proof, which player has a winning strategy.

2025 — 4

Record: `usa_2025_f546c3`. **Category:** contest. Source.

Let H be the orthocenter of acute triangle ABC, let F be the foot of the altitude from C to AB, and let P be the reflection of H across BC. Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y. Prove that C is the midpoint of XY.

VMO source statements

2003 — 1

Record: `vnm_2003_27553c`. **Category:** contest. Source.

Let be given an acute triangle ABC inscribed in a circle with center O and two points M, N on the line AC such that $\overline{MN} = \overline{AC}$. Let D be the orthogonal projection of M on the line BC, E be that of N on the line AB.

1/ Prove that the orthocenter H of triangle ABC lies on the circumcircle with the center O' of triangle BED.

2/ Prove that the midpoint of segment AN is symmetric to B with respect to the midpoint of segment OO'.

2003 — 2

Record: `vnm_2003_491dad`. **Category:** contest. Source.

In plane, let be given two fixed circles (O_1) and (O_2) touching each other at M, the radius of (O_2) is greater than that of (O_1). A is the point on (O_2) such that the points O_1, O_2, A are not collinear. Let AB and AC be the tangents to (O_1) with touching points B and C. The lines MB and MC cut again (O_2) respectively at E and F. Let D be the point of intersection of the line EF and tangent to (O_2) at A. Prove that D moves on a fixed line when A moves on (O_2) so that the three points O_1, O_2, A are not collinear.

2005 — Problem 3.

Record: `vnm_2005_05160d`. **Category:** contest. Source.

In plane, let be given an octagon $A_1A_2A_3A_4A_5A_6A_7A_8$ such that no three of its diagonals are concurrent. Each point of intersection of every two diagonals of the octagon is called a cross. Consider the convex quadrilaterals whose vertices are vertices of the given octagon. Each such quadrilateral is called a subquadrilateral. Find the least positive integer n such that one can colour n crosses so that for every $i, k \in \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $i \neq k$, if we denote by $s(i, k)$ the number of

subquadrilaterals admitting A_i, A_k as vertices and admitting a coloured cross as the point of intersection of the diagonals then the values $s(i, k)$ are all equal.

2005 — Problem 2.

Record: `vnm_2005_7f10fc`. **Category:** contest. Source.

In plane, let be given a fixed circle (O) with radius R, two fixed points A, B on the circle (O) such that A, B, O are not collinear. Let C be a point on (O) distinct from A and B. Construct the circle (O_1) passing through A, touching the line BC at C, construct the circle (O_2) passing through B, touching the line AC at C. The circle (O_1) and (O_2) intersect again at D distinct from C. Prove that

$$i)\, CD \leq R;$$

ii) The line CD passes through a fixed point when C moves on (O) so that C does not coincide with A and B. ((O) denotes a circle with center O.)

2006 — Problem 2

Record: vnm_2006_ab2541. **Category:** contest. Source.

Let be given a convex quadrilateral **ABCD**. A point **M** moves on the line **AB** but does not coincide with **A** and **B**. Let **N** be the second point of meeting (distinct from **M**) of the circles (**MAC**) and (**MBD**). Prove that

i) N moves on a fixed circle,

ii) The line MN passes through a fixed point.

(The symbol (XYZ) denotes the circle passing through the points X, Y, Z.)

2007 — 6

Record: vnm_2007_177129. **Category:** contest. Source.

Let $ABCD$ be a trapezium with the bottom edge BC ($BC \parallel AC$ and $BC > AD$) and inscribed in the circle (O) (O is the center of (O)). Let P be a point moving on the line BC outside the segment BC such that PA doesn't touch the circle (O). The circle with the diameter PD intersects (O) in E ($E \neq D$). Let M be the point of intersection of BC and DE, and N ($N \neq A$) be the second point of intersection of PA and (O). Prove that the line MN passes through a fixed point.

2007 — 3

Record: vnm_2007_5fc32b. **Category:** contest. Source.

Let $\triangle ABC$ be a triangle with two fixed vertices B, C and the vertex A is variable. Let H and G be the orthocenter and the centroid of the triangle ABC respectively. Find the locus of A such that the midpoint K of the segment HG moves on the line BC.

2009 — Problem 3

Record: vnm_2009_a85235. **Category:** contest. Source.

In the plane fix two points A, B ($A \neq B$). Consider a point C moving in the plane such that $\overrightarrow{ACB} = \alpha$, where α is a given angle ($0^\circ < \alpha < 180^\circ$). The circle with center in I, inscribed the triangle ABC touches the sides AB, BC and CA in the points D, E and F, respectively. The line AI and BI intersect EF in M and N, respectively. Show that: 1/ The segment MN has constant length; 2/ The circumcircle of the triangle DMN goes through a fix-point.

2010 — 3

Record: vnm_2010_282d47. **Category:** contest. Source.

In the plane given a circle (O) and two fixed points B, C on the circle, such that BC is not a diameter. Consider a point A moving on (O) in such a way that $AB = AC$ and A is not coincident with B, C. Denote by D and E the intersections of line BC with the internal bisector and the external bisector of angle BAC. Let I be the midpoint of DE. The line passing through

the orthocenter of ABC and perpendicular with AI intersects lines AD and AE in M and N respectively.

a/ Show that MN always passes through a fixed point.

b/ Determine the positions of A such that the triangle ABC has the largest area.

2011 — 6

Record: `vnm_2011_49c84a`. **Category:** contest. Source.

A triangle ABC is not isosceles at A, and its angles $\angle ABC$, $\angle ACB$ are acute. Consider a point D moving on edge BC, so that D does not coincide with B, C and with the perpendicular projection of A on BC. The line d, perpendicular with BC at D, intersects the lines AB and AC at E and F, respectively. Let M, N, and P be the incenters of the triangles AEF, BDE and CDF, respectively. Show that four points A, M, N, P lie on one circle if and only if the line d passes through the incenter of triangle ABC.

2011 — 3

Record: `vnm_2011_7581e4`. **Category:** contest. Source.

Given a circle (O) with diameter AB on the plane. A point P moves on the tangent at B to (O). The line PA intersects (O) in the second point C. Let D be the point symmetric to C with respect to O. The line PD intersects (O) in the second point E.

1/ Show that the lines AE, BC and PO pass through a common point. Call this point M.

2/ Determine the place of P such that the triangle ABC has maximum area. Compute that maximum area in terms of the radius of (O).

2011 — 4

Record: `vnm_2011_accee2`. **Category:** contest. Source.

Given a convex pentagon $ABCDE$, of which the length of each edge and of the diagonals AC, AD does not exceed $\sqrt{3}$. Choose 2001 arbitrary distinct points in the interior of that pentagon. Show that there exists a unit disk with center lying on the edges of the pentagon, which contains at least 403 of the chosen points.

2012 — Problem 2.

Record: `vnm_2012_2a1811`. **Category:** contest. Source.

Let ABC be a triangle inscribed in a circle (O). Let D be a point on the arc BC not containing A. Suppose that the tangent of (O) at D meets BC at E. The circumcircle of ODE meets AD at second point F (different to D). The line EF meets AB at M, the line DM meet (O) at the second point K (different to D). Prove that K, F, C are collinear.

2012 — Problem 3.

Record: `vnm_2012_65cc7c`. **Category:** contest. Source.

Let $\triangle ABC$ be a triangle inscribed in a circle (O). The point P is an arbitrary point on interior angle bisector AD of triangle ABC, D belongs to segment BC and P lies between A and D. The

line BP meets segment AC at E, the line CP meets segment AB at F. The projection of P on segment BC is M. Suppose that two lines PM, EF intersect at N and AN meets OP at L. Prove that the line ML always passes through a certain fixed point when P is moving on AD.

2012 — Problem 3.

Record: `vnm_2012_8a629d`. **Category:** contest. Source.

Let ABC be the isosceles triangle with $AB = AC$ and M, N are the midpoints of AB, AC, respectively. The circle (K) is tangent to line AB at B and tangent to AC at C. An arbitrary circle (I) tangents to (K), passes through A and cuts the line MN at D, E. Prove that:

a. Point I belongs to a fixed line and the angle $\angle MAN$ is a constant.

b. The circle with diameter DE is always tangent to two certain fixed circles and has the center belongs to the line AK.

2012 — Problem 5.

Record: `vnm_2012_99501a`. **Category:** contest. Source.

Let a, b, c be positive real numbers. It is known that the system of equations

$$\begin{cases} a^2x + b^2y + c^2z = 1 \\ xy + yz + zx = 1 \end{cases} \qquad \text{(where } a, b, c \text{ are parameters)}$$

has only one solution (x, y, z). Prove that a, b, c are the sides of a certain triangle.

2012 — 3

Record: `vnm_2012_e2d8cc`. **Category:** contest. Source.

Let $ABCD$ be a convex quadrilateral inscribed in a circle (O). Suppose that the line AB meets the line CD at M and the line AD meets the line BC at N. Let $P, Q, S,$ and T be the intersection points of the bisectors of the angles $\widehat{MAN}$ and $\widehat{MBN}$, $\widehat{MBN}$ and $\widehat{MCN}$, $\widehat{MCN}$ and $\widehat{MDN}$, $\widehat{MDN}$ and $\widehat{MAN}$, respectively. Suppose that P, Q, S and T are pairwise different.

1. Prove that four points P, Q, S and T are cyclic. Let I be the center of the cycle (P, Q, S, T).
2. Let E be the intersection of AC and BD. Prove that three points E, O, I are collinear.

2013 — Problem 6.

Record: `vnm_2013_c284da`. **Category:** contest. Source.

Given a $10 \times 10 \times 10$ box of 1000 unit white cubes. An and Binh play a game with this box. An selects some bands of size $1 \times 1 \times 10$ such that any two chosen bands have no common points and then changes all cells on these bands to black. Binh then selects some unit cubes and asks An what color of these cells. What is the least number of cells Binh must choose in order to determine all black cells based on An's answer?

2013 — Problem 4.

Record: `vnm_2013_f344b9`. **Category:** contest. Source.

The diagonals of a convex pentagon cut together divide the pentagon into one small pentagon and 10 sub-triangles. Find the maximum number of triangles among these sub-triangles such that they have the same area.

2014 — 4

Record: vnm_2014_80fcea. **Category:** contest. Source.

Given an acute, scalene triangle ABC ($AB < AC$) with the circumcircle (O). Let I be the midpoint of arc BC that does not contain A. Take K on AC ($K \neq C$) such that $IK = IC$. Line BK intersects (O) again at D and intersects AI at E. DI meets AC at F.

a) Prove that $EF = \frac{BC}{2}$.

b) Let M be the point on DI such that CM is parallel to AD. Line KM meets BC at N and the circumcircle of triangle BKN meets (O) again at P. Prove that PK passes through the midpoint of the segment AD.

2015 — 4

Record: vnm_2015_be292d. **Category:** contest. Source.

Given a circumcircle O and two fixed points B, C on O (BC is not the diameter of O). A point A is moving on O such that ABC is an acute triangle. Let E, F be the feet of the altitudes from B, C of triangle ABC. Let I be an arbitrary circle passing through E and F.

a) Let I touch BC at D. Prove that $\frac{DB}{DC} = \sqrt{\frac{\cot B}{\cot C}}$.

b) Suppose that I intersects BC at M and N. Let H be the orthocenter of triangle ABC and P, Q be the intersections of I and the circumcircle of triangle HBC. Let K be the circle passing through P, Q and touches O at T (T is on the same side with A with respect to PQ). Prove that the interior angle bisector of $\angle MTN$ passes through a fixed point.

2016 — 3

Record: vnm_2016_638715. **Category:** contest. Source.

Given a triangle ABC with fixed vertices B, C; point A moves such that triangle ABC is acute. Let D be the midpoint of BC and E, F be the projections of D to AB, AC respectively.

a) Let O be the circumcenter of triangle ABC. EF meets AO and BC at M, N respectively. Prove that the circumcircle of triangle AMN passes through a fixed point.

b) Suppose that the tangents of the circumcircle of triangle AEF at E, F intersect each other at T. Prove that T lies on a fixed line.

2016 — 6

Record: vnm_2016_787e4a. **Category:** contest. Source.

Let ABC be an acute, scalene triangle with the circumcircle (O) and angles ACB, ABC are acute. Let M be a point on arc BC that does not contain A and AM is not perpendicular to BC. The line AM meets the perpendicular bisector of BC at T and the circumcircle of triangle AOT meets O at N ($N \neq A$).

a) Prove that $\angle BAM = \angle CAN$.

b) Let I be the incenter of triangle ABC and G be the foot of the internal angle bisector of $\angle BAC$. Let AI, MI and NI intersect (O) at D, E and F respectively. Let P, Q be the intersections of DF and AM, DE and AN. The circle passing through P and touches AD at I meets DF at H ($H \neq D$). Similarly, the circle passing through Q and touches AD at I meets DE at K ($K \neq D$). Prove that the circumcircle of triangle GHK touches BC.

2018 — 4

Record: `vnm_2018_949f1a`. **Category:** contest. Source.

On the Cartesian plane, let C be the graph of the function $y = \sqrt[3]{x^2}$. A line d varies on the plane such that d always cuts C at three distinct points with x-coordinates x_1, x_2 and x_3.

a. Prove that the following value is a constant:

$$\sqrt[3]{\frac{x_1 x_2}{x_3^2}} + \sqrt[3]{\frac{x_2 x_3}{x_1^2}} + \sqrt[3]{\frac{x_3 x_1}{x_2^2}}$$

b. Show that:

$$\sqrt[3]{\frac{x_1^2}{x_2 x_3}} + \sqrt[3]{\frac{x_2^2}{x_3 x_1}} + \sqrt[3]{\frac{x_3^2}{x_1 x_2}} < -\frac{15}{4}$$

2018 — 3

Record: `vnm_2018_b1f4be`. **Category:** contest. Source.

An investor has two rectangular lands, each of size 120×100.

a. On the first land, she wants to build a house with a rectangular base of size 25×35 and nine circular flower pots with diameter 5 outside the house. Prove that for all positions of the flower pots, the remaining land is still sufficient to build the desired house.

b. On the second land, she wants to construct a polygonal fish pond such that the distance from an arbitrary point on the land, outside the pond, to the nearest pond edge is not over 5. Prove that the perimeter of the pond is not smaller than $440 - 20\sqrt{2}$.

2018 — Problem 3

Record: `vnm_2018_da4108`. **Category:** contest. Source.

A farmer has 2 rectangle lands of size 120 m $\times$ 100 m.

a. On the first land, there are 9 circle gardens of diameter 5 m. Prove that regardless to the position of gardens, he always can builds a rectangle garden of size 25 m $\times$ 35 m.

b. On the second land, he builds a convex polygon lake such that the shortest distance from any point on the boundary of the land to the lake is at most 5 m. Prove that the perimeter of the lake is at least $440 - 20\sqrt{2}$ m.

2021 — 3

Record: `vnm_2021_25d37d`. **Category:** contest. Source.

Let ABC be an acute, scalene triangle with orthocenter H and D, E, F be the feet of altitudes from vertices A, B, C respectively. Let (I) be the circumcircle of triangle HEF with centre I and

K, J be the midpoints of BC, EF respectively. HJ meets (I) again at G, GK meets (I) again at L.

a) Prove that AL is perpendicular to EF.

b) Let AL meet EF at M, IM meet the circumcircle of triangle IEF again at N and DN meet AB, AC at P, Q respectively. Prove that PE, QF and AK are concurrent.

2021 — 7

Record: `vnm_2021_7c1c6f`. **Category:** contest. Source.

Let D be the intersection of the two tangent lines of (O) at B and C. The circle passing through A and tangent to BC at B intersects the median passing A of the triangle ABC at G. Lines BG, CG intersect CD, BD at E, F respectively.

a) The line passing through the midpoints of BE and CF cuts BF, CE at M, N respectively. Prove that the points A, D, M and N lie on a same circle.

b) Lines AD, AG intersect the circumcircle of DBC, GBC at H, K respectively. The perpendicular bisectors of $HK, HE,$ and HF cut BC, CA and AB at R, P and Q respectively. Prove that the points R, P and Q are collinear.

2022 — 7

Record: `vnm_2022_03d689`. **Category:** contest. Source.

Let ABC be an acute triangle with fixed points B, C and point A moves on the big arc BC of (ABC) such that $AB \neq AC$. Incircle (I) of triangle ABC touches BC at D. Let I_a be the A-excenter of triangle ABC. I_aD cuts OI at L and E lies on (I) such that DE is parallel to AI.

a) LE cuts AI at F. Prove that $AF = AI$.

b) Let M be the point the circumcircle (J) of triangle I_aBC such that I_aM is parallel to AI. The line MD cuts (J) again at N. Prove that the midpoint T of MN lies on a fixed circle.

2022 — Problem 3

Record: `vnm_2022_3ef87d`. **Category:** contest. Source.

Let ABC be a triangle. Points E, F move on the opposite ray of BA, CA such that $BF = CE$. Let M, N be the midpoints of BE, CF. Suppose that BF cuts CE at D.

a) Let I, J be the centers of circumcircles of triangles DBE and DCF. Prove that MN is parallel to IJ.

b) Let K be the midpoint of MN and H be the orthocenter of triangle AEF. Prove that when E moves on the opposite ray of BA, line HK goes through a fixed point.

2023 — 7

Record: `vnm_2023_938036`. **Category:** contest. Source.

Let ABC be a scalene triangle with orthocenter H and circumcenter O. Incircle (I) of the ABC is tangent to the sides BC, CA, AB at M, N, P respectively. Denote Ω_A to be the circle passing through point A, external tangent to (I) at A' and cut again AB, AC at A_b, A_c respectively. The circles Ω_B, Ω_C and points $B', B_a, B_c, C', C_a, C_b$ are defined similarly.

a) Prove that $B_cC_b + C_aA_c + A_bB_a \geq NP + PM + MN$.

b) Suppose A', B', C' lie on AM, BN, CP respectively. Denote K as the circumcenter of the triangle formed by lines A_bA_c, B_cB_a, C_aC_b. Prove OH is parallel to IK.

2024 — 7

Record: `vnm_2024_9e7ac6`. **Category:** contest. Source.

In the space, there is a convex polyhedron D such that for every vertex of D, there are an even number of edges passing through that vertex. We choose a face F of D. Then we assign each edge of D a positive integer such that for all faces of D different from F, the sum of the numbers assigned on the edges of that face is a positive integer divisible by 2024. Prove that the sum of the numbers assigned on the edges of F is also a positive integer divisible by 2024.